Observable Compliance, Hidden Distortions: Cropland Protection under Incomplete Monitoring *

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November 2, 2026

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Abstract

This paper shows how multitasking bureaucrats generate within-task distortions under incomplete monitoring. I examine a tightening of cropland protection supervision in China, which strengthened enforcement of area targets in bureaucratic evaluations while leaving land quality weakly monitored. Using grid-level measures constructed from satellite imagery, I bring two identification strategies to the question. The first is a canonical difference-in-differences design that exploits cross-county variation in cropland protection pressure. The second is a spatial RD-DID design comparing pre-to-post changes across provincial borders separating higher- and lower-pressure jurisdictions. Both point to the same finding: higher-pressure jurisdictions expand cropland after the reform, but the newly reclaimed plots are less agronomically suitable and are used less intensively, with lower vegetation cover and persistently higher fallow rates. New cropland in higher-pressure jurisdictions is also more fragmented and comes disproportionately from converted water bodies and wetlands. Mechanism evidence indicates that the decline in reclaimed-land quality reflects distortions in local governments' reclamation choices rather than only movement down the land-quality frontier: reclamation shifts toward less-monitored and lower-cost sites, and quality losses are larger when officials face stronger career incentives.

Keywords: Multitasking; Imperfect monitoring; Cropland protection; Land misallocation; Bureaucratic incentives.

JEL classification: D73, D82, Q15, R52.

*I am especially grateful to my advisor, Yasuyuki Sawada, for his continued and insightful guidance. I also thank Hanming Fang, Jun Goto, Shiqi Guo, Yuxiao Hu, Shuhei Kitamura, Tong Liu, Yichuan Lou, Ming Lu, Yukichi Mano, Donnie Osborne, Yoshito Takasaki, Mari Tanaka, Shaoda Wang, Yujia Wang, Yilan Xu, and Noam Yuchtman, as well as seminar participants at the University of Tokyo and conference participants at Young JADE, ICCDS, Todai-HYI, SWET, and CES in 2026, for their constructive comments. I also acknowledge assistance from OpenAI's GPT-5.6. All errors are my own.

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1 Introduction

Fertile and accessible locations have long supported both agricultural production and dense settlement, leaving many cities surrounded by highly productive cropland. Yet this spatial overlap increasingly exposes cropland to competition from urban development. Bren d'Amour et al. (2017) estimate that urban expansion between 2000 and 2030 will convert 27 to 35 million hectares of cropland worldwide and that the affected land is 1.77 times as productive as the global average. For China, the corresponding estimate is 7.1 to 8.6 million hectares, equivalent to 8.2 to 9.8 percent of national crop production. Governments therefore face a central policy design problem: how to balance the land demands of urban development with the need to maintain cropland and sustain agricultural production.

China has responded to these pressures by strengthening cropland protection. Political authority in China is highly centralized, but policy implementation is decentralized to local governments, which retain substantial discretion over how central mandates are carried out (Xu, 2011; Zhang et al., 2018). Local bureaucrats implement multiple policy mandates and thus operate as multitasking agents (Holmstrom and Milgrom, 1991; Chen et al., 2018). Because implementation is decentralized and information is asymmetric between the central and local governments, the center cannot fully observe how these mandates are carried out. A natural response is to strengthen monitoring and performance-based assessments. Yet while monitoring remains incomplete within a task, with some dimensions observable and others not, local governments may concentrate compliance on the dimensions receiving greater scrutiny and shift distortions toward those that remain less observable.

This paper asks: when the center tightens supervision over one dimension of the cropland-protection task, does local implementation shift distortions toward less closely monitored dimensions of the same task? Cropland protection in China places local governments under two linked incentive tensions. The first is a multitask tension between urban expansion and cropland protection: local governments support urban development by converting cropland to construction use while remaining responsible for maintaining cropland quantity and quality. The second lies within cropland protection itself: the central government seeks to safeguard both cropland area and productive capacity, treating area as the basis and productive capacity as the core. I examine whether, when the conflict between urban expansion and cropland protection becomes more acute, compliance with the more closely monitored area requirement generates losses in less closely monitored land quality and actual cultivation.

Local governments in China have long been evaluated primarily on economic performance, particularly economic growth (Fan et al., 2025). To support industrialization and urban expansion, local officials have strong incentives to convert land for construction use, making land allocation a central instrument of local development policy. At the same time, concerns over food security led the central government to adopt the *Cropland Occupation–Compensation Balance Policy*, which requires each unit of cropland converted to construction use to be offset by an equal area of newly reclaimed cropland. Local governments thus operate in a multitask environment that requires balancing economic development with cropland protection.

Prior to 2017, however, cropland protection was weakly enforced despite the formal existence of quantitative targets. This changed with the 2017 reform of the cropland protection responsibility assessment system, under which compliance with cropland protection targets became a binding component of official evaluations and promotion criteria. While the reform aimed to strengthen the joint management of cropland quantity and productive capacity, anecdotal evidence suggests that implementation gave greater emphasis to the readily observable quantity target, with cropland quality and subsequent use remaining less closely monitored. This asymmetry raises a key concern: when quantitative targets become binding under incomplete monitoring, local governments may comply by expanding cropland area through the reclamation of lower-suitability plots, generating distortions in land allocation.

In this paper, I use satellite-based raster data to identify cropland and cultivation intensity. The unit of observation is a 500×500 meter grid cell. Aggregating raster measures to the grid level allows me to track changes in land allocation and land use at a spatial scale relevant for local policy implementation, while avoiding reliance on administrative statistics that may be subject to reporting biases. This study focuses exclusively on one Chinese province, Jiangsu. Only a few provincial governments, including Jiangsu's, explicitly report acute conflicts between urban expansion and cropland protection. Cropland-compensation obligations do not impose a binding constraint on economic development in every province.[1] In addition, Jiangsu is both a major agricultural producer and one of China's most economically developed provinces. Its predominantly flat terrain leaves much of the same land suitable for both cultivation and construction, sharpening competition between these uses. This setting allows me to study local responses to tighter enforcement where these obligations bind.

I leverage cross-county variation in cropland protection pressure to identify the effects of the 2017 enforcement tightening on land allocation and cultivation outcomes. Cropland protection pressure captures a county's predicted cropland-compensation burden relative to its available rural land reservoir. Under the *Cropland Occupation–Compensation Balance Policy*, compensation obligations generated by core urban districts can be met by non-core counties within the same prefecture. The prefecture-level obligation is assigned by higher-level governments, while its allocation across counties depends on each county's predetermined share of the prefecture's available rural land. Because these shares reflect long-established land endowments and administrative boundaries rather than contemporaneous county policy choices, the resulting cross-county variation in exposure is plausibly exogenous to local responses to the reform. I therefore use cropland protection pressure as a continuous measure of exposure to the common 2017 enforcement shock and compare changes before and after the reform across counties with different predetermined exposure in a difference-in-differences design.

The main difference-in-differences estimates reveal a quantity-quality trade-off in local policy implementation. Cropland shares followed similar trends before 2017 but rose more thereafter in counties facing greater cropland protection pressure. Yet newly reclaimed plots in these

[1] Section 2.5 discusses in greater detail the geographic distribution of land-use conflicts across China and the features of Jiangsu that motivate the sample choice.

counties are less suitable for cultivation, more likely to remain fallow, and exhibit lower vegetation coverage after reclamation. Tighter enforcement therefore increases the amount of land recorded as cropland while drawing less suitable and less intensively used plots into the cropland inventory.

Combining a border regression discontinuity (Black, 1999) with a difference-in-differences comparison (RD-DID; Grembi et al., 2016) across provincial borders yields the same pattern under a different source of identifying variation.[2] Along the provincial border, cropland expansion and fallowing both increase on the higher-pressure side, while pre-policy changes remain smooth at the border. This border design corroborates the extensive-margin expansion together with higher fallow and lower vegetation coverage found in the county-level analysis.

Additional evidence shows that post-reform reclamation in high-pressure counties moves farther from existing cropland and toward more isolated locations. The evidence does not support normal seasonality, temporary land preparation, or crop switching as alternative explanations for the utilization results: fallowing rises during the growing season, persists over time, and is not accompanied by clear shifts in crop composition or cropping frequency.

Mechanism evidence suggests that the decline in reclaimed-land quality reflects distorted site selection, not merely the exhaustion of suitable land. After 2017, reclamation shifts farther from provincial quality-monitoring stations, where newly reclaimed plots are less suitable for cultivation. Counties facing greater cropland protection pressure become more likely after the reform to reclaim cheaper but unsuitable land and to shift relatively away from more costly suitable land, using past cultivation as a proxy for lower reclamation costs. Bringing such land back into cultivation generally requires less effort than reclaiming land that has never been cultivated. Finally, I examine whether reclamation follows a sequence in which better land is used first and lower-quality land is reclaimed only as suitable sites become scarce. Although the suitability of the remaining land predicts the quality of newly reclaimed plots, cumulative reclamation does not consistently predict declining quality. These patterns provide indirect evidence that local governments respond to tighter acreage requirements by selecting sites with lower reclamation costs or weaker quality oversight, allowing losses along the less closely monitored quality margin.

Further evidence also points to multitask incentives operating under incomplete monitoring. The decline in the suitability of reclaimed land is larger when prefecture leaders face stronger promotion incentives. When pressure to sustain economic growth is greater, local governments expand cropland while continuing to allocate land to urban construction, and the additional cropland is less suitable. These patterns are consistent with multitask incentives shifting implementation toward dimensions of the mandate that remain less observable. The magnitudes show why so little restrains them. The agricultural losses from pressure-induced reclamation of inferior land are far smaller than the conveyance revenue raised by converting an equivalent hectare to construction use, so meeting the acreage target with poor land costs a local govern-

[2]Because the discontinuity is estimated in pre-to-post changes rather than in levels, cropland outcomes may differ across provinces, and identification requires only that expected changes evolve smoothly at the border. Section 5.2 tests this assumption. Fang et al. (2026) apply the same design to Chinese land policy.

ment almost nothing relative to what conversion earns it. This is why the pursuit of economic growth coexists with agricultural losses.

This paper contributes to the analysis of a central land-use conflict: balancing urban expansion with the need to protect cropland and sustain agricultural production. Yu (2026) also studies China's cropland protection policy and is closely related to this paper. Its analysis focuses on the extensive margin, examining how restrictions on cropland conversion to urban use constrain urban land supply and generate aggregate welfare losses. This paper extends the analysis of cropland protection to the intensive margin by studying the 2017 tightening of enforcement. The reform strengthened enforcement of cropland quantity while leaving quality and actual use less closely monitored. I show that this uneven tightening moves the distortion to the margins enforcement leaves unmeasured, with local governments meeting the observable area target by reclaiming lower-quality land that is cultivated less intensively.

Beyond this, the paper relates to three broader strands of the literature. First, this paper contributes to the multitasking principal–agent literature on performance measurement and incomplete monitoring (Holmstrom and Milgrom, 1991; Baker, 1992; Dixit, 2002). A large empirical literature studies how multitasking government agents reallocate effort across competing policy objectives when incentives or monitoring change (Chen et al., 2018; He et al., 2020; Fang et al., 2023; Axbard and Deng, 2024; Hong and Huang, 2025; Fan et al., 2025; Fisman et al., 2025; Cao et al., 2026). This paper instead focuses on distortions across differently observable dimensions of the same policy mandate. Such within-task distortions have been studied primarily in workplace and service-provider settings. In particular, Hong et al. (2018) show experimentally that quantity-based incentives increase workers' output but reduce unobserved product quality, providing direct evidence of the quantity–quality trade-off predicted by multitasking theory. This paper brings this mechanism into a central–local government principal–agent setting, and is, to my knowledge, the first to identify a within-task distortion arising from asymmetric monitoring in decentralized policy implementation: where monitoring remains incomplete, tightening enforcement on what the principal observes shifts rather than eliminates distortions.

Second, the paper relates to the literature on agricultural policy and land-use regulation (Adamopoulos and Restuccia, 2014; Cai et al., 2017; Restuccia and Santaeulalia-Llopis, 2017; Fu et al., 2021; Chari et al., 2021; Henderson et al., 2022; Chen et al., 2022; Adamopoulos et al., 2022; Chen et al., 2023; Kong et al., 2024; Adamopoulos et al., 2024; Hong et al., 2025; Chen and Li, 2025; Fang et al., 2026; Lu et al., 2026; Xing et al., 2026). Closest to this paper is Aspelund and Russo (2026), who study the Conservation Reserve Program, the largest US program paying landowners to retire cropland from production. The two settings are mirror images of one policy problem: there the government pays to move land out of farming to purchase environmental benefits, here it obliges local governments to move land into farming to protect food production. In both, what the planner scores omits the margin that determines the policy's return.

Finally, the paper connects with the literature on the Chinese economy, particularly research

on decentralization and local incentives in policy implementation (Montinola et al., 1995; Jin et al., 2005; Li and Zhou, 2005; Zhang, 2006; Xu, 2011; Zhang et al., 2018; Wang et al., 2020; Wang and Yang, 2025; Luo et al., 2025). This literature shows how decentralized implementation, top-down planning, and performance-based mandates shape local government behavior, with particular attention to information frictions and strategic responses. This paper enriches the literature by showing how institutional design itself can generate adverse effects even without collusion or deliberate manipulation. When monitoring is uneven and local governments face binding resource constraints, compliance with primary, observable targets can come at the expense of less observable environmental and agricultural outcomes.

The remainder of the paper is organized as follows. Section 2 describes the institutional background and China's cropland protection policy. Section 3 presents the data and measurement of cropland protection pressure. Section 4 documents the motivating facts. Section 5 presents the empirical strategy and main results. Section 6 examines the mechanisms underlying the land-use responses: whether the decline in reclamation quality reflects distorted implementation or costly compliance, whether the career and growth incentives facing local officials shape which parcels are reclaimed, and what agricultural losses these choices imply. Section 7 discusses the findings and concludes.

2 Institutional Background

2.1 Cropland Occupation–Compensation Balance Policy

To ensure food security and agricultural self-sufficiency, the Chinese government has long treated cropland protection as a national policy priority. China's modern cropland-protection framework dates to the 1990s, when rapid industrialization and urban expansion made limiting the loss of agricultural land increasingly urgent (State Council of the People's Republic of China, 1998).[3]

The principal policy instrument linking urban expansion to cropland protection is the *Cropland Occupation–Compensation Balance Policy*. Introduced nationally in the late 1990s, the policy imposes a one-for-one replacement rule: when a local government converts cropland to construction or another non-agricultural use, it must create an equal area of compensation cropland elsewhere. The rule therefore converts each unit of planned cropland occupation into an equal-area compensation obligation.

Compensation cropland created through land consolidation is not freely reversible. Under the *National Land Consolidation Plan (2016–2020)*, cropland consolidated under high-standard farmland construction must be designated as permanent basic farmland and protected from construction occupation (see Appendix Section G.5). Land brought into the cropland inventory as compensation therefore stays there, and any quality shortfall at the time of reclamation is

[3]The aggregate cropland-retention target within this broader framework is commonly referred to as the "cropland red line."

carried forward indefinitely.

Implementation operates through China's hierarchical land-use planning system. Upper-level governments assign a *planned land-use quota* to each lower-level jurisdiction, which caps the amount of agricultural land that may be converted into new construction land. The quota is therefore a land-conversion constraint, not a construction-land-supply target. Under the one-for-one rule, the amount of cropland planned for conversion under this quota simultaneously determines the jurisdiction's cropland-compensation obligation. The allocation cascades from the central government to provinces, prefectures, and counties.[4]

Local governments have limited spatial flexibility in meeting the resulting obligation. If a county cannot create sufficient compensation cropland within its own jurisdiction, compensation may be arranged across counties within the same prefecture. If prefecture-level adjustment is infeasible, the obligation may be met across prefectures within the same province, typically through a paid transfer mechanism. These arrangements separate the jurisdiction that generates a compensation obligation from the jurisdiction that supplies the replacement cropland, a feature central to the empirical design.

2.2 Policy Revision and Monitoring Tightening in 2017

Despite formal protection rules, cropland area continued to decline and only became relatively stable after 2017 (Appendix Figure A.1, Panel A). Both the pre-2017 decline and the post-2017 rebound are more pronounced among the six province-level units with the highest GDP per capita in 2021 (Panel B).

To strengthen enforcement of the Cropland Occupation–Compensation Balance Policy, the State Council issued the *Opinions on Strengthening Cropland Protection and Improving the Balance of Cropland Occupation and Compensation* in January 2017 (hereafter, the 2017 Opinions) (State Council of the People's Republic of China, 2017). The Opinions retained the policy's one-for-one structure but made principal local leaders accountable and linked cropland-protection performance to cadre and ecological civilization assessments. Revised provincial assessment methods followed in January 2018; provincial and prefectural governments then adopted corresponding measures for lower-level jurisdictions. Appendix Figure A.2 summarizes this administrative rollout.

The central government framed the revision as protecting both cropland quantity and productive capacity. In practice, however, contemporaneous accounts and government reports indicate that the quantitative cropland-area target was systematically assessed, while the quality of compensation cropland received much less scrutiny (see Appendix Section G.2). Writing in *People's Daily*, Wang (2023) documents low-quality reclamation and risks of abandonment and subsequent loss of newly created cropland.

[4]Under China's administrative hierarchy, county-level jurisdictions include counties, county-level cities, autonomous counties, and municipal districts. These units occupy the same formal administrative tier; the paper therefore uses "county-level" as an institutional classification rather than a sample-specific convention.

2.3 Local Bureaucrats and Urban Expansion

China's local officials implement multiple policy objectives under a performance-based personnel system. Promoting economic growth is their primary task. The evaluation system also encompasses resource use and environmental protection. Cropland protection forms part of this broader evaluation framework.

Land conversion links these objectives. Additional construction land accommodates manufacturing, infrastructure, and urban development. Conveying state-owned construction land also generates fiscal revenue. Official fiscal rules require land-conveyance receipts to be paid into local treasuries and budgeted through local government-fund accounts; permitted uses include land development and urban infrastructure (see Appendix Section G.1). These production and fiscal channels link urban expansion to local economic performance and sharpen the conflict with cropland protection.

2.4 Government-Led Land-Use Allocation

China's land-use system is organized around legally defined use controls. The Land Administration Law classifies land as agricultural, construction, or unused, requires land-use planning, and restricts the conversion of agricultural land to construction use through an approval process (see Appendix Section G.1). Transactions therefore allocate use rights only within the legally and administratively permitted category. The institutional sequence differs across the two categories.

For construction land, city or county authorities announce the location, permitted use, planning conditions, and conveyance terms of a state-owned parcel and assign its land-use right through invitation to bid, auction, or public listing; the winning user then signs a fixed-term conveyance contract. Agricultural land, by contrast, generally remains collectively owned and is contracted to rural households. Once a parcel, including reclaimed land, is designated for agricultural use, the contractor may cultivate it directly or transfer the operating right through lease or other legally permitted arrangements, but the transaction may not change its agricultural use. Thus, both construction and agricultural transactions occur within an administratively determined use category, while the relevant market instrument differs: land-use-right conveyance for construction land and transfer of operating rights for cropland (see Appendix Section G.1).

2.5 Jiangsu Province

The analysis focuses on Jiangsu because the cropland-compensation constraint is not uniformly binding across China. Appendix Figure A.3 maps provincial compensation obligations per unit of land area and shows that high-intensity obligations are concentrated in the eastern coastal region. Official reports and policy documents from Jiangsu, Zhejiang, and Guangdong explicitly state that these provinces face acute land-use conflicts and difficulty meeting cropland-compensation obligations locally (see Appendix Section G.3). By contrast, in provinces with

slower urbanization, weaker growth pressure, persistent population outflows, or more abundant reserve land, local governments face a more limited trade-off between urban expansion and cropland retention. A nationwide analysis would therefore combine jurisdictions in which the constraint is binding with many in which it is not, obscuring the mechanism of interest.

Jiangsu provides a sharp setting for studying a binding cropland-compensation constraint. Located in the Yangtze River Delta, it is both a major agricultural province, with a long history of intensive cultivation, and one of China's most economically developed provinces, with high GDP per capita and rapid urbanization. Its predominantly flat terrain makes almost all land physically suitable for cultivation, while sustained industrialization and urban expansion generate strong and persistent demand for construction land. Local officials therefore face an acute conflict between economic development and cropland protection. The aim is not to estimate nationwide aggregate effects but to observe how local governments resolve this conflict where the compensation constraint binds most tightly. Taken together, these considerations motivate the use of Jiangsu data.

3 Data and Measurement

3.1 Data Sources and Measurement

This study primarily relies on satellite-based raster data to measure land characteristics at a fine spatial resolution. To harmonize datasets with different spatial and temporal resolutions and to facilitate empirical analysis, I overlay all raster layers onto a uniform 500×500 meter grid covering the study area. For each grid cell, I compute average values of the underlying raster measures, which constitute the main unit of observation in the analysis. Appendix Table B.1 summarizes the data sources.

3.1.1 Cropland

As summarized in Appendix Table B.1, cropland information is obtained from the 30-meter annual land cover dataset for China (CLCD; Yang and Huang (2021)). I use the CLCD land-cover class labeled "cropland" as the empirical measure of cropland throughout the analysis and use annual files from 2012 through 2022. The dataset distinguishes among cropland, forest, shrubland, grassland, barren land, impervious surface, water bodies, snow, and ice. It therefore allows me to identify both cropland and built-up areas, with the latter captured by the impervious-surface category.

3.1.2 Fractional Vegetation Cover and Measurement of Fallow Rate

Fractional Vegetation Cover (FVC) data are obtained from the National Tibetan Plateau Data Center (Zhao et al., 2023). The original dataset provides spatiotemporally continuous vegetation cover for China at a 30-meter resolution and a bimonthly frequency from 2010 to 2022.

Following standard practice in the remote sensing literature, I aggregate the bimonthly observations to annual averages at the grid-cell level, which serve as a measure of vegetation coverage for each grid-year.

I use FVC to measure land-use intensity and to define fallow land. A grid cell is classified as fallow in a given year if its annual average FVC is below 0.2, which corresponds to the bottom 25 percent of the FVC distribution across cropland grids. This threshold captures land with very low vegetation cover and limited cultivation activity. In Appendix Figure A.6c, I examine the sensitivity of the results to alternative FVC cutoffs. The estimates remain stable when the threshold is chosen sufficiently low, indicating that the main findings are not driven by the specific choice of cutoff.

3.1.3 Soil Type and Measurement of Soil Suitability

The first soil measure comes from the 1:1,000,000 Soil Map of China compiled by the National Soil Survey Office in 1995. The second comes from the Harmonized World Soil Database v2.0, released in 2023 at an approximately 1-kilometer resolution. For both measures, I map soil types to a common five-category suitability scale using the FAO Land Evaluation Framework and China's National Land Capability Standard (GB/T 28407–2012). Appendix Table B.2 presents the mapping. Each soil type receives a score from 1 to 5: permanently unsuitable (1), currently unsuitable (2), marginally suitable (3), moderately suitable (4), or highly suitable (5). The assignment reflects the agricultural suitability of each soil group; for example, alluvial soils receive a score of 5, whereas saline and sodic soils receive a score of 2. Because the frameworks provide qualitative guidance instead of a one-to-one correspondence between soil types and numerical scores, the mapping provides a coarse proxy for inherent land quality.

I construct two grid-level outcomes from each soil measure. The continuous outcome, which I call the *soil suitability score*, is the area-weighted average suitability score within each 500×500 meter grid cell and ranges from 1 to 5. To construct the second outcome, I first assign a suitable-soil indicator to each source pixel, equal to one if its suitability score is at least 3 and zero otherwise. I then take the area-weighted average of these pixel-level indicators within each grid cell. I refer to this second outcome as *soil suitability*. It ranges from zero to one and measures the share of mapped land classified as suitable. This cutoff separates soils that are marginally suitable or better from soils classified as currently or permanently unsuitable, including saline, sodic, and waterlogged soils.

3.1.4 Administrative and Socioeconomic Data

County-level socioeconomic variables are obtained from the China County Statistical Yearbooks. The data include information on economic activity, agricultural production, population, and fiscal conditions. Cropland compensation obligations and the classification of core urban districts are collected from prefecture-level Land Use Master Plans, which specify each jurisdiction's targets for urban expansion and cropland compensation through 2020. Administrative

boundaries come from mainland China county-level shapefiles distributed through Harvard Dataverse. Appendix Table B.1 summarizes the remaining data sources.

3.2 Measuring Cropland Protection Pressure

This subsection constructs a county-level measure of exposure to the prefecture-level cropland compensation burden. The measure captures the cropland protection pressure faced by local officials, which reflects the intensity of multitask pressure arising from the need to satisfy cropland protection targets while accommodating urbanization and development demands. It serves as the key explanatory variable in the empirical analysis.

The construction of the cropland protection pressure measure exploits institutional features of China's Cropland Occupation–Compensation Balance Policy. Two policy rules are central to this design:

1. **One-for-one occupation–compensation rule.** Any conversion of cropland to construction land must be offset by the creation of an equivalent amount of compensation cropland.

2. **Non-local compensation with prefecture-level priority.** Compensation is not necessarily required to occur within the converting county. When core urban districts lack sufficient land, cropland compensation is, in principle, fulfilled through *cross-county adjustment within the same prefecture*. Only if prefecture-level adjustment is infeasible may compensation be arranged across prefectures within the same province, typically through a paid transaction mechanism.

Together, these rules imply that cropland compensation is organized around prefecture-level obligations and the non-core available-land reservoir. Because core urban districts are primarily tasked with meeting economic growth targets, their cropland compensation obligations are frequently met by counties within the same prefecture.[5] As a result, non-core counties absorb compensation obligations generated by urban expansion, creating variation in cropland protection pressure across counties.[6]

Based on this structure, I define cropland protection pressure at the county level as

$$\text{Pressure}_c = \left(\frac{\text{Obligation}_p}{\text{TotalLand}_p} \right) \times \text{LandShare}_c \times 1000, \tag{1}$$

where p denotes the prefecture to which county c belongs. Obligation_p measures the total cropland compensation obligation assigned to the core urban districts of prefecture p under

[5] This practice is documented in official planning materials. For example, Lianyungang City's Land Use Master Plan states that "districts such as Haizhou District and Lianyun District that are unable to achieve local cropland reclamation—replacement balance due to insufficient land reserves may, upon approval, implement off-site reclamation—replacement within the municipal jurisdiction, provided that the aggregate reclamation obligation remains unchanged."

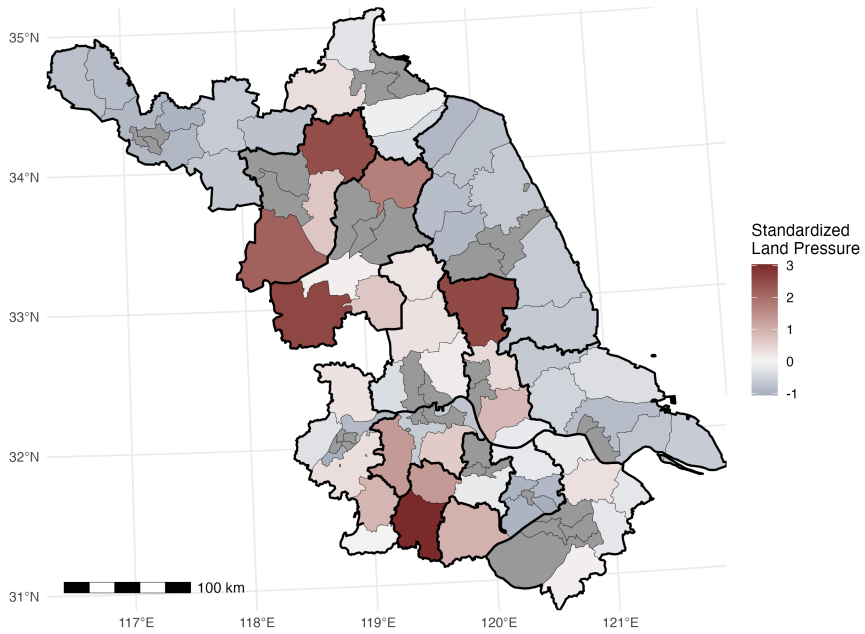
[6] The definition of the core urban area is specified in each prefecture's Land Use Master Plan (2006—2020).

provincial land-use planning targets. TotalLand_p captures the total area of available land outside core urban districts within prefecture p, excluding impervious surfaces and water bodies, and represents the effective land reservoir for cropland compensation. LandShare_c denotes county c's share of this reservoir, defined as the ratio of available land in county c to the total available land summed across all non-core counties within the same prefecture.

This construction implies that counties with larger shares of non-core, available land are more exposed to cropland compensation pressure when prefecture-level obligations increase. Importantly, LandShare_c is determined by pre-existing land endowments and administrative boundaries, and is fixed prior to the tightening of cropland protection in 2017. As a result, conditional on prefecture-level obligations, cross-county variation in cropland protection pressure reflects an exogenous allocation of the delegated compensation burden rather than endogenous policy choices by county governments. Conceptually, this structure resembles a shift–share design, in which prefecture-level compensation obligations act as the common shock, while county-level land shares determine differential exposure. This feature underpins the empirical strategy used in subsequent sections.

Figure 1 maps the standardized cropland protection pressure index across Jiangsu counties, while Appendix Figure A.7 summarizes the distribution of the index across counties by prefecture. The map shows substantial spatial heterogeneity in exposure. The box plot shows both differences in prefecture-level medians and dispersion across non-core counties within the same prefecture. Together, the two figures show that the pressure index has sufficient variation both across and within prefectures.

Figure 1: Standardized Cropland Protection Pressure in Jiangsu



Notes: The map shows county cropland protection pressure in Jiangsu, standardized to mean zero and unit standard deviation across non-core counties. Raw pressure equals the prefecture's reclamation obligation divided by available non-core land, multiplied by the county's share of that land and by 1,000. Available land excludes impervious surfaces and water bodies. Grey areas are core urban districts excluded from standardization. Thin and thick black lines mark county and prefecture boundaries, respectively.

To formally distinguish the within- and between-prefecture components of variation in this key variable, I conduct a simple variance decomposition based on the ANOVA identity.[7] Specifically, total variation in $Pressure_c$ is decomposed into variation across prefecture-level means and variation across counties within the same prefecture. The results indicate that 39.8% of the total variation in the pressure index arises from within-prefecture differences across counties, while the remaining 60.2% reflects differences across prefectures.

4 Motivating Facts

4.1 Land-Use Area Trends by Cropland Protection Pressure

Descriptive analysis reveals a central contrast between high- and low-pressure prefectures. Following the 2017 tightening of cropland protection, urban expansion trajectories remain similar across prefectures, whereas cropland adjustment differs sharply by cropland protection pressure. Figure 2 presents prefecture-demeaned trends in impervious surface area and cropland area, separately for core and non-core areas and by cropland protection pressure.

[7] $\sum_{c,p}\left(X_{cp}-\bar{X}\right)^2=\sum_p n_p\left(\bar{X}_p-\bar{X}\right)^2+\sum_p\sum_{c\in p}\left(X_{cp}-\bar{X}_p\right)^2$.

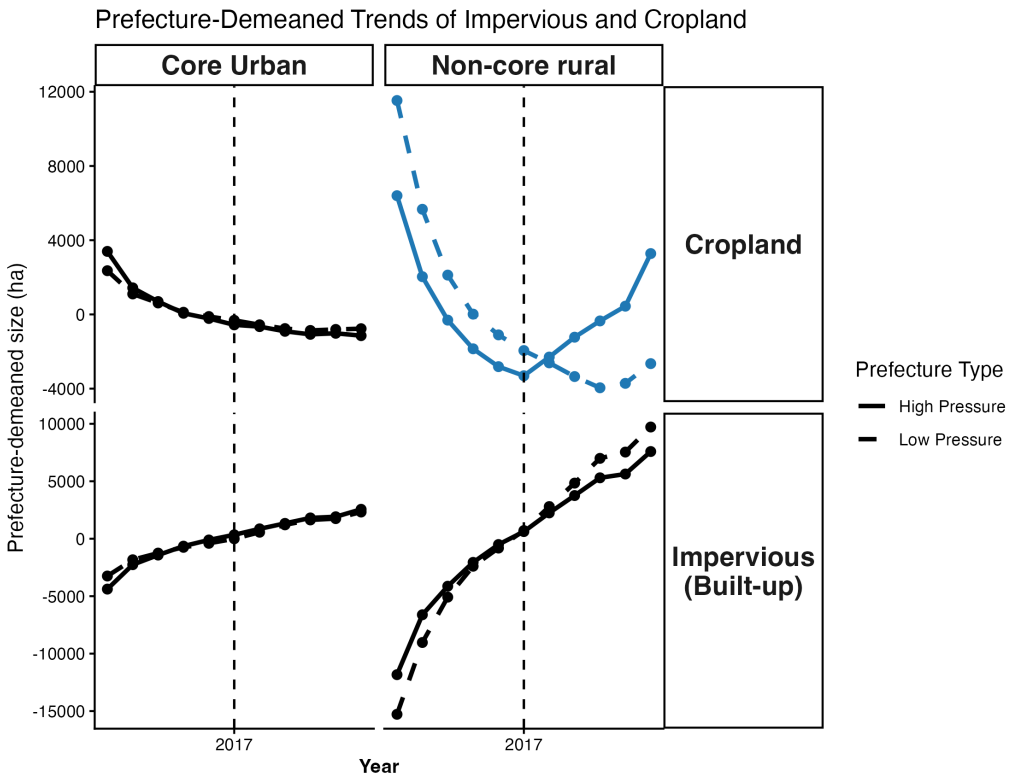


Figure 2: Trends in impervious area and cropland area by land type

Notes: Lines report prefecture averages of annual cropland and impervious-surface area, in hectares, after demeaning each prefecture-by-land-type series over time. Core urban and non-core areas are shown separately. High-pressure prefectures are above the median prefecture pressure; the remaining prefectures are low-pressure. The vertical line marks 2017.

Across both core and non-core areas, trends in impervious surface expansion are broadly comparable between high- and low-pressure prefectures, with steady growth and no visible divergence around 2017. This pattern indicates that urban development proceeded along similar paths regardless of cropland protection pressure, in line with the continued prioritization of economic growth objectives.

In contrast, cropland dynamics diverge markedly by cropland protection pressure after 2017. Prior to the policy tightening, cropland area declines in both high- and low-pressure prefectures. After 2017, however, cropland area rebounds and increases in high-pressure prefectures, while continuing to decline and exhibiting only a limited and delayed recovery in low-pressure prefectures. This divergence is concentrated in non-core areas; cropland trends within core urban areas remain similar across prefecture types throughout the period.

These patterns suggest that the 2017 policy tightening primarily constrained high-pressure prefectures, while having limited effects in low-pressure prefectures. At the same time, urban expansion in core areas appears insulated from these constraints, implying that the burden of meeting binding cropland protection targets was shifted toward non-core areas within high-pressure prefectures.

4.2 Cropland Quality under the Occupation–Compensation Balance

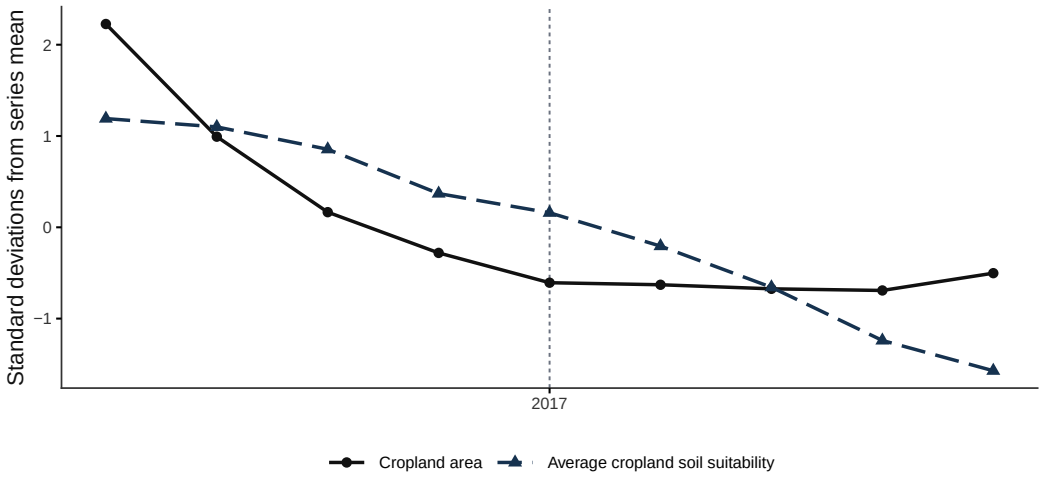
Figure 3a shows cropland quantity and cropland quality moving apart in Jiangsu. Cropland area declines before 2017 and remains broadly stable after the tightening of cropland protection, while average soil suitability among cropland-dominant grids continues to decline throughout. A stable cropland inventory thus coincides with a falling average quality of the land that remains. This raises the concern that the cropland created to compensate for urban expansion is not as good as the cropland that expansion destroys.

China's occupation–compensation policy requires each unit of cropland converted to construction use to be offset by newly reclaimed cropland of equal area and of no lower quality. The two are not monitored equally: the area target is verified and reported upward, quality only loosely. Whether the quality requirement binds in practice is therefore an empirical question. Figure 3b compares the land actually converted in each direction: cropland that becomes impervious surface, and non-cropland that becomes cropland. Occupied cropland is more suitable than newly reclaimed cropland in every year and in both pressure groups, and its suitability is stable throughout, near 93 percent in high-pressure prefectures and 89 percent in low-pressure ones. The adjustment falls entirely on the compensating land. After 2017 the suitability of newly reclaimed land declines in high-pressure prefectures, from 76 to 69 percent, while in low-pressure prefectures it rises from 62 to 67 percent. The gap between occupied and newly reclaimed land therefore widens by about eight percentage points where compensation pressure is high and narrows by about five where it is low. These patterns are consistent with the asymmetry in monitoring: where the area obligation binds, it is met with land of lower quality.

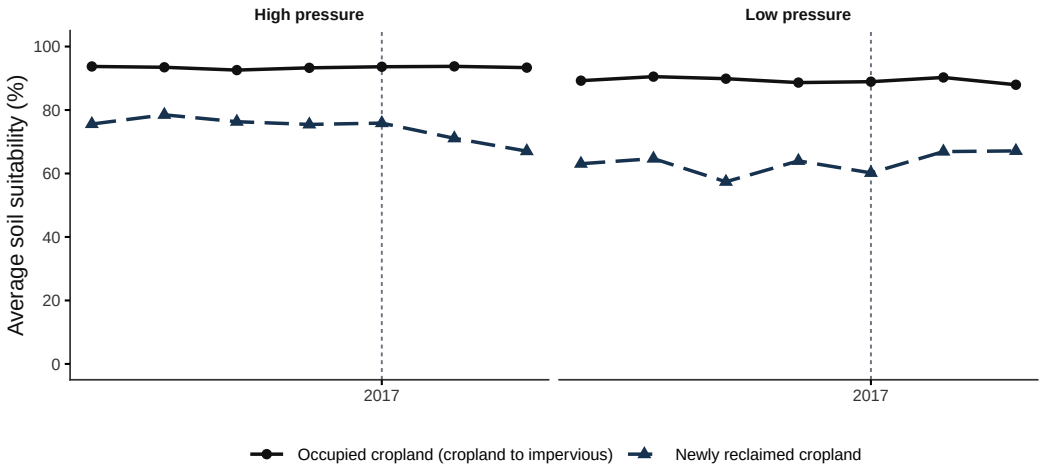
4.3 Vegetation Coverage over the Reclamation Cycle

Figure 4a illustrates the evolution of fractional vegetation coverage on newly reclaimed land as a function of years since reclamation. Pooling all newly reclaimed grids in the sample, mean FVC increases monotonically over time. This pattern indicates that vegetation coverage improves gradually following reclamation, indicating that FVC captures the intensity of agricultural use and land cultivation in the years after cropland creation.

Figure 3: Cropland Area, Quality, and the Occupation–Compensation Balance



(a) Cropland area and average soil suitability



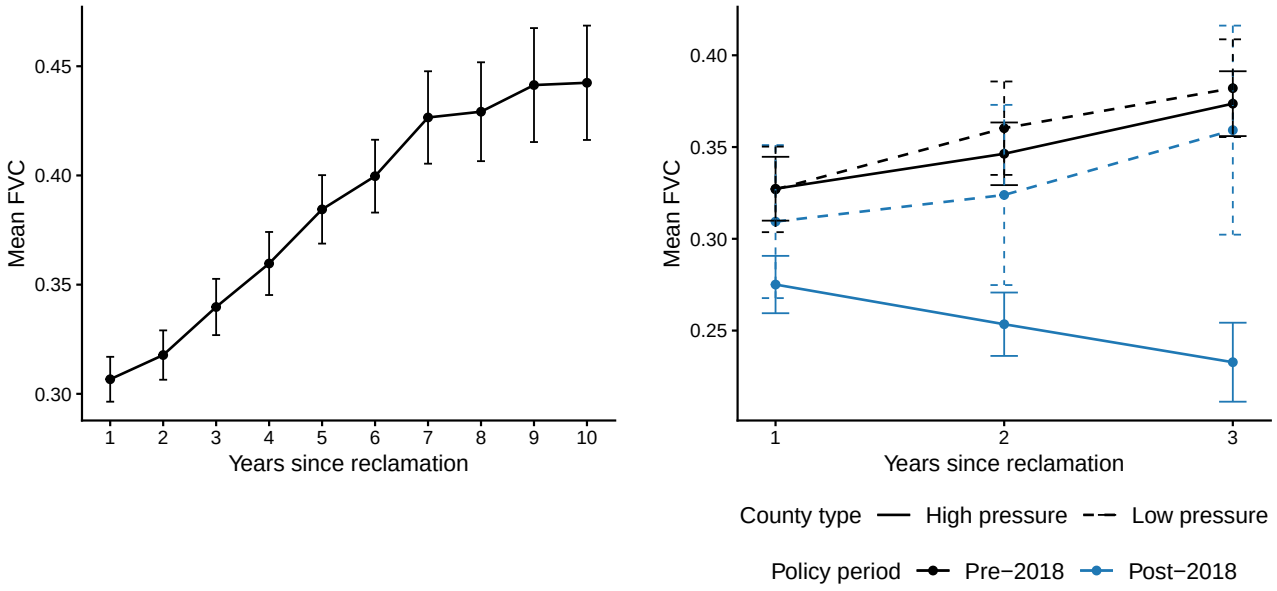
(b) Soil suitability of occupied and newly reclaimed cropland

Notes: Panel (a) reports cropland area and average soil suitability across cropland-dominant 500-meter grids, each series demeaned and divided by its own standard deviation. Panel (b) reports the share of 30-meter pixels that actually change class whose 1995 soil suitability score is at least three. Occupied cropland is cropland that becomes impervious surface; newly reclaimed cropland is non-cropland that becomes cropland. A pixel enters only if its land cover is stable over the four years on each side of the transition, which ends panel (b) in 2019. Shares pool all transition pixels within a series-by-pressure-by-year cell. High-pressure prefectures are above the prefecture median of cropland protection pressure. The dashed line marks the 2017 tightening of cropland-protection enforcement. Both panels are descriptive.

Figure 4b further disaggregates these patterns by cropland protection pressure and policy period. For grids reclaimed prior to 2018, as well as for grids located in low-pressure prefectures, FVC exhibits a similar upward trajectory to that observed in the full sample, increasing with years since reclamation. In contrast, newly reclaimed land in high-pressure prefectures after the policy tightening displays a markedly different pattern. These grids exhibit substantially lower initial FVC and show little subsequent improvement over time, with mean vegetation coverage remaining persistently low.

In a nutshell, these descriptive patterns suggest that while newly reclaimed land typically becomes increasingly cultivated over time, this process is disrupted for cropland created after the policy tightening in high-pressure prefectures. The persistently low vegetation coverage observed for these grids is consistent with lower post-reclamation cultivation or temporary fallow, whereas similar patterns are not observed in low-pressure prefectures. These motivating facts point to heterogeneous responses to the tightening of cropland protection, concentrated in regions with higher assigned compensation burdens.

Figure 4: Vegetation Coverage after Reclamation



(a) All newly reclaimed land

(b) By cropland protection pressure and policy period

Notes: Points show mean fractional vegetation coverage (FVC) in non-core newly reclaimed grids by years since first reclamation. Reclamation requires cropland shares below 0.60 in the preceding year and above 0.70 in the reclamation and following years. Panel (b) separates cohorts reclaimed before 2018 and from 2018 onward and counties above versus at or below the median pressure across non-core grid-year observations. Bars are descriptive 95 percent confidence intervals, calculated as the mean plus or minus 1.96 standard errors without clustering.

5 Empirical Analysis

The empirical analysis proceeds in four parts. First, the difference-in-differences analysis estimates effects on three primary outcomes: cropland expansion, soil suitability, and cultivation intensity. It combines an event-study design for cropland share with cohort difference-in-differences designs for the characteristics and subsequent use of newly reclaimed land. Second, a spatial RD-DID design compares changes across the higher- and lower-pressure sides of provincial borders. Third, the subsection on additional results and alternative explanations reports other spatial and cropping patterns and tests whether seasonality, temporary land preparation, or crop switching can account for the cultivation results. Finally, the robustness analysis varies

the construction of cropland protection pressure, newly reclaimed land, and fallow status.

5.1 Difference-in-Differences

5.1.1 Event-Study DiD: Cropland Area

The 2017 policy tightening constitutes a common shock across all counties, while cross-county differences in cropland protection pressure capture each locality's differential exposure to the tightening. A standard difference-in-differences design exploits this variation to identify the policy's effects on cropland expansion.

I estimate the following two-way fixed effects event-study specification:

$$\text{Cropshare}_{ict} = \sum_{k \neq -1} \beta_k \big(\text{Pressure}_c \times \text{Event}_{k,t}\big) + \mu_i + \lambda_t + \varepsilon_{ict}, \tag{2}$$

where Cropshare_{ict} denotes the cropland share of grid cell i in county c in year t. Pressure_c measures county-level cropland protection pressure, and $\text{Event}_{k,t}$ is an indicator for event time k, indexed relative to the first post-policy year, 2018. The omitted category is $k = -1$, corresponding to the 2017 reference year. Grid fixed effects μ_i absorb time-invariant spatial heterogeneity, while year fixed effects λ_t control for common shocks affecting all locations. The estimation sample excludes core urban districts, and standard errors are clustered by county.

The coefficients β_k trace out how the change in cropland share responds over time to the policy tightening as a function of cropland protection pressure. Identification relies on the assumption that, absent the policy change, cropland trends would have evolved similarly across counties with different levels of cropland protection pressure. The pre-policy period of the event study allows this parallel-trends assumption to be assessed directly.

5.1.2 Cohort DiD: Cropland Quality

While the event-study design captures aggregate changes in cropland area, it does not directly answer how the intensive margin of newly created cropland, namely its suitability and subsequent use, is affected. Before a grid cell meets the reclamation criterion, post-reclamation suitability and utilization are not defined, so no comparable pre-reclamation outcome can be observed. An event study based on outcome levels before and after reclamation is therefore inapplicable. I instead study post-reclamation outcomes in a cohort-based difference-in-differences design, where a cohort is the set of grid cells first classified as reclaimed in a given year.

The sample comprises non-core grid cells first classified as reclaimed between 2013 and 2021. Each grid cell i, located in county c, is characterized by a reclamation year t_0. For time-varying outcomes, cells are observed one to four years after reclamation. The outcome variable $Y_{i,s}$ captures post-reclamation outcomes along two intensive-margin dimensions: the suitability of newly created cropland and its subsequent utilization. The baseline empirical model is:

$$Y_{i,s} = \alpha_c + \mu_{t_0} + \delta_s + \beta\big(\mathbf{1}\{t_0 \geq 2018\} \times \text{Pressure}_c\big) + \varepsilon_{i,s}, \tag{3}$$

where α_c are county fixed effects, μ_{t_0} are reclamation-cohort fixed effects, and δ_s are years-since-reclamation fixed effects, which enter only for time-varying outcomes and absorb common post-reclamation dynamics. The coefficient β multiplies the interaction of a post-2017 reclamation indicator $\mathbf{1}\{t_0 \geq 2018\}$ with county-level cropland protection pressure Pressure_c, capturing differential exposure to the policy tightening among land reclaimed after the reform. Additional specifications interact baseline county population, GDP, land area, and government revenue with a linear calendar-time term. Standard errors are clustered by county.

Identification relies on a parallel-trends assumption: absent the policy tightening, the outcomes of newly reclaimed land in high- and low-pressure counties would not have diverged across reclamation cohorts. Appendix Table B.3 probes this assumption within the pre-policy sample by assigning false first-treated cohorts in 2014, 2015, and 2016 and re-estimating the baseline specification; none of the placebo estimates is statistically distinguishable from zero.

5.1.3 DiD Results

Figure 5 first establishes the extensive-margin response. Beginning in 2018, counties facing greater cropland protection pressure respond more strongly to the policy tightening and experience larger increases in cropland share than counties facing less pressure. All five post-policy estimates are positive and statistically significant. Before 2017, the estimates show no detectable differential pre-trend. None of the pre-policy coefficients is statistically significant. The data therefore provide no evidence that the parallel-trends assumption is violated. The contrast between the pre- and post-policy estimates supports interpreting the post-policy divergence as an effect of the tightening.

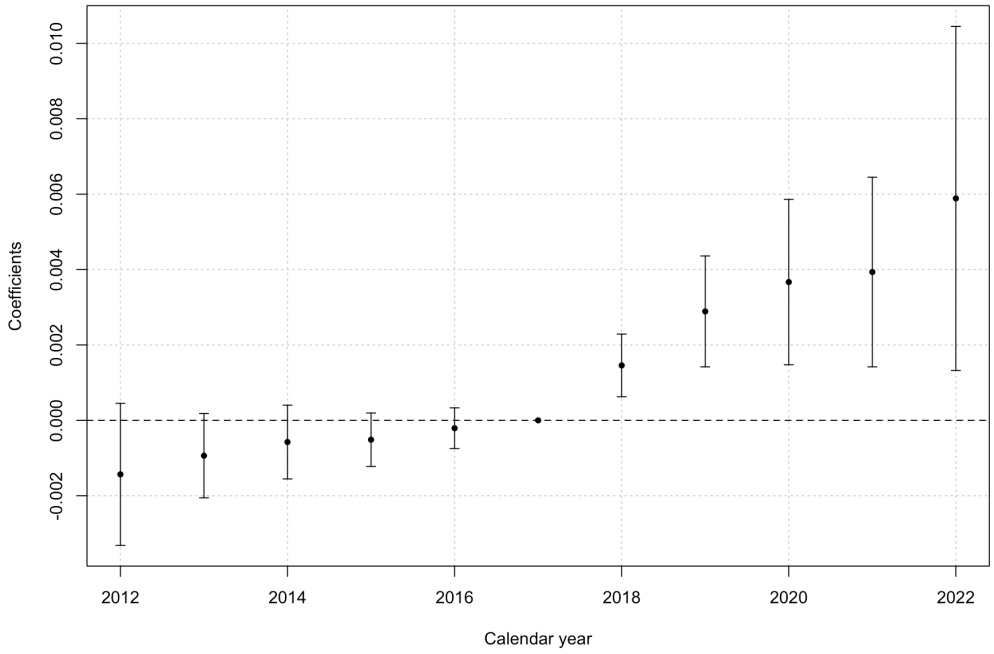


Figure 5: Effect on Cropland Share

Notes: The figure plots coefficients on interactions between cropland protection pressure and calendar-year indicators in a panel of 500 × 500 meter grid cells in non-core counties. The outcome is cropland share; the specification includes grid-cell and year FE, with 2017 omitted. Vertical bars show 95 percent confidence intervals based on standard errors clustered by county.

The estimates in Figure 5 are small in percentage-point terms because cropland reclamation is a rare event at the grid-year level. In any given year, only a small fraction of Jiangsu's land is converted to cropland, so the policy-induced increase in reclamation is smaller still when expressed as a change in cropland share averaged across all grid cells. To assess its economic significance, Appendix Table B.4 standardizes cropland protection pressure to mean zero and unit standard deviation and benchmarks the resulting policy effect against historical cropland expansion. In 2018, a one-standard-deviation increase in pressure raises cropland share by 0.16 percentage points, equivalent to approximately 13,700 hectares in the non-core sample. This increase is 84 percent of the sample's average annual gross cropland expansion during 2013–2017, approximately 16,200 hectares per year.

Table 1 turns from how much cropland counties created to what kind of land they created and whether they subsequently used it. Across both the 1995 China Soil Map and HWSD v2 2023, greater exposure to the tightening predicts a lower share of land suitable for cultivation within newly reclaimed grids. The estimates remain similar when baseline county characteristics are allowed to generate differential time paths, and the same post-policy cohorts are also more likely to remain fallow. The two outcomes capture distinct margins: soil suitability reflects the underlying agricultural quality of the selected location, whereas fallow status captures realized use after reclamation. Their joint movement therefore indicates that the response was not limited to expanding cropland onto less suitable land. High-pressure counties also brought into the cropland inventory plots that were less likely to be actively cultivated. The agreement

across two soil classifications and a separate utilization measure makes it unlikely that the result is driven by a particular land-quality dataset or outcome definition.

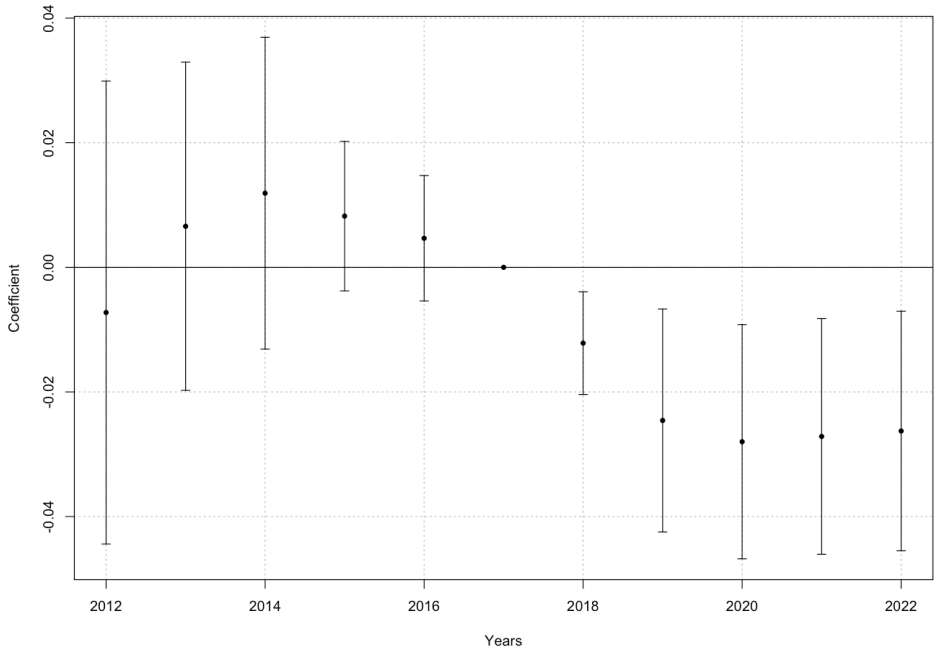
Table 1: Pressure, Land Suitability, and Fallow Status

	Soil suitability				**1**(Fallow)	
	China Soil Map 1995		HWSD v2 2023			
	(1)	(2)	(3)	(4)	(5)	(6)
Pressure × Post	-0.051**	-0.057***	-0.032*	-0.037*	0.055***	0.056***
	(0.020)	(0.021)	(0.018)	(0.019)	(0.015)	(0.017)
Control Mean	0.593	0.593	0.573	0.573	0.198	0.200
County controls × year		✓		✓		✓
County & Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE					✓	✓
Cluster level	County	County	County	County	County	County
Number of counties	52	52	52	52	54	53
Num. Obs.	939	939	939	939	3,244	3,222
R^2	0.452	0.457	0.490	0.491	0.232	0.230

Notes: The table reports cohort difference-in-differences estimates from Model 3. Columns (1)–(4) observe each newly reclaimed grid once. Columns (1)–(2) use the 1995 China Soil Map, and Columns (3)–(4) use HWSD v2 2023. For each source, suitability is the area-weighted mean within a 500-meter grid of source-pixel or soil-component indicators equal to one when the suitability score is at least three. The resulting suitable-land share ranges from zero to one; the threshold is applied before aggregation. Columns (5)–(6) observe newly reclaimed grid cells by post-reclamation year and measure fallow status (fractional vegetation coverage below 0.20) in the first four post-reclamation years. Control Mean is the mean for pre-2018 reclamation cohorts within each column's estimation sample. All specifications include county and reclamation-cohort FE; the fallow specifications also include years-since-reclamation FE. Even-numbered columns add baseline county population, GDP, government revenue, and land area interacted with a linear calendar-time term. Standard errors clustered by county are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fallow status is defined only after reclamation, so the same binary outcome cannot provide a conventional pre-trend test. Figure 6 instead uses county-year vegetation coverage among grids that are ever reclaimed during the sample period. Before 2017, FVC follows similar paths across counties with different levels of cropland protection pressure. After the tightening, it declines more in high-pressure counties. This break provides an outcome available on both sides of the reform and therefore offers complementary support for the parallel-trends assumption underlying the cohort design. It also corroborates the fallow result with a continuous measure of realized vegetation, reducing the concern that the utilization finding is an artifact of the threshold used to classify land as fallow.

Figure 6: Effect of Cropland Protection Pressure on County-Level FVC among Ever-Reclaimed Grids



Notes: The figure plots interactions of county cropland protection pressure with year indicators, omitting 2017. The outcome is county-year mean fractional vegetation coverage among ever-reclaimed grids whose current cropland share exceeds 0.60. The sample excludes core counties. The specification includes county and year FE; bars show 95 percent confidence intervals based on county-clustered standard errors.

County-level output data provide complementary evidence on realized agricultural production. I divide the sum of grain, oilseed, and cotton output by satellite-measured cropland area and estimate a county-year event study using the paper's cropland protection pressure measure, with county and year fixed effects. Appendix Figure A.8 plots the year-specific coefficients relative to 2017. The pre-policy estimates are small and statistically insignificant, and a joint test cannot reject that the 2012–2016 coefficients equal zero ($p = 0.741$). The estimate is -0.012 log points in 2018 and is marginally significant at the 10 percent level ($p = 0.085$). From 2019 through 2022, the estimates range from -0.025 to -0.030 log points and are statistically significant at the 5 percent level in every year. Crop output per cropland hectare therefore begins to decline with greater policy exposure in 2018, with a larger and persistent gap from 2019 onward. Because this outcome combines county-reported production with satellite-measured cropland area, I interpret it as complementary to the grid-level evidence on vegetation and fallowing.

The results show two linked responses to the policy tightening. More exposed counties expanded cropland rapidly after 2017, but the newly created land was less agriculturally suitable and less intensively cultivated. Tighter enforcement therefore increased the amount of land recorded as cropland while newly reclaimed plots were more likely to remain fallow and exhibited lower vegetation coverage.

5.2 Spatial RD-DID

5.2.1 Design

The county-level difference-in-differences design identifies differential responses to the 2017 policy tightening from cross-county variation in land protection pressure. I complement that analysis with a spatial RD-DID design—combining a border regression discontinuity with a difference-in-differences comparison—that asks whether the same response appears among locations that are geographically close but fall under jurisdictions with different protection pressure. The design compares changes in cropland outcomes among grid cells located on opposite sides of a provincial administrative border.

Appendix Figure A.9 depicts the difference in provincial cropland protection pressure across the border between Jiangsu and Anhui. Jiangsu faces greater provincial cropland protection pressure than neighboring Anhui. The main RD-DID comparison is therefore between nearby grid cells on the Jiangsu and Anhui sides of their shared provincial border.

The analysis compares changes in cropland outcomes across adjacent counties along the border corridor between Jiangsu and Anhui. I construct pairs of counties that share the provincial border and form the analysis pool from non-mountain grid cells located within a 50 km corridor on either side of the border. The running variable is the signed distance from each grid-cell centroid to the provincial border, with positive values on the Jiangsu side and the cutoff at zero. The outcome, ΔY_i, is the cell's mean post-policy outcome minus its mean pre-policy outcome. I estimate the local linear specification

$$\Delta Y_i = \alpha + \tau J_i + \beta_1 d_i + \beta_2 (J_i \times d_i) + \boldsymbol{P}_i' \gamma + \varepsilon_i, \qquad |d_i| \leq h, \tag{4}$$

where J_i equals one for a cell on the Jiangsu side, d_i is signed distance to the border, and $\boldsymbol{P}_i$ is a vector of cross-border county-pair indicators. The coefficient τ is the local Jiangsu-minus-Anhui discontinuity in the outcome change at the border. For each kernel, the county-pair indicators enter the local regression additively, and the MSE-optimal bandwidth is selected for that same covariate-adjusted estimator (Calonico et al., 2014, 2019; Cattaneo et al., 2024).

A central feature of the design is that the discontinuity is estimated in changes rather than in levels. For each grid cell, I define the dependent variable as the post-policy outcome relative to its pre-policy benchmark and estimate whether this change is discontinuous at the provincial border. The resulting estimator is thus a spatial RD-DID—a regression discontinuity in outcome changes—and allows cropland outcomes to differ in levels across the two provinces.

The identifying assumption is that, absent the 2017 tightening, expected changes in cropland outcomes would have evolved smoothly at the border. The design accordingly assumes parallel counterfactual changes among nearby grid cells on opposite sides of the border. Cell locations and the provincial boundary are fixed, so they cannot be chosen in response to the 2017 tightening. The main identification concern is that another contemporaneous factor may also have changed discontinuously at the border.

I probe the parallel-change assumption by estimating the same local linear RD-DID sepa-

rately for changes relative to 2012 in each year from 2013 through 2016. Appendix Table B.6 reports that none of the pre-policy cropland-share or fallow-rate discontinuities is statistically different from zero across the three kernels. The falsification evidence therefore supports the parallel-change assumption. Appendix Table B.7 further reports falsification tests for discontinuities in outcomes that the policy should not affect. Neither the pre-policy outcomes nor the policy invariant outcomes exhibit a statistically significant discontinuity at the border in the full sample of eligible grid cells.

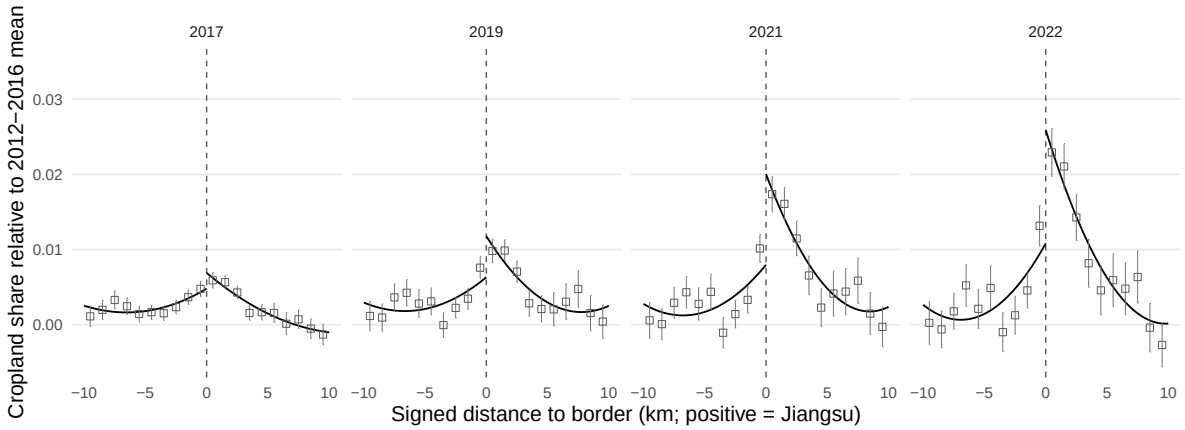
5.2.2 Results

Figure 7a plots cropland-share changes at the Jiangsu and Anhui border. In each panel, the outcome is the cropland share in the indicated year minus the same cell's average cropland share from 2012 to 2016.[8] The discontinuity is small in 2017 and widens in later years. By 2022, the change in cropland share exhibits a positive discontinuity at the border, with a larger increase on the Jiangsu side than on the Anhui side. This spatial RD-DID result is consistent with the county-level DiD estimates: after the 2017 policy tightening, cropland expands more where cropland protection pressure is greater.

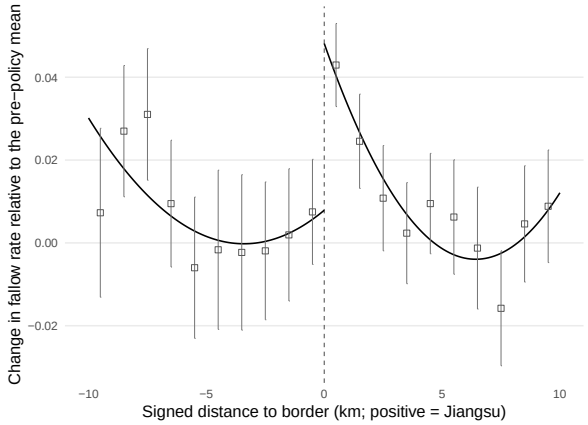
[8]The figure ends the benchmark in 2016 so that the 2017 panel shows whether cropland-share changes were comparable across the border before full implementation. The formal RD-DID estimates use 2012–2017 as the pre-policy benchmark, consistent with the main analysis, which treats 2017 as the reference year.

Figure 7: Spatial RD-DID at the Jiangsu and Anhui Border

(a) Cropland-share changes



(b) Change in fallow rate



Notes: The figure shows spatial RD-DID patterns at the Jiangsu–Anhui border. Panel (a) plots each cell's cropland share in 2017, 2019, 2021, and 2022 minus its 2012–2016 pre-policy mean. It uses non-mountain cells in 13 cross-border county pairs within the 50 km border corridor. Panel (b) plots the change in mean fallow rate in 2018–2022 relative to the 2012–2017 pre-policy mean for potentially reclaimable cropland: non-mountain, ex ante marginal cells with 2012–2017 mean cropland share below 0.6, within cross-border county pairs. Outcomes are residualized by county-pair FE in both panels and additionally by year FE in panel (a). In both panels, the running variable is signed distance to the border in kilometers, positive on the Jiangsu side. Points are 1 km bin means within a ±10 km display window; vertical bars show 90% confidence intervals. Solid lines are separate quadratic fits on either side of the border for visualization. Table 2 reports the formal benchmark estimates.

Table 2: Spatial RD-DID Estimates at the Jiangsu and Anhui Border

	(1)	(2)	(3)
Panel A: Δ Cropland share			
RD in Δ cropland share	0.006**	0.006**	0.007**
(Jiangsu − Anhui)	(0.003)	(0.003)	(0.003)
Panel B: Δ Fallow rate			
RD in Δ fallow rate	0.037***	0.038***	0.037***
(Jiangsu − Anhui)	(0.010)	(0.011)	(0.014)
County-pair FE	✓	✓	✓
Kernel	Triangle	Epanech.	Uniform

Notes. This table reports the formal local-linear RD-DID estimates at the Jiangsu–Anhui border. The outcome in Panel A is the cell's mean cropland share in 2018–2022 minus its mean in 2012–2017. The outcome in Panel B is the corresponding change in mean fallow rate, defined by fractional vegetation coverage (FVC) below 0.20. Changes are expressed in share units. Panel A uses eligible non-mountain cropland cells in matched cross-border county pairs within the ±50 km corridor. Panel B uses the subset with complete FVC and restricts the sample to ex ante marginal cells with mean cropland share below 0.60 in 2012–2017. The running variable is signed distance to the provincial border, with positive values on the Jiangsu side. County-pair FE enter the local regression additively. Columns (1)–(3) use triangular, Epanechnikov, and uniform kernels, respectively; each column selects an MSE-optimal bandwidth. Entries are robust bias-corrected estimates. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Figure 7b plots the change in fallow rate among cells that could potentially be used for cropland reclamation. The increase in fallowing is larger on the Jiangsu side, producing a positive discontinuity at the border. Thus, the higher pressure side exhibits both greater cropland expansion and a larger increase in fallowing among land that could be used to meet cropland protection requirements.

Table 2 reports the formal RD-DID estimates. Across all three kernels, the Jiangsu-minus-Anhui discontinuities in cropland-share and fallow-rate changes are positive and statistically significant. Under the triangular kernel, the estimated discontinuities are 0.006 for cropland share and 0.037 for the fallow rate. These estimates confirm the patterns in Figures 7a and 7b and align with the DiD results: both designs show greater cropland expansion and increased fallowing where cropland protection pressure is higher.

As a robustness assessment, I examine whether the direction of the discontinuity follows the pressure ranking at the Jiangsu and Shanghai border. As shown in Appendix Figure A.9, Shanghai faces greater pressure than Jiangsu. Because the coefficients are defined as Jiangsu minus the comparison side, this reversed pressure ranking implies negative coefficients. Appendix Table B.8 reports the estimates for both outcomes across all three kernels. The coefficients are negative in every specification, as predicted by the reversed pressure ranking. The spatial RD-DID and DiD estimates imply that higher-pressure locations expand cropland more after the policy but also experience greater fallowing among ex ante marginal cells.

5.3 Additional Results and Alternative Explanations

5.3.1 Additional Characteristics of Newly Reclaimed Land

Spatial Allocation and Fragmentation. Soil suitability and vegetation coverage may miss costs created by the location of new fields. If policy pressure pushes expansion toward the agricultural frontier, newly created cropland should be farther from existing fields and more isolated. I assess fragmentation at the expansion margin by examining where new cropland is placed relative to existing cultivation. Plots located near and connected to existing cultivation support more contiguous operations, reduce machinery movement between separated parcels, and facilitate mechanized, large-scale farming. China's high-standard farmland criteria likewise emphasize land leveling, parcel consolidation, and field-road access as conditions for mechanized and large-scale production.[9]

To estimate whether the policy shifted where counties place newly created cropland, I measure the amount of newly created cropland in each 500-meter grid as the positive annual change in its pixel-derived cropland share.[10] I then characterize the location of newly created cropland relative to the prior-year cropland landscape. The first outcome is the area-weighted mean distance from newly created cropland to the nearest grid with at least 1 percent cropland in the prior year. The second and third outcomes are the shares of newly created cropland located on grids with no cropland in the prior year and on isolated grids, respectively. I define an isolated grid as one for which the focal grid and all rook neighbors within the same county had cropland shares below 1 percent in the prior year. I construct all three outcomes at the county-year level and estimate a parsimonious $\text{Pressure}_c \times \text{Post}_t$ specification with county and year fixed effects and county-clustered standard errors, where Post_t equals one for annual transitions ending in each year from 2018 through 2022.

Table 3 shows that stronger policy pressure shifts newly created cropland toward locations farther from and less connected to existing cultivation after 2017. Column (1) shows that, for a one standard deviation increase in pressure, newly created cropland is located 4.662 meters farther from the nearest grid with existing cropland ($p = 0.021$, BH-adjusted $q = 0.035$). Columns (2) and (3) show, respectively, that the share of newly created cropland located on grids with no cropland in the prior year rises by 0.52 percentage points ($p = 0.013$, BH-adjusted $q = 0.032$) and the share located on isolated grids rises by 0.29 percentage points ($p = 0.036$, BH-adjusted $q = 0.045$). These placement shifts indicate that stronger policy pressure redirects newly created cropland beyond the existing cultivated footprint and toward less connected locations.

[9] See Appendix Section G.2 for the national high-standard farmland construction standard and the Ministry of Agriculture and Rural Affairs' upgrading guidance.

[10] This measure differs from the main analysis, which identifies newly reclaimed land when a grid's cropland share rises persistently from below 0.60 to above 0.70. The raw-change measure retains smaller within-grid changes omitted by that threshold definition. When constructing the outcomes, I weight each grid by the amount of newly created cropland.

Table 3: Spatial Allocation and Environmental Outcomes of Newly Created Cropland

	Spatial allocation			Environmental outcomes		
	Dist. to existing cropland (m)	Zero-cropland grids	Isolated grids	Water	Forest	Wetland
	(1)	(2)	(3)	(4)	(5)	(6)
Pressure × Post-2017	4.222**	0.470**	0.263**	0.082***	-0.010	0.038***
	(1.782)	(0.184)	(0.123)	(0.018)	(0.008)	(0.009)
	[0.035]	[0.032]	[0.045]	[< 0.001]	[0.238]	[< 0.001]
Magnitudes:						
1-SD Pressure	4.662**	0.519**	0.291**	0.090***	-0.011	0.042***
Outcome mean	11.011	1.189	0.311	0.214	0.046	0.087
County FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Observations	640	640	640	576	576	576
R-squared	0.336	0.324	0.200	0.690	0.478	0.587
Number of counties	64	64	64	64	64	64
Pre-policy joint p-value	0.317	0.600	0.288	0.101	0.366	0.335

Notes: All columns report county-year regressions for 64 counties. Columns (1)–(3) span 2013–2022; Columns (4)–(6) span 2013–2021 because the environmental measures end in 2021. Each regression includes county and year FE; standard errors are clustered by county. Post-2017 equals one from 2018 onward. Column (1) is the mean distance from newly created cropland to the nearest grid with at least 1% prior-year cropland, weighting each expanding grid by its newly created cropland area. Column (2) is the percentage-point share of newly created cropland on grids with zero prior-year cropland. Column (3) is the percentage-point share on isolated grids, defined as grids whose own and all within-county rook neighbors' prior-year cropland shares are below 1%. Columns (4)–(6) measure annual $t-1$ to t source-to-cropland transition area as a percentage of observed valid CLCD area, averaged across grids within each county-year. The wetland measure uses GAWD and may overlap the CLCD water and forest measures. Outcome mean is the sample mean of the dependent variable. The pre-policy row reports the joint Wald-test p-value for the 2013–2016 pressure-by-year coefficients, with 2017 omitted. Standard errors appear in parentheses, and BH-adjusted q-values appear in brackets. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Environmental Outcomes. Cropland compensation may impose environmental costs in addition to creating less productive or more fragmented farmland. Because expansion converts land previously classified as noncropland, it can replace water bodies, forests, and wetlands that provide ecological functions. I therefore examine whether stronger policy pressure increases the conversion of these land types into cropland. The annual CLCD data used to construct the paper's cropland measure also identify each pixel's land cover before conversion. I define a strict annual conversion when a 30 meter pixel is classified as water or forest in year $t-1$ and as cropland in year t. I use GAWD to measure wetland to cropland conversion (see Appendix Table B.1 for the data source). Because the wetland measure may overlap the CLCD water and forest categories, the three measures should be interpreted as distinct outcomes based on different classification systems, not as additive components of total land conversion. I average

grid level conversion shares within each county and year and estimate the same parsimonious specification as above. The environmental sample covers annual transitions ending in 2013 through 2021 to maintain common data support across the three outcomes.

Columns (4), (5), and (6) of Table 3 show that the effects are concentrated in the conversion of water and wetlands. A one standard deviation increase in cropland protection pressure raises water to cropland conversion by 0.090 percentage points ($p < 0.001$, BH-adjusted $q < 0.001$) and wetland to cropland conversion by 0.042 percentage points ($p < 0.001$, BH-adjusted $q < 0.001$). The joint tests for coefficients before the policy do not reject for water ($p = 0.101$) or wetland ($p = 0.335$). In contrast, the estimated change in forest to cropland conversion is small and statistically insignificant at negative 0.011 percentage points ($p = 0.232$, BH-adjusted $q = 0.238$). These estimates indicate that cropland compensation imposes an environmental cost by redirecting expansion toward water and wetland areas.

5.3.2 Alternative Explanations

Three alternative explanations could weaken the interpretation of the main fallow result. First, low annual vegetation coverage may reflect normal seasonal variation rather than non-cultivation when crops should grow. Second, newly reclaimed plots may require a temporary preparation period before entering production. Third, they may be planted with crops that produce different vegetation profiles. Each explanation would weaken the conclusion that tighter cropland protection reduced post-reclamation cultivation. Appendix Sections C.1 and C.2 test these alternative explanations.

The first alternative is normal seasonality. To distinguish seasonal vegetation from fallow, I use pre-policy vegetation profiles and agronomic evidence to recover Jiangsu's crop calendar, then ask whether post-policy fallow rises when fields are normally bare or during the growing season, when crop canopies should be dense. The latter pattern appears in the data. Relative to reclaimed land from pre-policy cohorts, post-policy reclaimed land has higher fallow rates in both high- and low-pressure counties, but the growing-season increase is consistently larger in high-pressure counties. This evidence supports weaker cultivation when crops should be growing, rather than unusually sparse vegetation during the lean season (Appendix Section C.1).

The second alternative is temporary land preparation. If newly reclaimed land requires time to enter production, the fallow difference should be largest immediately after reclamation and then decline. Cropping pattern maps allow me to follow newly reclaimed land through the first three years and distinguish detected cultivation from fallow.[11] Cultivation and fallow do not differ clearly during the first two years, whereas lower detected cultivation and greater fallow emerge in the available third-year comparison. The delayed emergence of these gaps is inconsistent with a temporary land-preparation account (Appendix Section C.2).

The third alternative is crop switching. If a different crop mix accounts for lower vegetation

[11] Fallow here refers to the fallow category in the cropping pattern maps, not the FVC-based definition used above. The maps partition newly reclaimed land into detected cultivation, fallow, and unclassified pixels. These categories are mutually exclusive and exhaustive.

coverage, the land would still be classified as cropland, but its cropping frequency or crop composition would differ from the pre-policy average. Conditional on detected cropland, I find no clear shift in cropping-frequency or crop-specific shares. Nor do the data indicate that crop switching accounts for lower vegetation coverage (Appendix Section C.2).

5.4 Robustness Checks

5.4.1 Sensitivity to Alternative Definitions

I first present a series of sensitivity checks using alternative definitions of cropland protection pressure, newly reclaimed land, and fallow status.

First, I examine whether the findings depend on how cropland protection pressure is measured. I consider alternative constructions of the pressure index by (i) redefining core-urban districts in the obligation component and (ii) modifying the definition of available land in the total-land component. These exercises test whether the results are driven by specific coding choices in the pressure measure.

Second, I evaluate the sensitivity of the results to the classification of newly reclaimed land and fallow status. I vary the threshold used to define reclamation based on pre-reclamation cropland share and allow for a range of alternative cutoffs. I also consider alternative definitions of fallow land by varying the FVC threshold used to identify uncultivated plots. All robustness exercises are implemented for the main outcomes and are reported in the Appendix.

5.4.2 Robustness of the Cropland-Share Event Study

I then assess the robustness of the cropland-share event-study estimates to violations of parallel trends and county-specific linear time trends. I first apply the relative-magnitude restriction proposed by Rambachan and Roth (2023), which bounds changes in post-treatment deviations from parallel trends relative to the largest corresponding change observed before treatment. For the coefficient on a one-standard-deviation increase in cropland protection pressure, the break value of $\overline{M}$ lies between 0.75 and 0.80 (Appendix Figure A.4). The positive 2018 effect remains statistically significant when post-treatment changes in the deviation from parallel trends reach 75 percent of the largest corresponding pre-treatment change. The absence of detectable pre-trends and this bound make it unlikely that differential pre-policy trends explain the post-policy increase in cropland share.

Counties may follow different underlying time trends that could account for part of the estimated post-policy response. To address this concern, I re-estimate the event study with county-specific linear calendar-year trends, which absorb gradual differences in the evolution of cropland share across counties. This specification normalizes the 2016 and 2017 pressure-by-year interactions to zero: 2017 fixes the level, while 2016 anchors the linear slope. Appendix Figure A.5 shows that all five post-policy estimates remain positive and statistically significant at the 1 percent level. After rescaling pressure to one standard deviation, the estimated effect

is 0.14 percentage points in 2018 and rises to 0.53 percentage points in 2022. County-specific linear trends therefore do not account for the post-policy increase in cropland share.

6 Mechanisms

Following the 2017 policy tightening, cropland expansion in high-pressure counties was accompanied by lower agronomic suitability and less intensive subsequent cultivation. The mechanism analysis addresses two questions. First, does the decline in reclamation quality reflect distorted reclamation choices? Costly compliance remains one possible explanation: local governments may expand sequentially along a land-quality frontier, so that the exhaustion of more suitable sites mechanically pushes additional reclamation toward marginal land. Distorted implementation, by contrast, arises when local governments depart from the best feasible reclamation path and select lower-quality land than resource constraints alone would require, potentially because identifying and developing better land requires greater effort or exposes implementation to closer quality monitoring. Section 6.1 presents several pieces of indirect evidence supporting the presence of distorted reclamation choices. Second, do the incentives facing local officials shape which parcels are reclaimed? Because China's performance-evaluation system requires officials to pursue multiple targets, reclamation choices may reflect both career concerns and pressure from competing economic-growth objectives. Section 6.2 tests these two incentive channels. Section 6.3 quantifies the agricultural losses associated with these reclamation choices and discusses their implications for local government incentives.

6.1 Evidence on Distorted Reclamation Choices

6.1.1 Strategic Avoidance of Quality Monitoring?

If the allocation of reclamation sites is shaped by local bureaucratic incentives, including performance-evaluation concerns, or is otherwise distorted, the locations selected for reclamation should vary with spatial differences in land-quality monitoring intensity. I examine this possibility using distance to the nearest province-level cropland-quality monitoring station as a proxy for monitoring exposure. The analysis asks whether governments avoid parcels near monitoring stations, where land quality is more observable, and whether the 2017 tightening of cropland-protection enforcement mitigated such avoidance. These tests provide indirect evidence on whether quality monitoring shapes where land is reclaimed and whether tighter area enforcement weakens this avoidance pattern.

Cropland-quality monitoring stations are long-term fixed sites that produce standardized information on soil and crop conditions for provincial monitoring, spot checks, databases, and annual reports.[12] Because monitoring is spatially local (see Appendix Section G.2), greater distance indicates weaker exposure to higher-level quality oversight. I restrict the analysis to

[12]See Department of Agriculture and Rural Affairs of Jiangsu Province (2021).

stations operating before 2017, so their locations predate the reform.[13] Their siting follows explicit representativeness and density criteria: national rules require monitoring stations to represent local soil types and cropping systems, while Jiangsu imposes minimum coverage-density requirements.[14] Station placement therefore centers on local agro-ecological representativeness and coverage density, not officials' short-run reclamation choices. To examine whether these siting decisions confound the estimated relationship between monitoring exposure and reclamation choices, I also report a coefficient-stability test that adds a broad set of covariates potentially related to station siting.

I test whether, before and after the policy tightening, distance to the nearest monitoring station predicts two patterns: (i) the extensive margin—grid cells farther from a station are more likely to be reclaimed; and (ii) the intensive margin—among newly reclaimed grid cells, soil suitability declines with distance. I estimate:

$$Y_{it} = \theta_E \log(d_i) + \beta_E[\log(d_i) \times \text{Post}_t] + X_i'\gamma_E + \alpha_{c(i)} + \lambda_t + \varepsilon_{it}, \tag{5a}$$

$$Y_i = \theta_I \log(d_i) + \beta_I[\log(d_i) \times \text{Post}_{t_0(i)}] + X_i'\gamma_I + \alpha_{c(i)} + \mu_{t_0(i)} + \varepsilon_i. \tag{5b}$$

Here d_i is the distance in kilometers from grid i to the nearest pre-2017 province-level monitoring station. In equation (5a), Y_{it} indicates whether grid i is newly reclaimed in year t: θ_E measures the pre-policy relationship between reclamation probability and monitoring distance, while β_E measures how that relationship changes after the policy tightening. In equation (5b), Y_i is soil suitability for land first reclaimed in cohort $t_0(i)$: θ_I measures the pre-policy distance–quality gradient, while β_I measures how that gradient changes after the policy tightening.

Both equations control for baseline cropland and vegetation conditions, core-area status, county differences, and urban-center distance; equation (5a) includes year fixed effects, and equation (5b) reclamation-cohort fixed effects. Avoidance of monitoring implies $\theta_E > 0$, stronger post-tightening avoidance implies $\beta_E > 0$, and a negative distance–quality gradient implies $\theta_I < 0$; β_I captures its post-tightening change.

Table 4 reports the results. Even before the policy tightening, grid cells farther from monitoring stations were more likely to be reclaimed ($p < 0.001$). The policy tightening strengthens this pattern ($p < 0.001$). Columns (3)–(4) replace the level of log distance with grid fixed effects, leaving the interaction estimates unchanged. Among newly reclaimed grid cells, greater distance predicts lower soil suitability under both the 1995 and 2023 measures (both $p < 0.001$). The policy tightening does not alter this distance–quality gradient ($p = 0.439$ and $p = 0.708$, respectively). This persistence is consistent with continued limitations in quality monitoring for reclaimed land farther from stations after the tightening.

[13] Not all Jiangsu jurisdictions publicly report monitoring-station coordinates. The analysis therefore uses only grid cells in the reporting jurisdictions and excludes grid cells elsewhere. Appendix Figures A.10 and A.11 show, respectively, the sample's geographic coverage and the distribution of distances from every 100-meter grid cell in these jurisdictions to its nearest station.

[14] See Ministry of Agriculture of the People's Republic of China (2016) and Department of Agriculture and Rural Affairs of Jiangsu Province (2021).

Table 4: Cropland Quality Monitoring and Land Misallocation

	Extensive margin				Intensive margin: Soil suitability			
	1(Reclaimed in year)				1995 China Soil Map		2023 HWSD v2	
	Level gradient		Grid FE					
	Baseline (1)	+ Urban (2)	Baseline (3)	+ Urban (4)	Baseline (5)	+ Urban (6)	Baseline (7)	+ Urban (8)
$\log(d_i) \times$ Post	0.080^{***}	0.066^{***}	0.080^{***}	0.066^{***}	0.010	0.011	−0.004	−0.006
	(0.007)	(0.007)	(0.007)	(0.007)	(0.014)	(0.014)	(0.016)	(0.016)
$\log(d_i)$	0.011^{**}	0.014^{***}			-0.064^{***}	-0.067^{***}	-0.135^{***}	-0.130^{***}
	(0.004)	(0.004)			(0.013)	(0.013)	(0.015)	(0.015)
Grid FE			✓	✓				
Year FE	✓	✓	✓	✓				
County FE	✓	✓			✓	✓	✓	✓
Reclamation-cohort FE					✓	✓	✓	✓
Baseline land-use controls	✓	✓			✓	✓	✓	✓
Urban-center distance control		✓		✓		✓		✓
Core grids included	✓	✓	✓	✓	✓	✓	✓	✓
Mean dependent variable	0.002	0.002	0.002	0.002	0.742	0.742	0.623	0.623
Observations	15,512,250	15,512,250	15,512,250	15,512,250	30,891	30,891	30,892	30,892
R^2	0.007	0.007	0.101	0.101	0.345	0.346	0.407	0.408

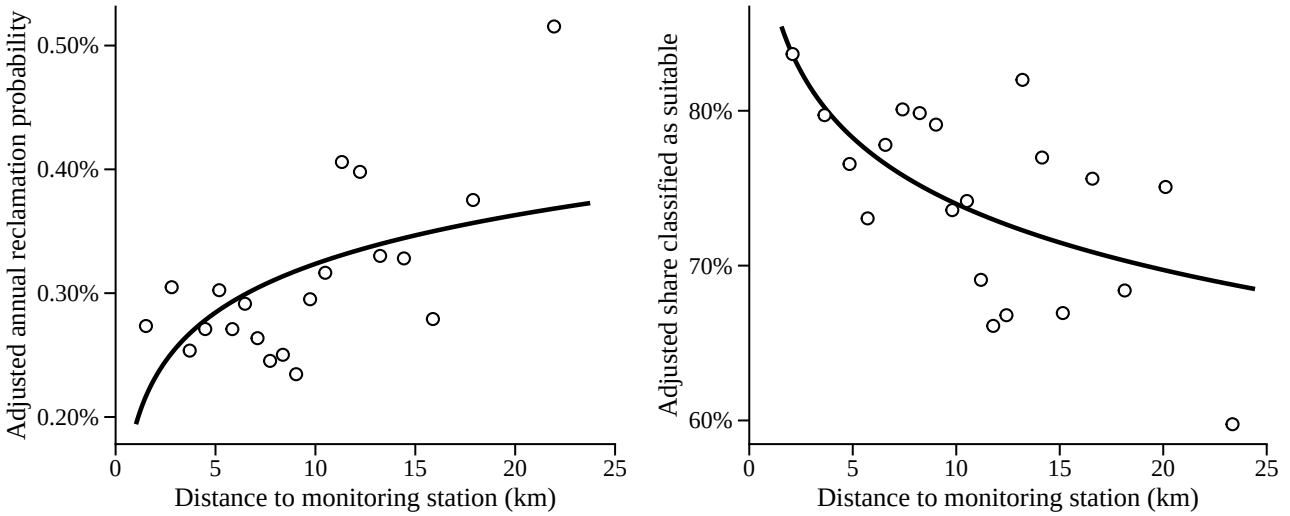
Notes: d_i is the distance in kilometers from 100-meter grid i to the nearest pre-2017 province-level cropland-quality monitoring station. Columns (1)–(4) use annual grid-year observations for 2013–2022, with Post = 1 in 2018–2022. Columns (1)–(2) estimate equation (5a) with county and year FE and baseline land-use controls; columns (3)–(4) use grid and year FE. Columns (5)–(8) use grids first reclaimed in 2013–2022 with county and reclamation-cohort FE. Even-numbered columns add log urban-center distance, based on county centers constructed from 2012–2017 impervious-surface density: its level and Post interaction in column (2), its Post interaction in column (4), and its level in columns (6) and (8). Columns (1)–(4) report coefficients and standard errors in percentage points; columns (5)–(8) use outcome-share units. For each soil measure, suitability is first defined at the source-pixel or soil-component level as a score of at least 3 and then area-weighted within each 500-meter grid. The resulting suitable-land share ranges from zero to one and is assigned to all 100-meter cells within that parent grid; it is not a separate 100-meter soil measurement. Baseline controls include 2012 cropland share and vegetation conditions. Standard errors are clustered at the parent 500-meter grid level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

To assess whether post-policy reclamation outcomes increase or decrease with monitoring-station distance, Figure 8 plots residualized outcomes against physical distance to the nearest station. The outcomes are residualized with respect to baseline cropland and vegetation conditions, core-area status, urban-center distance, and county fixed effects. Figure 8 shows that the residualized annual reclamation probability increases with distance, whereas the residualized suitable-soil share among newly reclaimed grid cells decreases. These patterns reinforce Table 4 by showing directly that both reclamation and the quality of selected land vary with monitoring-station distance.

A potential concern is that monitoring-station siting may be correlated with other spatial conditions that also affect reclamation choices. To address this concern, Appendix Figure A.12 examines coefficient stability after adding combinations of ten covariate groups covering administrative remoteness (county-boundary and government-seat distances); soil and baseline land cover around the nearest station (soil suitability and cropland, impervious-surface, forest, and water shares within 5 kilometers); terrain (elevation and ruggedness); and local economic activity (nighttime lights). The coefficients retain their signs across the displayed combinations, and

the fully adjusted estimates remain statistically significant. The monitoring-distance gradients are therefore robust to a broad set of factors potentially related to station siting.

Figure 8: Monitoring-Station Distance and Post-Policy Reclamation Choices



(a) Reclamation probability

(b) Suitable share of reclaimed land

Notes: The horizontal axis measures distance to the nearest cropland-quality monitoring station. Panel (a) uses all 100-meter grid cells and plots annual reclamation probability during 2018–2022. Panel (b) uses grid cells reclaimed during 2018–2022 and plots the suitable-land share from the 1995 China Soil Map. Source-pixel indicators equal to one for scores of at least three are averaged within each parent 500-meter grid using area weights; each 100-meter observation inherits this parent-grid share. The vertical axis reports outcomes residualized with respect to baseline cropland and vegetation conditions, core-area status, urban-center distance, and county FE; panel (b) also includes reclamation-cohort FE. Circles show means within 20 equal-count distance bins, and solid lines show fitted gradients. Standard errors are clustered by parent 500-meter grid. The coefficients on monitoring proximity, $-\log(d_i)$, are -0.057 in panel (a) and 6.158 in panel (b), expressed in percentage points; both $p < 0.001$.

6.1.2 Do Governments Favor Low-cost, Unsuitable Land?

A government can meet a binding cropland-replacement target at lower cost by favoring parcels that are easier to bring into production. If local officials respond along this margin, the 2017 enforcement tightening should change not only total reclamation but also the composition of the land selected: jurisdictions facing greater land pressure should shift more strongly toward low-cost, unsuitable sites. This sorting response would allow governments to satisfy the monitored acreage target while limiting the effort required per unit of newly created cropland. It therefore provides a separate test of whether implementation tilts toward cheaper but unsuitable sites instead of expanding uniformly along the available land frontier.

Prior evidence supports using cultivation history as a proxy for reclamation cost. A United Nations report estimates that restoring previously cultivated but abandoned land costs about $100 per dunum, compared with $180–200 for never-cultivated marginal land. A USDA report likewise puts clearing costs at $15–30 per acre for lightly vegetated, recently abandoned fields, compared with $95–160 for woodland converted to crops (see Appendix Section G.4). Because

parcel-level reclamation costs are not observed, I use prior cultivation to capture this cost margin. Using the same annual land-cover data as in the main analysis, I reconstruct each source pixel's land-use history over 2001–2011. I classify a pixel as previously cultivated if it is classified as cropland for at least two consecutive years during this period but is no longer classified as cropland in 2012. The no-history group consists of pixels classified as noncropland throughout 2001–2012. Low-cost, unsuitable land combines prior cultivation with soil unsuitability; high-cost, suitable land combines no historical cultivation with soil suitability.

I test this prediction by asking whether reclamation shifts toward low-cost, unsuitable pixels after 2017 and, if so, whether this shift is concentrated in high-pressure counties. The specification includes Post × Pressure and an interaction between low-cost, unsuitable status and the post-policy dummy, together with its further interaction with cropland protection pressure. I estimate the same specification separately for high-cost, suitable land. The latter triple interaction captures whether any post-policy shift toward low-cost, unsuitable land occurs disproportionately in high-pressure counties.[15]

Table 5 reports results for low-cost, unsuitable land in Panel A and high-cost, suitable land in Panel B, using China Soil Map 1995 and HWSD v2 2023. A 1-SD increase in Pressure raises the relative post-policy reclamation probability of low-cost, unsuitable land by 2.38–2.78 percentage points across both soil sources (both $p < 0.01$). Across both soil sources, higher-pressure counties therefore become more likely to reclaim low-cost, unsuitable land after 2017, consistent with greater selection of lower-cost parcels with poor soil suitability. Panel B shows the opposite shift: a 1-SD increase in pressure lowers the relative post-policy reclamation probability of high-cost, suitable land by 0.77–0.86 percentage points (both $p < 0.01$). Together, these results support the hypothesis that reclamation costs help explain why local governments shift toward cheaper but unsuitable land and away from more costly suitable land under tighter enforcement.

[15] Specifically, I estimate $Y_{ict} = \beta_0(\text{Post}_t \times \text{Pressure}_c) + \beta_1(G_i \times \text{Post}_t) + \beta_2(G_i \times \text{Post}_t \times \text{Pressure}_c) + \alpha_i + \lambda_t + \varepsilon_{ict}$, where G_i indicates the group in each panel. Persistent reclamation requires noncropland in $t-1$ and cropland in $t, t+1$; pixels exit after their first event.

Table 5: Selection of Low-cost, Unsuitable and High-cost, Suitable Land

	China Soil Map 1995 (1)	HWSD v2 2023 (2)
Panel A. Low-cost, unsuitable land		
Low-cost, unsuitable × Post × Pressure	2.514***	2.153***
	(0.832)	(0.774)
Low-cost, unsuitable × Post	1.333	1.800*
	(1.082)	(0.970)
Post × Pressure	0.590*	0.594*
	(0.330)	(0.323)
Additional effect per 1-SD pressure	2.776***	2.377***
	(0.918)	(0.854)
Pixel FE	✓	✓
Year FE	✓	✓
Observations	60,743,493	61,414,714
R^2	0.332	0.333
Panel B. High-cost, suitable land		
High-cost, suitable × Post × Pressure	-0.700***	-0.783***
	(0.175)	(0.208)
High-cost, suitable × Post	-0.906	-0.962*
	(0.552)	(0.563)
Post × Pressure	1.037***	1.084***
	(0.348)	(0.355)
Additional effect per 1-SD pressure	-0.773***	-0.864***
	(0.193)	(0.229)
Pixel FE	✓	✓
Year FE	✓	✓
Observations	60,743,493	61,414,714
R^2	0.332	0.333
Number of counties	64	64
Cluster level	County	County

Notes: The unit of observation is a pixel-year (2013–2021). The outcome equals 100 if the pixel is first persistently reclaimed that year and zero otherwise; coefficients measure changes in reclamation probability in percentage points. Panel A identifies previously cultivated, unsuitable pixels; Panel B identifies suitable pixels with no historical cultivation. Historical cultivation requires at least two consecutive cropland years in 2001–2011; all pixels are noncropland in 2012. Post equals one from 2018. All models include pixel and year FE. County-clustered standard errors are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.1.3 Do Local Governments Move Sequentially Down the Land-Quality Frontier?

A decline in the quality of newly reclaimed land does not by itself establish distorted implementation. Suppose a county follows a land-quality frontier: it reclaims the highest-quality feasible parcels first and turns to progressively less suitable land only as this reservoir is depleted. A larger acreage obligation would then reduce the quality of newly reclaimed land even if the local government made the best feasible choice at each step. The resulting decline would reflect

costly compliance rather than a distorted reclamation choice.

I use two exercises to assess whether reservoir depletion can account for the policy response: controlling for the remaining land stock in the main regressions, and decomposing the decline in reclamation quality into a frontier component reflecting costly compliance and a selection component capturing distorted reclamation choices.

First, I add depletion and the remaining suitable share to the parsimonious pooled specifications.[16] Greater pressure still predicts more cropland, lower suitability, and more fallowing after conditioning on depletion and the remaining suitable share; all three coefficients remain significant at the 5 percent level (Appendix Table B.13). The main results are therefore robust to controlling for depletion and remaining suitability: these measured constraints do not account for the entire response.

Second, I construct a historical reference for the quality of newly reclaimed land. The best feasible parcel cannot be identified directly because the complete set of costs and constraints governing site selection is unobserved. I instead use the pre-policy land-cover history to relate each cohort's average suitability to the state of its county's reservoir. If governments reclaim better land first, the remaining suitable share should predict the quality of the next cohort, while depletion and a larger reclamation requirement should push selection toward poorer land. I estimate this relationship over the pre-policy period, 2002–2017, and apply it to the observed reservoir states and reclamation intensity in the post-policy period, 2018–2021. The resulting reference holds actual reclamation quantities fixed and asks what quality the historical relationship predicts. I call the decline from the county's historical mean to this reference the *frontier component*, and the shortfall of actual quality below the reference the *selection wedge*, which serves as the empirical proxy for distortion in this decomposition. Appendix Section D details the construction.

Table 6 shows that the selection wedge accounts for 73–78 percent of the aggregate suitability decline across the three reference specifications. In the full specification, the total decline is 10.156 percentage points: the frontier component accounts for 2.718 points and the selection wedge for 7.438 points, or 73.2 percent. The selection wedge accounts for 87.5–99.7 percent of the additional quality loss in higher-pressure counties relative to the historical reference (Appendix Table D.1, Panel C).

[16] I estimate a single pressure-by-post coefficient, retaining grid and year FE for cropland share, county and reclamation-cohort FE for suitability, and county, cohort, and years-since-reclamation FE for fallow status. The controls describe the reservoir at the beginning of the calendar year for cropland share and the reclamation year for the other outcomes. These states precede the outcome but can respond to earlier policy exposure, so conditioning on them can absorb part of the policy response operating through the reservoir. The coefficients are conditional rather than total effects; their differences from the baseline are not separately identified mediation effects.

Table 6: Historical-Reservoir Decomposition of Reclaimed-Land Suitability

Reference predictors	Total loss	Frontier component	Selection wedge	Frontier share (%)	Selection share (%)
Remaining suitable share	10.156	2.413	7.743	23.8	76.2
Remaining suitable share + depletion	10.156	2.204	7.952	21.7	78.3
Remaining suitable share + depletion + reclamation intensity	10.156	2.718	7.438	26.8	73.2

Notes: Reclaimed-area-weighted accounting for 92 county–cohort observations in 37 counties, 2018–2021. For both newly reclaimed land and the remaining reservoir, quality is the suitable-land percentage under the 1995 soil classification: source pixels are classified as suitable when their score is at least three, before area-weighted aggregation to 500-meter grids. Total loss is the county historical mean minus observed quality; the frontier component is the historical mean minus predicted reference quality; the selection wedge is reference quality minus observed quality. The first three numeric columns are in percentage points. Remaining suitable share measures suitable land as a percentage of the beginning-of-year remaining reservoir. Depletion measures the percentage of the initial 2001 reservoir currently in cropland. Reclamation intensity is current reclaimed area as a percentage of the beginning-of-year remaining reservoir. References use 2002–2017 county and cohort FE models and county historical means; actual reclamation quantities are held fixed. Appendix Section D gives definitions and calculations.

The two exercises support distorted reclamation choices over an account based solely on reservoir depletion. Policy effects persist after conditioning on the remaining land stock, and the distortion proxy accounts for most of the quality loss relative to the historical reference, particularly its pressure-related component. This is a back-of-the-envelope decomposition, not a precise welfare calculation: the benchmark assumes that historical selection patterns remain informative about feasible choices, and unobserved changes in costs or constraints may also enter the wedge. Subject to that benchmark, the estimates point to distorted selection as a major source of the decline.

6.2 Incentives Facing Local Officials

Section 6.1 provides several pieces of indirect evidence consistent with distorted reclamation choices. The following analysis asks whether the incentive scheme facing local officials helps explain these patterns. Officials are evaluated on multiple objectives, including cropland area compliance and economic growth, while the quality of newly reclaimed land is monitored less closely. The hypothesis is that officials prioritize observable and rewarded outcomes at the expense of the weakly monitored quality margin when career incentives are stronger or multiple objectives are more difficult to satisfy jointly. I test this hypothesis by asking whether land misallocation is greater when promotion incentives are stronger and when economic growth pressure is higher.

6.2.1 Career Incentives

The career-incentive hypothesis predicts greater hidden quality distortions in higher-pressure counties when bureaucrats face stronger career incentives. Under China's cadre evaluation

system, a good performance record on assigned tasks is crucial for future advancement. The hypothesis implies that prefecture party secretaries with stronger promotion motives would place more weight on observable tasks such as economic growth and cropland protection. They would in turn press subordinate counties more strongly to create compensation cropland while preserving urban expansion. Higher-pressure counties bear more of this delegated obligation and are thus expected to respond more strongly after 2017. When acreage is closely monitored but quality is not, this differential response is expected to appear as a larger decline in reclaimed-land suitability.

In China's cadre promotion system, promotion prospects fall sharply after age 57 (Huang et al., 2024). I measure high promotion incentives with an indicator equal to one when the prefecture party secretary is not older than 57 and the year precedes the 2020 nationwide land inspection. Table 7 tests this prediction by interacting the promotion-incentive indicator with county-level cropland protection pressure and the post-2017 indicator.[17] The triple-interaction coefficient asks whether the post-2017 suitability gap between higher- and lower-pressure counties within a prefecture widens when the party secretary's incentives are stronger. All specifications include county and reclamation-cohort fixed effects, county-specific linear trends, and controls for the party secretary's age, a male indicator, and Jiangsu birthplace.

[17] For newly reclaimed grid cell i in county c, prefecture p, and reclamation cohort t, I estimate $Y_{icpt} = \beta_1 P_c D_t + \beta_2 P_c D_t M_{pt} + X'_{cpt}\gamma + \alpha_c + \lambda_t + \kappa_c t + \varepsilon_{icpt}$, where P_c is county pressure, D_t is the post-2017 indicator, and M_{pt} is the promotion-incentive indicator. X_{cpt} collects party-secretary controls; α_c, λ_t, and $\kappa_c t$ denote county fixed effects, reclamation-cohort fixed effects, and county-specific linear trends. The coefficient of interest is β_2.

Table 7: Multitask Incentives and Rural Cropland Suitability

	China Soil Map	HWSD v2
	1995	2023
	Soil suitability	Soil suitability
	(1)	(2)
Pressure × Post	-0.097^{***}	-0.067^{**}
	(0.021)	(0.022)
	[0.006]	[0.089]
Party-secretary incentive × Pressure × Post	-0.058^{***}	-0.067^{***}
	(0.013)	(0.021)
	[0.022]	[0.067]
Control Mean	0.593	0.573
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & Cohort FE	✓	✓
Number of counties	52	52
Num. Obs.	939	939
R^2	0.498	0.532

Notes: This table reports mechanism tests relating political incentives to the quality of newly reclaimed cropland. The unit of observation is a newly reclaimed grid cell. Columns (1) and (2) use the area-weighted mean of pixel- or component-level indicators equal to one when the suitability score is at least 3, based on the 1995 China Soil Map and HWSD v2 2023, respectively. The promotion-incentive indicator equals one when the prefecture party secretary is aged 57 or younger and the reclamation year precedes 2020. Party-secretary controls include age, a male indicator, and an indicator for Jiangsu birthplace. County linear trends are county-specific linear trends in reclamation cohort. Parentheses report conventional standard errors clustered by prefecture, and stars are based on them. Square brackets report two-sided null-imposed, studentized Rademacher wild-cluster bootstrap p-values based on 9,999 random draws at the prefecture level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The triple-interaction estimates are negative and significant at the 1 percent level in both soil datasets, with standard errors clustered by prefecture. Under the more conservative prefecture-level wild-cluster bootstrap, they remain significant at the 5 percent level for the 1995 China Soil Map and the 10 percent level for HWSD v2 2023. Higher-pressure counties experience larger post-2017 suitability losses when the party secretary has stronger promotion incentives. Appendix Table B.11 also shows that the pressure-related increase in fallowing is larger on lower-quality reclaimed land. This pattern suggests that higher fallowing rates may partly arise because lower-quality land selected at the reclamation stage is less likely to be cultivated afterward. These results support a career-incentive mechanism in which compliance shifts toward observable acreage and distortion toward the less monitored quality margin.

6.2.2 Economic Growth Pressure

Prefecture leaders face a multitask problem between economic growth, fueled by urban expansion, and cropland protection. Urban expansion converts cropland to construction use and thereby raises compensation obligations, so prefectures at greater risk of missing their growth targets have stronger incentives to preserve construction land while meeting cropland requirements through reclamation. As pressure on competing observable tasks rises, less effort is left for the unobserved quality margin. The conflict predicts larger cropland expansion but lower-quality reclaimed land.

Economic growth burden is measured at the prefecture-year level as the shortfall between target growth and predicted annual growth based on first-quarter GDP performance. A prefecture-year is classified as high-burden when this shortfall is positive and ranks in the top tercile among Jiangsu prefectures that year. The specification asks whether the post-policy land-use response to cropland protection pressure is larger in high-burden prefecture-years.[18] Each regression includes prefecture fixed effects, the relevant year or reclamation-cohort fixed effects, and party-secretary controls; the cropland-share specification also includes grid fixed effects.

Appendix Table B.12 is consistent with this prediction. After the tightening, high-burden prefecture-years with greater cropland protection pressure supply more construction land, in both the number and the area of transactions and including transactions on land converted from cropland. Cropland expands more in their non-core rural areas, and the newly reclaimed land is less likely to be suitable for cultivation. With standard errors clustered by prefecture these estimates are significant at conventional levels. The design has only thirteen prefecture clusters, however, and under the wild-cluster bootstrap none of the construction-land coefficients remains significant. The evidence in this subsection is therefore suggestive rather than conclusive.

6.3 Agricultural Losses and Local Incentives

The central government's objectives for cropland protection extend beyond economic output to food security and cereal self-sufficiency (Appendix Section G.1). For local governments, these objectives enter the incentive framework indirectly through cropland-area requirements and GDP performance. When subsequent grain production is not directly measured, suitability and cultivation affect measured performance through GDP once the area requirement is met. Local governments can convert cropland to construction land and meet the compensation requirement through reclamation, or retain cropland and forgo conveyance revenue. I compare the capitalized agricultural losses implied by pressure-induced changes in reclaimed land with the gross revenue at stake in this choice.

I estimate annual agricultural value-added losses implied by pressure-induced changes in the suitability and cultivation of reclaimed land. An agricultural production function maps the estimated suitability and fallowing effects into proportional losses, which I value using baseline value added per hectare. This calculation captures both lower productivity on cultivated land and forgone output on fallow land. The policy requirement to designate consolidated farmland for permanent protection motivates an infinite-horizon valuation (Appendix Section G.5); I hold annual losses constant. Table 8 reports the results. A one-standard-deviation increase

[18] For outcome observation i in prefecture p and year or reclamation cohort t, I estimate $Y_{ipt} = \beta_1(P_p \times D_t) + \beta_2(P_p \times B_{pt}) + \beta_3(D_t \times B_{pt}) + \beta_4(P_p \times D_t \times B_{pt}) + \rho B_{pt} + X'_{pt}\gamma + \alpha_i + \mu_p + \lambda_t + \varepsilon_{ipt}$, where P_p is the prefecture-level cropland protection measure, D_t is the post-2017 indicator, and B_{pt} is the high-economic-growth-burden indicator. X_{pt} includes party-secretary controls. The prefecture and period fixed effects absorb the main effects of P_p and D_t. Here α_i is included only for the grid-year cropland-share outcome, while μ_p and λ_t denote prefecture and outcome-specific year or reclamation-cohort fixed effects. The coefficient of interest is β_4.

in pressure implies an annual agricultural value-added loss of 11.1 percent: RMB 3,456 per hectare per year under the agricultural baseline and RMB 5,530 per hectare per year under the broader primary-industry baseline. The capitalized value of these perpetual losses is RMB 59,583 and RMB 95,346 per hectare, respectively. Appendix Section E provides the derivation, calibration, provincial aggregation, and sensitivity analysis.

Table 8: Annual and Capitalized Agricultural Losses

Panel A. Annual loss per hectare	
Loss relative to baseline agricultural value (%)	11.1
Agricultural value added (RMB/ha/year)	3,456
Primary-industry value added (RMB/ha/year)	5,530
Panel B. Perpetual loss per hectare at 5.8%	
Agricultural value added (RMB/ha)	59,583
Primary-industry value added (RMB/ha)	95,346
Panel C. Jiangsu losses (RMB million)	
Agricultural value added, annual	28.6
Agricultural value added, perpetual present value	493.6
Primary-industry value added, annual	45.8
Primary-industry value added, perpetual present value	789.8
Panel D. Comparison with gross land-conveyance revenue	
Local conveyance revenue (RMB/ha)	42,072,508
Agricultural loss PV / conveyance revenue (%)	0.14
Primary-industry loss PV / conveyance revenue (%)	0.23

Notes: Panels A and B report calibrated losses from a one-standard-deviation increase in county cropland protection pressure. Annual losses are capitalized at 5.8%, assuming constant losses forever; present values are measured at end-2022, with the first future loss at end-2023. Panel C uses 8,283.565 pressure-weighted hectares reclaimed in 2018–2022. Panel D uses 712 same-year, five-kilometer nearby conveyances of newly added construction land, totaling 1,887.97 hectares. Revenue per hectare is the ratio of total payments to total area, after deflating each payment to 2017 RMB. Gross receipts exclude deductions for acquisition and development costs. The ratios compare per-hectare magnitudes, not matched development projects or net social benefits. Appendix Section E provides inputs, spatial-data limitations, and sensitivity.

For comparison, newly added construction-land conveyances within five kilometers of land reclaimed in the same year yield gross revenue of RMB 42.1 million per hectare, measured as total deflated payments divided by total area. The perpetual agricultural losses are only 0.14–0.23 percent of this benchmark; all amounts are in 2017 RMB. Their small monetary scale is consistent with weak local incentives to avoid unsuitable reclamation and fallowing, even when these distortions compromise national food-security objectives.[19]

[19]The loss estimates measure only the monetary value of forgone agricultural value added. They do not monetize food security or cereal self-sufficiency, both central-government objectives. The revenue benchmark consists of gross conveyance receipts, not net fiscal gains.

7 Discussion and Conclusion

This paper examines how tighter enforcement of cropland protection shapes local land-use behavior in a centralized administrative system characterized by multitask incentives and imperfect monitoring. Leveraging cross-county differences in exposure to cropland protection pressure, the analysis shows that the post-2017 policy tightening generates responses along both the extensive and intensive margins of land use. Counties subject to higher pressure are more likely to expand cropland area, yet the land brought into cultivation is of lower soil suitability and is cultivated less intensively, with elevated fallow rates that persist over time. A spatial RD-DID design corroborates this pattern, while additional results document more fragmented reclamation and greater conversion of water bodies and wetlands. Mechanism evidence points to distorted reclamation choices shaped by local incentives under imperfect quality monitoring. These patterns suggest that stricter enforcement of observable quantity targets can induce distortions in land allocation and land use when monitoring of land quality and utilization remains limited.

These findings partly reflect differences in land endowments across jurisdictions and have implications for policy implementation. Where suitable land is scarce and urban demand is high, meeting local compensation obligations creates stronger incentives for unsuitable reclamation. Broader trading of cropland compensation indicators could ease these pressures by allowing compensation to occur where suitable land is more abundant. Yet under China's current institutional framework, trading remains restricted, potentially reflecting a trade-off between efficiency and spatial equity: supplying jurisdictions must preserve land for agriculture and forgo urban development opportunities (discussed in Appendix Section F). Expanding trade therefore requires addressing how these opportunity costs are shared across jurisdictions.

The analysis leaves open how newly reclaimed land enters cultivation. The evidence is consistent with lower suitability contributing to subsequent fallowing, but I do not observe rural land-rental contracts or the operators of reclaimed plots. Once acreage targets are met, local governments may have limited incentives to facilitate leasing; alternatively, farmers, cooperatives, or other operators may lease the land but face weak incentives to cultivate it. The available data cannot distinguish limited government effort to place land with cultivators from operators' subsequent production decisions. Linking reclamation records to rental contracts and operator-level outcomes would help identify these channels and clarify how China's rural land markets mediate the effects of cropland protection.

Conflict of interest

The author declares no conflicts of interest.

Funding

This research received support for travel and conference participation from Yasuyuki Sawada's International Leading Research project.

Data availability

The data used in this study are publicly available or obtainable by application to the respective data providers, subject to their access requirements. Section 3 and Appendix Table B.1 describe the data sources. Detailed access channels and application procedures will be documented in the replication package README. Redistribution of third-party data is subject to the respective providers' terms.

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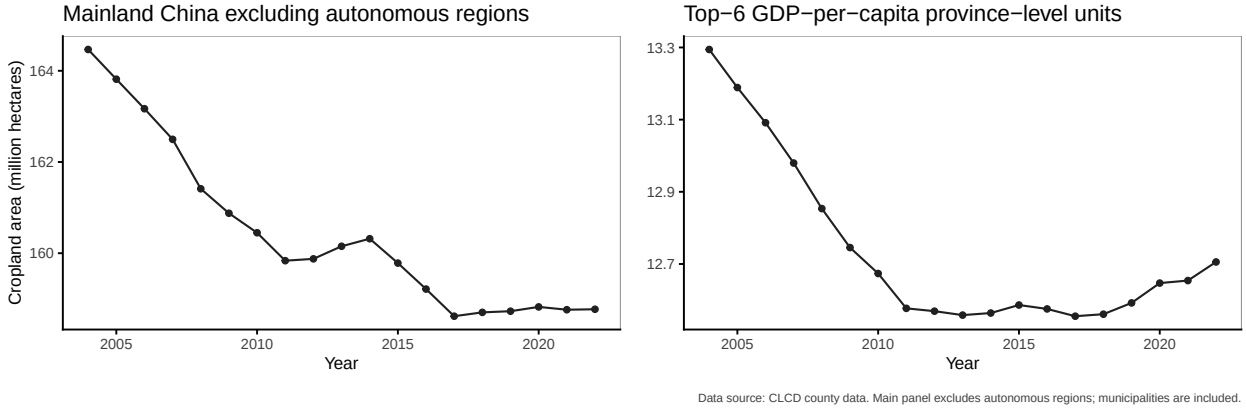
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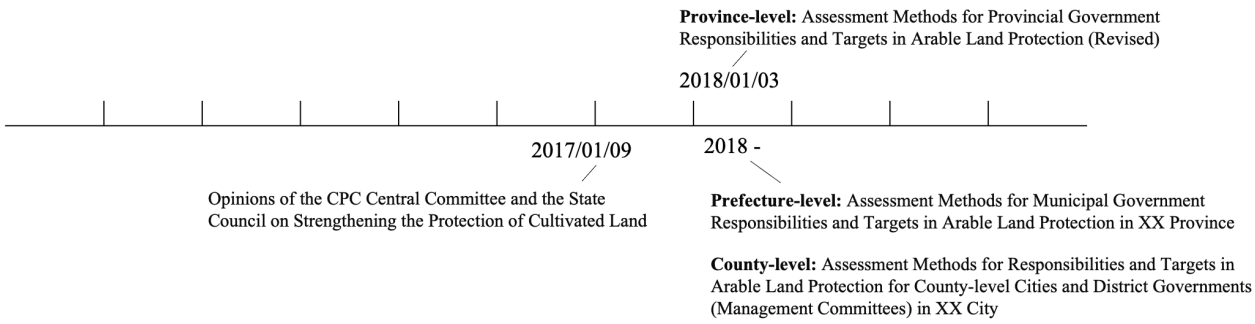
A Figures

Figure A.1: Satellite-Based Cropland-Area Trends in Mainland China



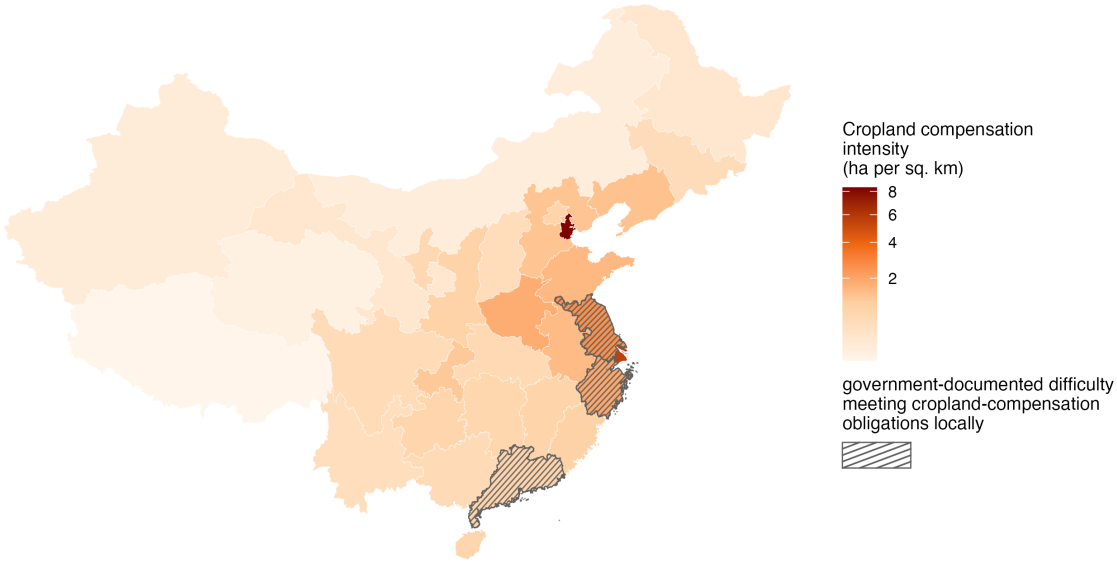
Notes: The figure reports cropland area aggregated from annual 30-meter China Land Cover Dataset classifications (Yang and Huang, 2021). CLCD is derived from satellite imagery. Panel A reports mainland China excluding the five autonomous regions (Inner Mongolia, Guangxi, Tibet, Ningxia, and Xinjiang); municipalities remain included. Panel B reports the six province-level units with the highest GDP per capita in 2021. Cropland area is measured in million hectares.

Figure A.2: Administrative Rollout of the 2017–2018 Policy Tightening



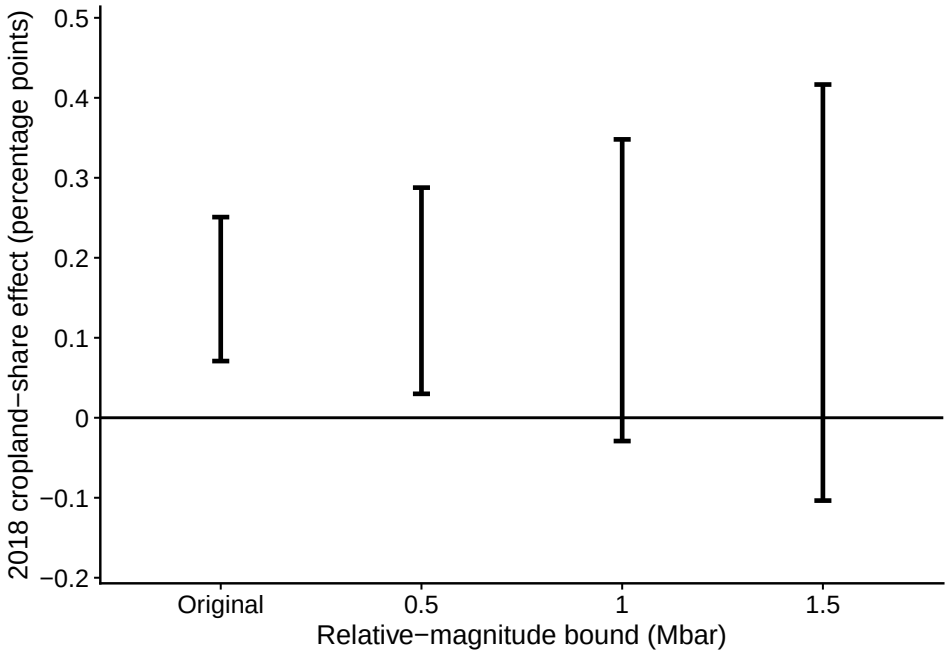
Notes: The 2017 Opinions strengthened cropland-protection accountability. Revised provincial assessment methods followed in January 2018, after which provincial and prefectural governments adopted corresponding measures for lower-level jurisdictions.

Figure A.3: Cropland Compensation Intensity



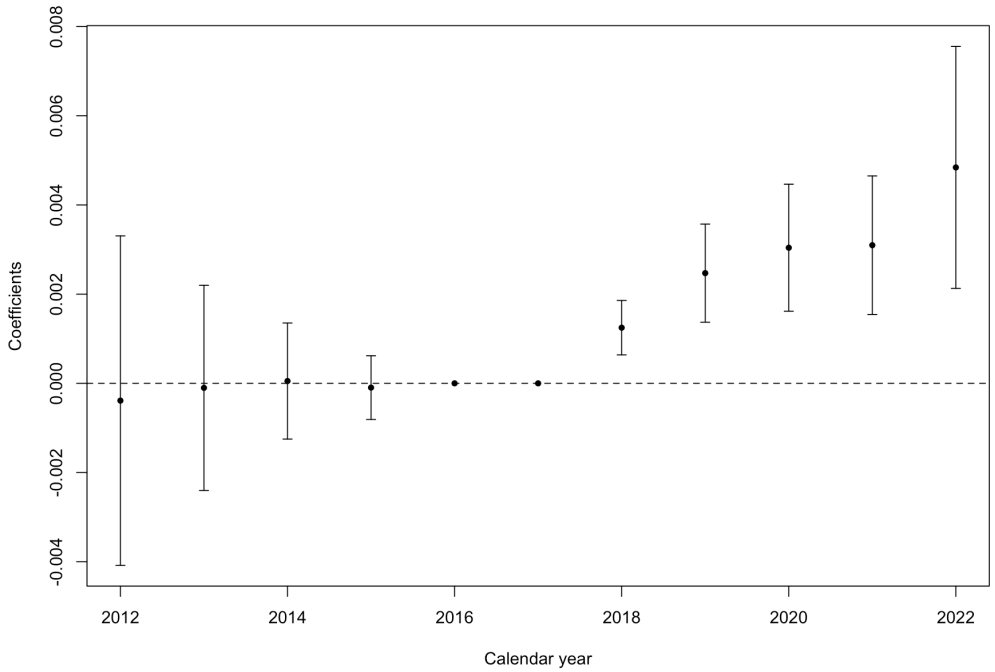
Notes: The map shows assigned cropland compensation obligations per square kilometer of provincial land area, in hectares. Darker shades indicate greater intensity; grey areas have missing obligation data. Jiangsu, Zhejiang, and Guangdong are marked because government reports identify acute land-use constraints or difficulty meeting compensation obligations locally (Appendix Section G.3).

Figure A.4: Rambachan–Roth Bounds for the Cropland-Share Event Study

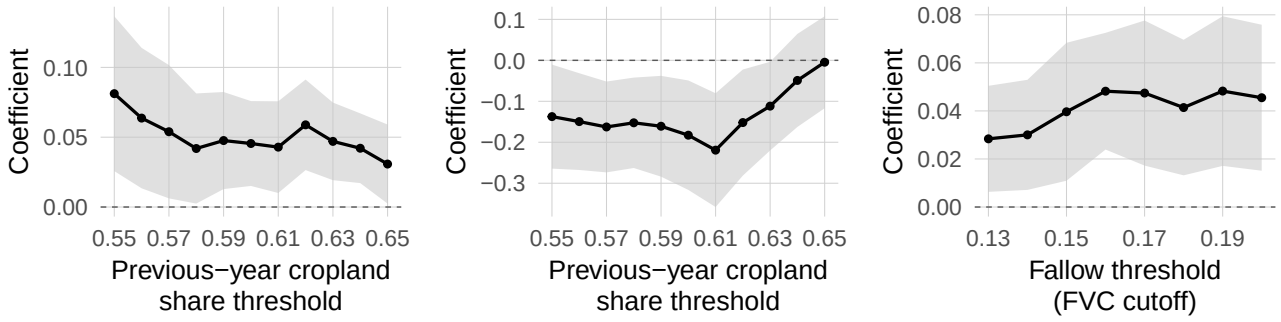


Notes: The figure reports 95 percent confidence intervals for the coefficient on standardized county cropland protection pressure interacted with the 2018 indicator in the event-study specification, relative to the omitted 2017 period. "Original" reports the conventional confidence interval based on standard errors clustered by county. The remaining bars report robust confidence intervals at the indicated values of $\overline{M}$ under the relative-magnitude restriction of Rambachan and Roth (2023). Cropland protection pressure is standardized to mean zero and standard deviation one across the 64 non-core counties, so each interval represents the effect of a one-standard-deviation increase in pressure. The outcome is grid-level cropland share, measured in percentage points. The specification includes grid-cell and year FE and uses 3,743,982 grid-year observations. The bound $\overline{M}$ limits changes in post-treatment deviations from parallel trends relative to the largest corresponding pre-treatment change. The lower endpoint remains above zero at $\overline{M} = 0.75$ and includes zero at $\overline{M} = 0.80$, placing the break value in $(0.75, 0.80]$.

Figure A.5: Cropland-Share Event Study with County-Specific Linear Trends



Notes: The figure plots cropland-protection-pressure-by-year coefficients for cropland share in non-core 500-meter grids, controlling for grid and year FE and county-specific linear calendar-year trends. The 2016 and 2017 interactions are omitted to normalize the slope and level. Bars show 95 percent confidence intervals based on county-clustered standard errors.



(a) Alternative reclaimed-land threshold: fallow

(b) Alternative reclaimed-land threshold: soil suitability

(c) Alternative fallow definition

Figure A.6: Sensitivity to Alternative Definitions

Notes: Each panel plots the coefficient on Pressure × Post from the corresponding cohort difference-in-differences specification. Shaded bands report 95% confidence intervals. Panels (a) and (b) vary the previous-year cropland-share cutoff used to define newly reclaimed land. Panel (c) varies the FVC cutoff used to classify a newly reclaimed grid cell as fallow. Panel (b) uses the continuous grid-average 1995 soil suitability score (1–5).

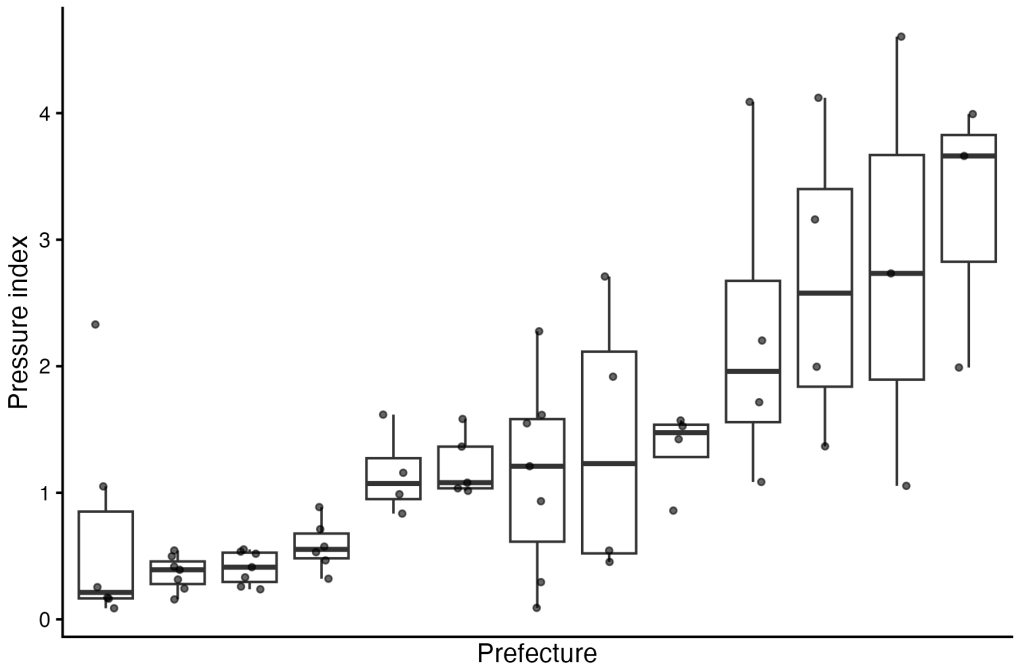
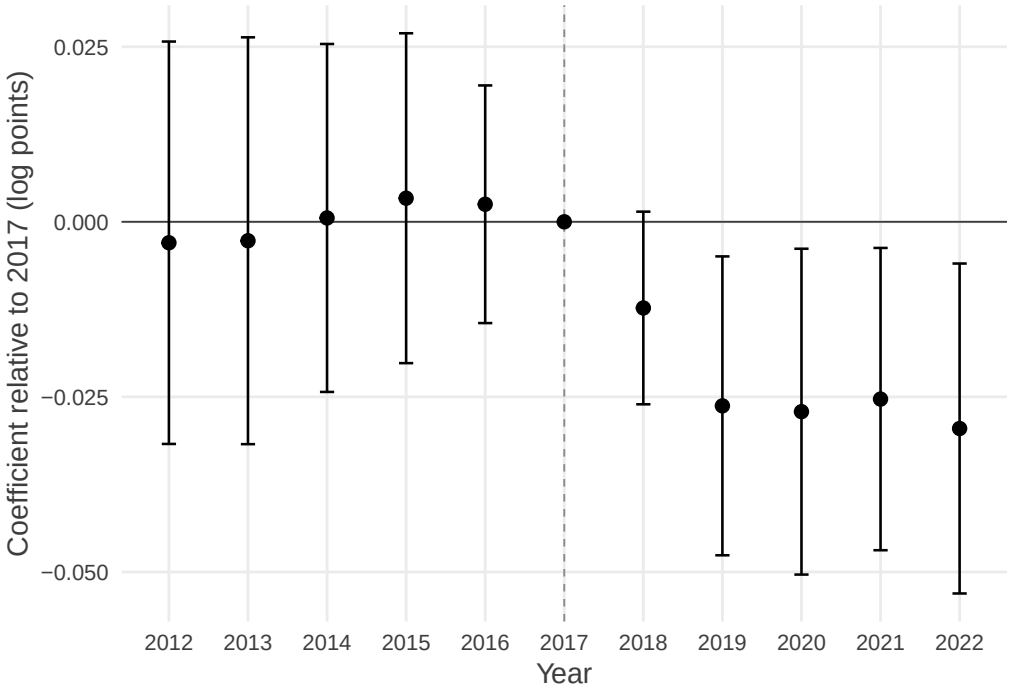


Figure A.7: Distribution of Cropland Protection Pressure within and across Prefectures

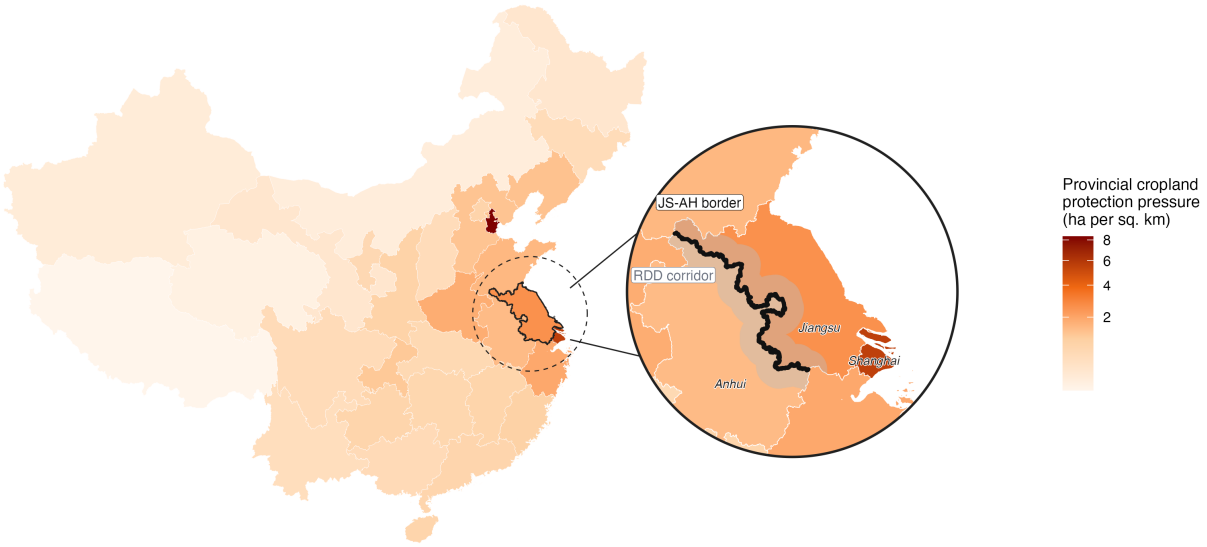
Notes: This figure plots the distribution of the cropland protection pressure index ($Pressure_c$) across counties within each prefecture. Each box summarizes the distribution of $Pressure_c$ among non-core counties within a prefecture: the box represents the interquartile range (25th–75th percentiles), and the thick horizontal line indicates the median. Dots denote individual counties.

Figure A.8: Effect of Cropland Protection Pressure on Crop Output per Cropland Hectare



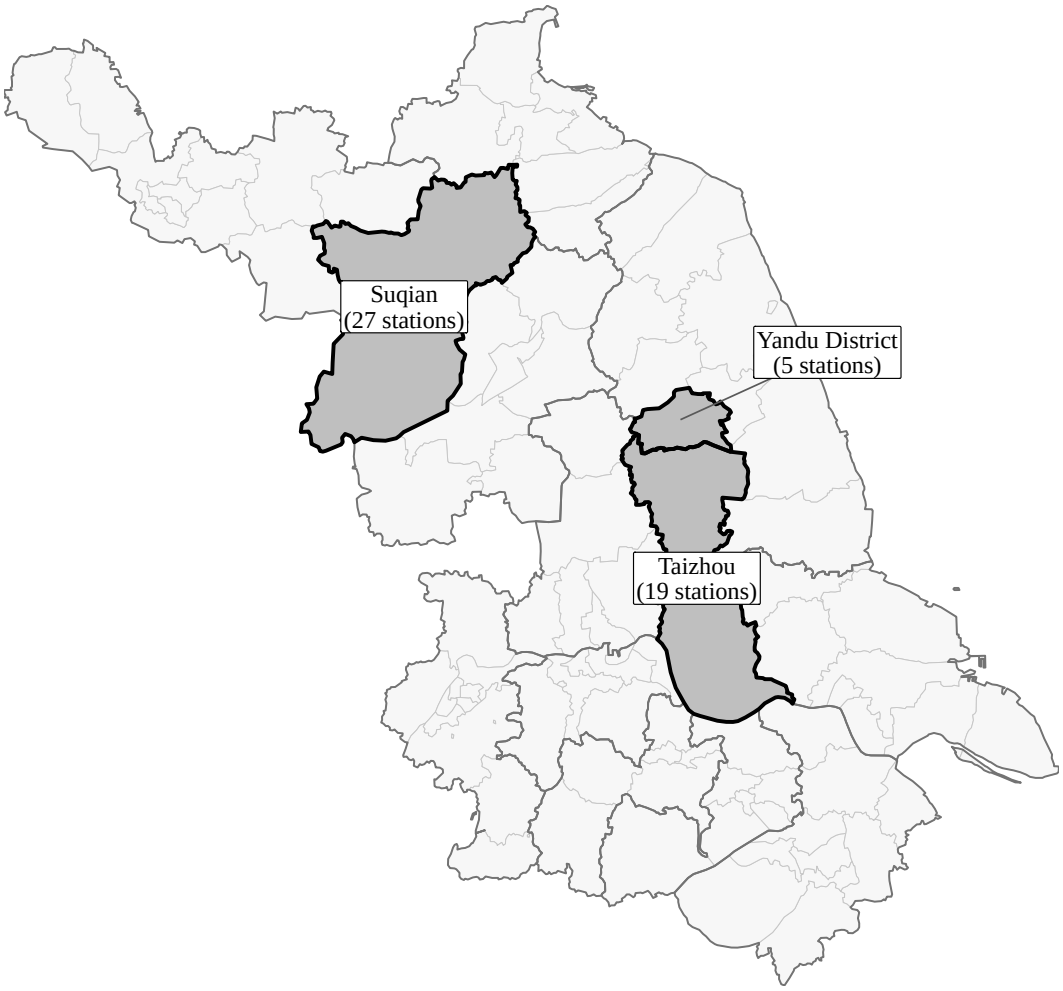
Notes: The figure plots coefficients on interactions between the county-level cropland protection pressure index and calendar-year indicators in a county-year panel, with 2017 omitted. Cropland protection pressure is standardized to mean zero and standard deviation one across the 64 non-core counties, so each coefficient reports the change associated with a one-standard-deviation increase in pressure relative to 2017. The outcome is the log of total grain, oilseed, and cotton output divided by satellite-measured cropland hectares. The specification includes county and year FE. Vertical bars report 95 percent confidence intervals based on standard errors clustered by county. The sample contains 533 county-year observations from 58 counties. A joint test of the 2012–2016 coefficients yields $p = 0.741$.

Figure A.9: Provincial Cropland Protection Pressure at the Border between Jiangsu and Anhui



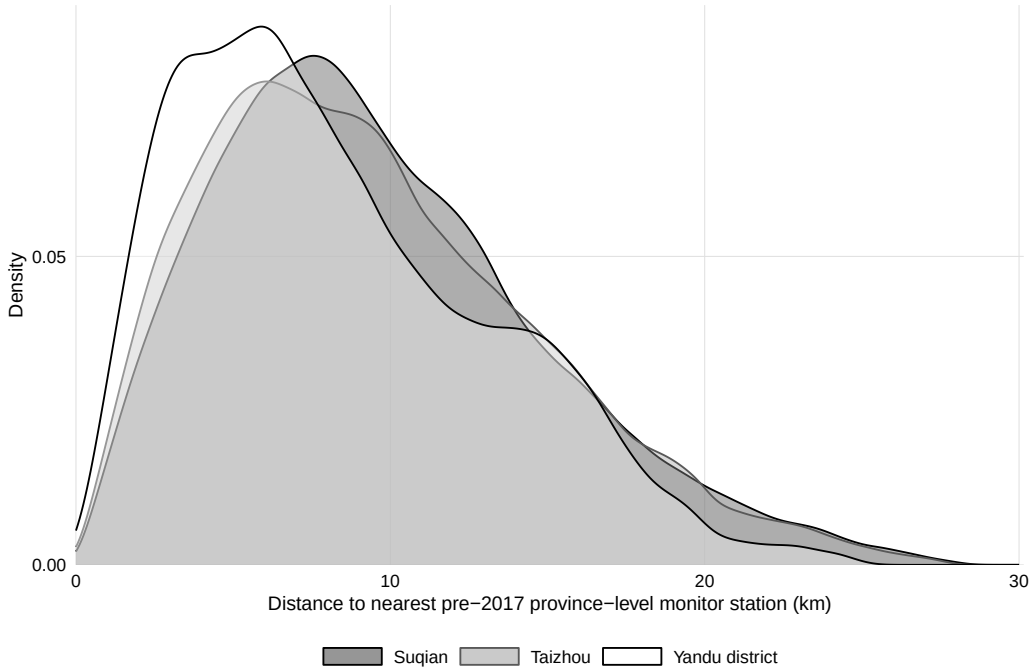
Notes: The national map shades mainland province-level units by provincial cropland protection pressure, measured as hectares of assigned cropland compensation per square kilometer of land area. Darker shades indicate greater pressure. The inset enlarges the region around Jiangsu and marks the spatial RD-DID corridor along its border with Anhui. Jiangsu faces greater provincial cropland protection pressure than Anhui, while Shanghai faces greater pressure than Jiangsu.

Figure A.10: Monitoring-Station Sample Areas in Jiangsu



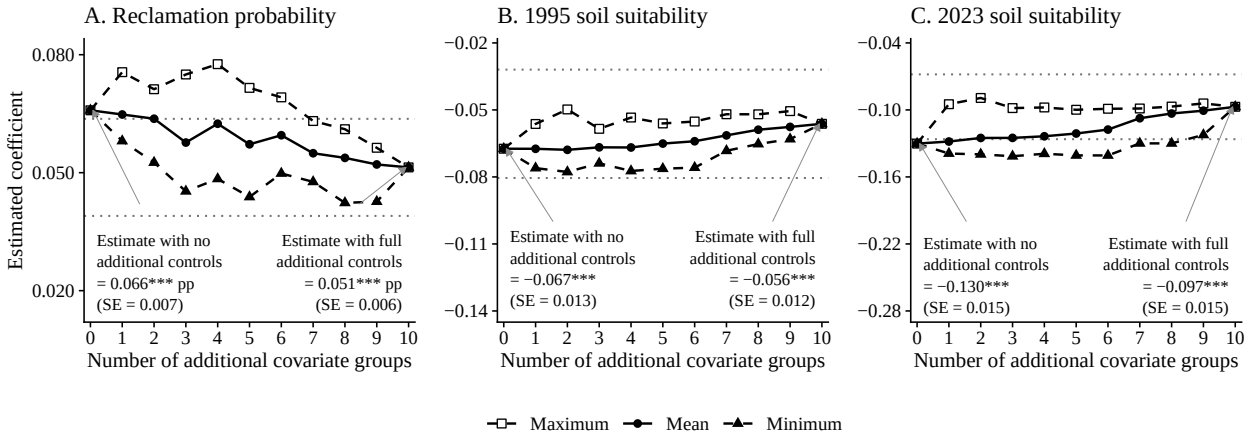
Notes: Shaded areas identify the jurisdictions covered by the monitoring-station sample used in Table 4: Suqian Prefecture, Taizhou Prefecture, and Yandu District of Yancheng Prefecture. Labels report the number of pre-2017 province-level cropland-quality monitoring stations retained in each area: 27 in Suqian, 19 in Taizhou, and 5 in Yandu. Thin lines show county boundaries, and thicker lines show prefecture boundaries.

Figure A.11: Distance Distribution in the 100m Monitor-Station Sample

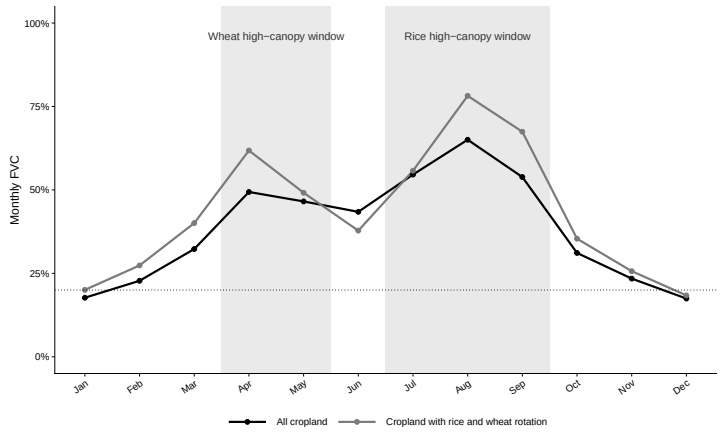


Notes: This figure plots the distribution of distances from 100-meter grid-cell centroids to the nearest pre-2017 province-level cropland-quality monitoring station. The station set matches Table 4: province-level stations in Suqian, Taizhou, and Yandu district. The public reports provide 51 pre-2017 province-level stations: 27 in Suqian, 19 in Taizhou, and 5 in Yandu. Across the 1.55 million 100-meter grids in this subsample, the mean distance to the nearest province-level station is 9.47 km, the median is 8.70 km, and the interquartile range is 5.52–12.71 km. Distances are measured in kilometers.

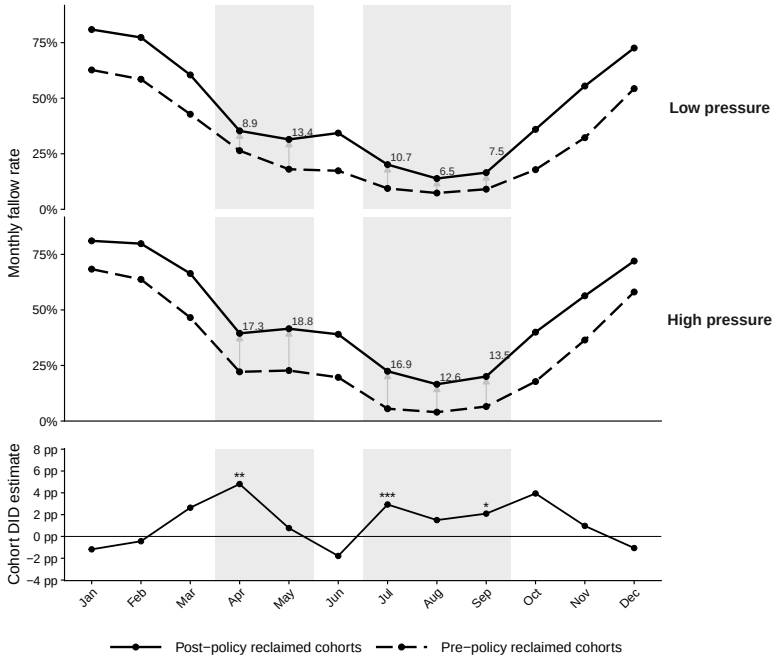
Figure A.12: Sensitivity of Monitoring-Station Distance Estimates to Additional Covariates



Notes: Each panel holds the corresponding Table 4 specification fixed and adds subsets of ten covariate groups: county-boundary distance; soil suitability, cropland, impervious surface, forest, and water conditions within 5 kilometers of the nearest station; government-seat distance; elevation; terrain ruggedness; and nighttime lights. Panel (a) reports, in percentage points, the coefficient on $\log(d_i) \times \text{Post}$ for annual reclamation; panels (b) and (c) report the level coefficient on $\log(d_i)$ for the 1995 and 2023 soil suitability measures in outcome-share units. The zero-additional-group estimates reproduce Table 4 columns (2), (6), and (8). In panel (a), each added time-invariant covariate enters both in levels and interacted with Post; in panels (b) and (c), it enters in levels. For each number of added groups, the solid and dashed series report the mean, maximum, and minimum coefficient across the displayed specifications. The figure uses the unique specifications at zero and ten groups, all ten subsets at one and nine groups, and ten fixed random subsets at each count from two through eight; the same subsets are used across panels. Dotted lines show the parent-500-meter-clustered 95 percent confidence interval for the ten-group specification, and endpoint standard errors use the same clustering. Each 100-meter cell inherits its parent-grid share. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



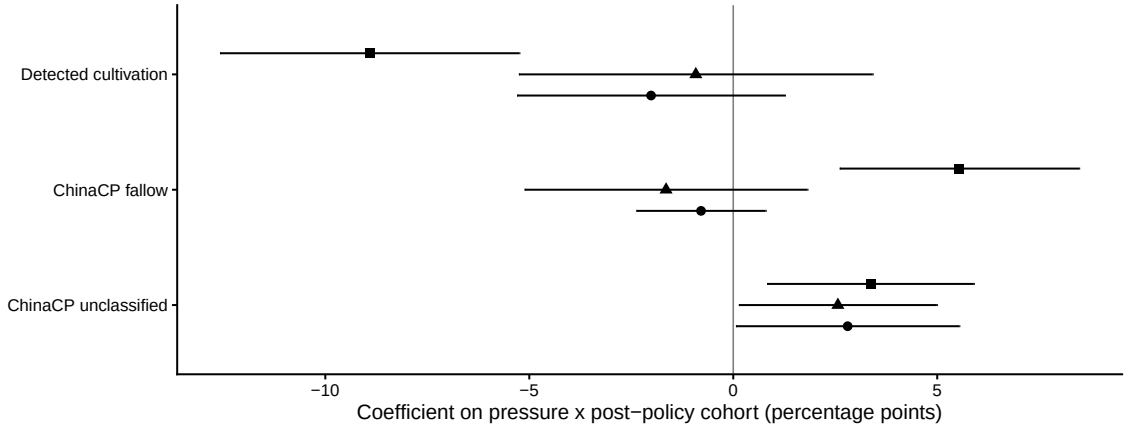
(a) Pre-policy monthly FVC profile



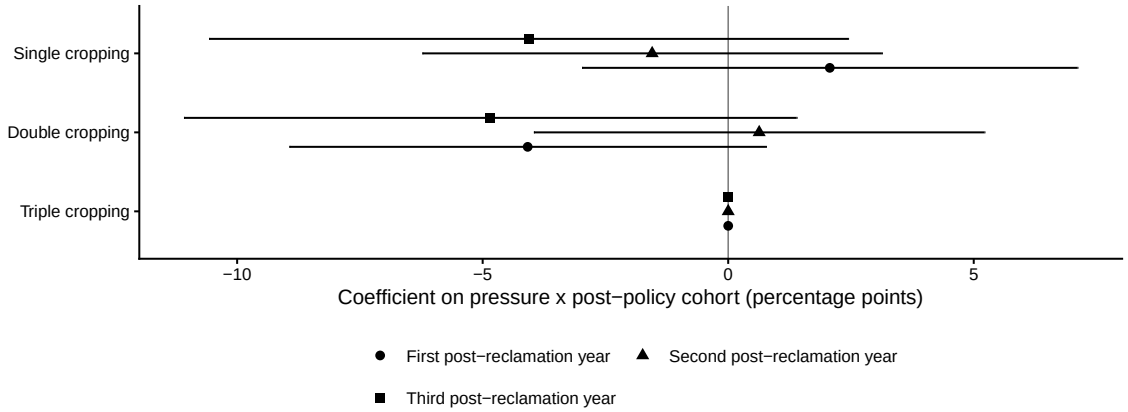
(b) Monthly fallow-rate profiles and cohort DID estimates

Figure A.13: Seasonal FVC Benchmarks and Monthly Fallow-Rate Effects

Notes: Panel A plots mean monthly FVC for all cropland in 2012–2017 and for cropland with rice and wheat rotation in 2015–2017, the years for which crop-pattern data are available. The gray shaded regions mark crop-specific high canopy windows: April–May for winter wheat and July–September for rice. The dotted horizontal line marks FVC = 0.20, the threshold used to define monthly fallow. Panel B uses newly reclaimed grids observed in the first four years after reclamation, $s \in \{1, 2, 3, 4\}$. The upper two plots report mean monthly fallow rates for pre-policy and post-policy reclaimed cohorts separately in low- and high-pressure counties; high-pressure counties are those above the sample county median of the pressure index. Light-gray arrows connect the pre-policy and post-policy values in the high canopy windows, and the adjacent labels report the increase in percentage points. The lower plot reports month-specific cohort DID coefficients on pressure by post-policy cohort. The cohort DID specifications include county, reclamation-cohort, and years-since-reclamation FE; standard errors are clustered by prefecture. Monthly estimation N ranges from 3,925 to 4,004. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.



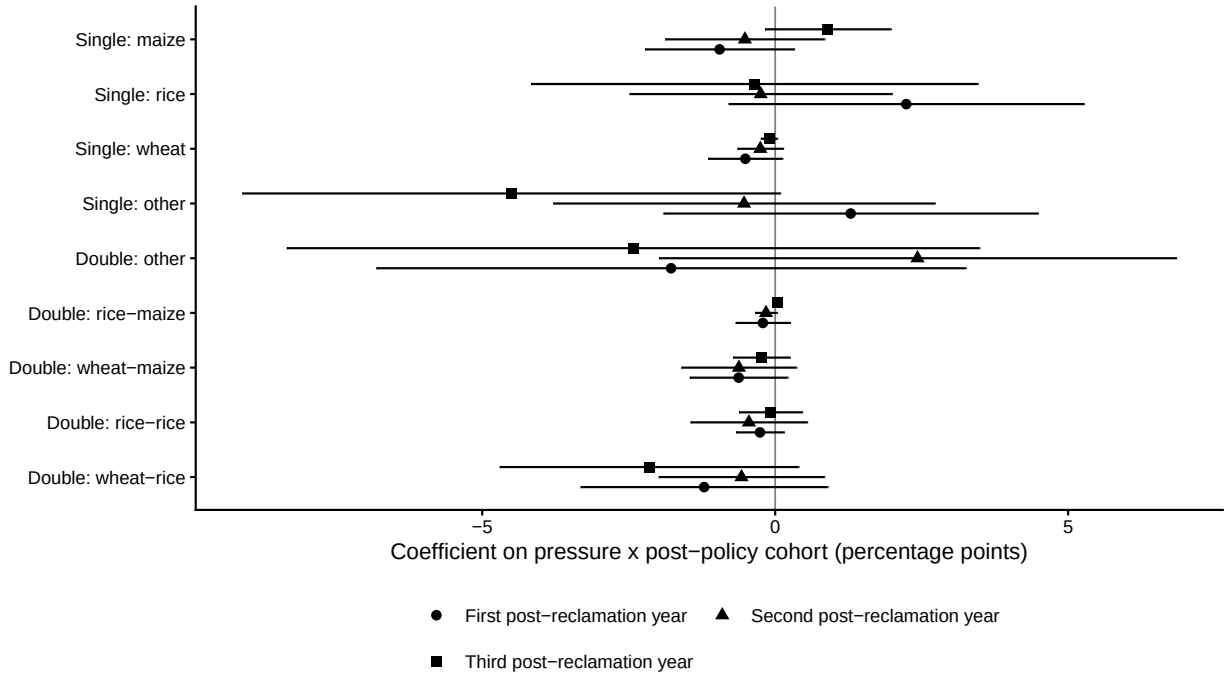
(a) Cultivation status



(b) Decomposition of detected cultivation

Figure A.14: Crop Patterns among Newly Reclaimed Pixels

Notes: Panel A partitions newly converted CLCD pixels into detected cultivation, the ChinaCP fallow category, and ChinaCP unclassified; Panel B decomposes detected cultivation into mutually exclusive single-, double-, and triple-cropping shares. Newly converted pixels are non-cropland in $t-1$ and remain cropland through $t+2$, subject to right censoring. ChinaCP unclassified means no detected crop-pattern code and is not treated as observed fallow. Points report Model 3 cohort DID estimates separately by post-reclamation year; horizontal lines show 95 percent confidence intervals. Standard errors are clustered by county. The third-year estimate is identified by the 2018 post-policy cohort.



(c) Crop-specific decomposition

Figure A.14: Crop Patterns among Newly Reclaimed Pixels (continued)

Notes: Panel C decomposes the single- and double-cropping shares in Panel B into nine mutually exclusive crop-specific ChinaCP classes. Triple cropping remains an aggregate category because ChinaCP does not identify its constituent crops. Points report the same Model 3 cohort DID estimates and 95 percent confidence intervals as Panels A and B. Within each post-reclamation year, p-values are adjusted across the nine classes using the Benjamini–Hochberg false discovery rate procedure; no adjusted q-value is below 0.10.

B Tables

Table B.1: Data Sources

Data	Source and coverage
Cropland compensation obligations and core-urban status	Prefecture Land Use Master Plans (2006–2020)
Land-cover data and county land shares	China Land Cover Dataset (CLCD), 30 m annual, 2012–2022; mainland China county boundaries, Harvard Dataverse
Fractional vegetation cover	National Tibetan Plateau Data Center, 30 m, 15-day, 2010–2022
Soil data	Soil Map of China, 1:1,000,000, National Soil Survey Office (1995); Harmonized World Soil Database v2.0, approximately 1 km (2023)
County socioeconomic data	China County Statistical Yearbooks
Officials' biographies	Publicly available biographies of prefecture- and province-level officials
Government work reports and quarterly GDP	Municipal Government Work Reports and compiled prefecture-level quarterly GDP releases and statistical summaries, 2012–2022
Construction-land transactions	China Land Market Network primary-market transaction records for Jiangsu, 2012–2022
Cropland-quality monitoring stations	Public cropland-quality monitoring reports for Suqian, Taizhou, and Yandu District; 51 province-level stations established before 2017
Cropping patterns	China Cropping Pattern maps (ChinaCP), 500 m annual, 2015–2021
Crop-specific harvested area	ScienceDB gridded maps for 16 crops, 1 km, five-year intervals, 1990–2020
Annual wetland classifications	Global Annual Wetland Dataset at 30 m with a Fine Classification System from 2000 to 2022 (GAWD) (Zhang et al., 2024). Annual files are archived as Time-Series Global 30 m Wetland Maps from 2000 to 2022 on Zenodo (Liu and Zhang, 2023). Analysis coverage is 2012 to 2021
Elevation and terrain ruggedness	NASA digital elevation model, 30 m
County agricultural values	China County Statistical Yearbooks
Consumer Price Index	World Bank World Development Indicators, China

Table B.2: Mapping Soil Types to Suitability Classes

Score	FAO Class	Representative Soil Groups
5 (S1)	Highly suitable	Fluvisols, Anthrosols, alluvial soils
4 (S2)	Moderately suitable	Cambisols, Luvisols, Ferralsols, Andosols
3 (S3)	Marginally suitable	Regosols, Leptosols, Cambic soils, hardpan soils
2 (N1)	Currently unsuitable	Saline soils, sodic soils, waterlogged paddy soils
1 (N2)	Permanently unsuitable	Urban land, marsh soils, water bodies, sandbars

Table B.3: Parallel-Trends Tests for the Cohort Difference-in-Differences Design

	China Soil Map 1995 Soil suitability (1)	HWSD v2 2023 Soil suitability (2)	Fallow **1**(Fallow) (3)
Pressure × false post (2014)	0.000	-0.022	0.063
	(0.028)	(0.025)	(0.052)
Pressure × false post (2015)	0.009	-0.004	0.057
	(0.025)	(0.015)	(0.051)
Pressure × false post (2016)	0.001	-0.023	0.057
	(0.019)	(0.024)	(0.046)
County FE	✓	✓	✓
Reclamation-cohort FE	✓	✓	✓
Cluster level	County	County	County
Number of counties	47	47	49
Num. Obs.	448	448	460
R^2 (2014 cutoff)	0.598	0.566	0.348
R^2 (2015 cutoff)	0.598	0.565	0.347
R^2 (2016 cutoff)	0.598	0.566	0.348

Notes: The sample contains the pre-policy reclamation cohorts 2012, 2014, 2015, and 2016. Each row reports a separate regression that defines false post as an indicator for a reclamation cohort at or after the year shown and interacts it with cropland protection pressure. Each newly reclaimed grid is observed in its first post-reclamation year, so realized outcomes fall in calendar years 2013–2017. For both soil maps, source-pixel or soil-component indicators equal to one for suitability scores of at least three are averaged using area weights within each 500-meter grid. These suitable-land shares range from zero to one, and the two soil columns retain the same common pre-policy soil sample. Fallow equals one when fractional vegetation cover is below 0.20. All regressions include county and reclamation-cohort FE. Standard errors clustered by county are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.4: Standardized Pressure and Cropland Share

	Cropland share
2012 ($k = -6$) × Standardized Pressure	-0.002
	(0.001)
2013 ($k = -5$) × Standardized Pressure	-0.001*
	(0.001)
2014 ($k = -4$) × Standardized Pressure	-0.001
	(0.001)
2015 ($k = -3$) × Standardized Pressure	-0.001
	(0.000)
2016 ($k = -2$) × Standardized Pressure	-0.000
	(0.000)
2018 ($k = 0$) × Standardized Pressure	0.002***
	(0.000)
2019 ($k = 1$) × Standardized Pressure	0.003***
	(0.001)
2020 ($k = 2$) × Standardized Pressure	0.004***
	(0.001)
2021 ($k = 3$) × Standardized Pressure	0.004***
	(0.001)
2022 ($k = 4$) × Standardized Pressure	0.007**
	(0.003)
Grid and year FE	✓
Number of counties	64
Num. Obs.	3,743,982
R^2	0.981

Notes: Pressure is standardized among non-core counties. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.5: Standardized Pressure: Newly Reclaimed Cropland

	China Soil Map 1995				HWSD v2 2023		Fallow	
	Soil suitability score (1–5) (1)	Soil suitability score (1–5) (2)	Soil suitability (3)	Soil suitability (4)	Soil suitability (5)	Soil suitability (6)	Fallow (7)	Fallow (8)
Standardized Pressure × Post	-0.202***	-0.223***	-0.057**	-0.063***	-0.035*	-0.041*	0.061***	0.061***
	(0.064)	(0.071)	(0.022)	(0.023)	(0.019)	(0.021)	(0.017)	(0.019)
Control Mean	3.455	3.455	0.593	0.593	0.573	0.573	0.198	0.200
County controls × year		✓		✓		✓		✓
County & Cohort FE	✓	✓	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE							✓	✓
Cluster level	County	County	County	County	County	County	County	County
Number of counties	52	52	52	52	52	52	54	53
Num. Obs.	939	939	939	939	939	939	3,244	3,222
R^2	0.414	0.417	0.452	0.457	0.490	0.491	0.232	0.230

Notes: Pressure is standardized to mean zero and standard deviation one among non-core counties. Columns (1)–(2) use the continuous grid-average 1995 soil suitability score (1–5). In columns (3)–(6), soil suitability is the area-weighted fraction of source pixels or soil components with a suitability score of at least 3, aggregated to each 500-meter grid; it ranges from zero to one. Soil outcomes are observed in the first post-reclamation year. Fallow equals one when fractional vegetation coverage is below 0.20 and is measured during the first four post-reclamation years. Control Mean is the pre-2018 reclamation-cohort mean within each column's estimation sample. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.6: Spatial RD-DID Placebo Tests Assigning the Policy Change to Pre-Policy Years

	(1)	(2)	(3)
Panel A: Cropland-share change			
RD in 2013-2012 change	-0.002	-0.002	-0.001
(Jiangsu − Anhui)	(0.002)	(0.002)	(0.002)
RD in 2014-2012 change	-0.001	-0.001	-0.001
(Jiangsu − Anhui)	(0.002)	(0.002)	(0.002)
RD in 2015-2012 change	-0.001	-0.001	0.000
(Jiangsu − Anhui)	(0.002)	(0.002)	(0.003)
RD in 2016-2012 change	0.000	0.001	0.001
(Jiangsu − Anhui)	(0.003)	(0.003)	(0.003)
Eligible cells	161,770	161,770	161,770
Panel B: Fallow-rate change (ex ante marginal cells)			
RD in 2013-2012 change	0.004	0.003	0.000
(Jiangsu − Anhui)	(0.008)	(0.009)	(0.010)
RD in 2014-2012 change	-0.015	-0.017	-0.025
(Jiangsu − Anhui)	(0.014)	(0.015)	(0.017)
RD in 2015-2012 change	0.015	0.012	0.022
(Jiangsu − Anhui)	(0.018)	(0.019)	(0.016)
RD in 2016-2012 change	0.006	0.004	0.006
(Jiangsu − Anhui)	(0.017)	(0.018)	(0.017)
Eligible cells	30,600	30,600	30,600
County-pair FE	✓	✓	✓
Kernel	Triangle	Epanech.	Uniform

Notes. Each row reports a separate spatial RD-DID falsification estimate at the Jiangsu–Anhui border. Outcomes are changes relative to 2012 for 2013–2016. The falsification window ends before the 2017 policy announcement. The running variable is signed distance to the provincial border in kilometers, with positive values on the Jiangsu side. Panel A uses all eligible non-mountain cropland cells in matched cross-border county pairs within the ±50 km corridor and is not restricted by FVC availability. Panel B uses cells with complete FVC in 2012–2022 and restricts the sample to cells with 2012–2017 mean cropland share below 0.6. Each column is a covariate-adjusted local-linear RD-DID that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Eligible cells are counted before local bandwidth selection. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Table B.7: Spatial RD-DID Falsification Checks Using Alternative Outcomes

	N	(1)	(2)	(3)
Geography, soil, and baseline characteristics				
Mean cropland share, 2012–2016	161,770	0.010	0.007	0.009
(Jiangsu − Anhui)		(0.018)	(0.017)	(0.020)
Mean elevation (m)	160,565	0.350	0.444	0.578
(Jiangsu − Anhui)		(2.615)	(2.651)	(2.699)
Mean terrain ruggedness index (m)	160,565	-0.001	0.004	0.003
(Jiangsu − Anhui)		(0.255)	(0.263)	(0.277)
Soil suitability	161,770	0.004	0.006	0.011
(Jiangsu − Anhui)		(0.030)	(0.031)	(0.033)
Mean annual precipitation, 1970–2000 (mm)	161,770	1.289	1.621	1.823
(Jiangsu − Anhui)		(3.271)	(3.310)	(3.364)
Policy invariant outcomes				
Change in mean annual precipitation (mm)	161,770	-3.792	-4.021	-3.989
(Jiangsu − Anhui)		(3.650)	(3.676)	(3.762)
Change in mean annual temperature (°C)	161,770	-0.000	-0.000	-0.000
(Jiangsu − Anhui)		(0.001)	(0.001)	(0.001)
County-pair FE		✓	✓	✓
Kernel		Triangle	Epanech.	Uniform
Cluster level		County	County	County
Number of clusters		26	26	26

Notes. Each row reports a separate border-discontinuity falsification estimate at the Jiangsu–Anhui border. Coefficients are Jiangsu-minus-Anhui discontinuities. Mean cropland share is measured over 2012–2016. Soil suitability is the area-weighted fraction of source soil units with suitability scores of at least three within each 500-meter grid. Both provinces use HWSD v2 and the same suitability crosswalk; the threshold is applied before aggregation. Mean annual precipitation is the sum of the 12 WorldClim 2.1 monthly normals for 1970–2000, bilinearly interpolated at each grid centroid. The first five rows test level discontinuities in geography, soil, and baseline characteristics; the final two test discontinuities in pre-to-post changes. The soil database release date does not identify a common survey year. Changes in mean annual precipitation and temperature are the 2018–2022 average minus the 2012–2017 average from TerraClimate v1.1. Mean annual temperature is the day-weighted mean of monthly $(T_{\max} + T_{\min})/2$. These final two rows are policy invariant outcomes because the weather measures should not respond to the 2017 enforcement tightening. The running variable is signed distance to the provincial border in kilometers, with positive values on the Jiangsu side. The sample uses all eligible non-mountain cropland cells in matched cross-border county pairs within the ±50 km corridor. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. N is the outcome-specific complete-case count before local bandwidth selection. A regression R^2 is not reported for the robust bias-corrected local RD estimator. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.8: Cropland-Share and Fallow-Rate Discontinuities Across Provincial Borders

	Main comparison			Robustness comparison		
	Jiangsu–Anhui			Jiangsu–Shanghai		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ Cropland share						
RD in Δ cropland share	0.006**	0.006**	0.007**	-0.027*	-0.027*	-0.029**
(Jiangsu − comparison side)	(0.003)	(0.003)	(0.003)	(0.014)	(0.015)	(0.012)
Panel B: Δ Fallow rate						
RD in Δ fallow rate	0.037***	0.038***	0.037***	-0.256***	-0.266***	-0.266***
(Jiangsu − comparison side)	(0.010)	(0.011)	(0.014)	(0.035)	(0.031)	(0.042)
Panel C: Δ Fallow rate (excluding high pre-policy built-up share)						
RD in Δ fallow rate	0.037***	0.034***	0.039***	-0.205***	-0.206***	-0.214***
(Jiangsu − comparison side)	(0.009)	(0.011)	(0.011)	(0.025)	(0.024)	(0.019)
Panel D: Δ Fallow rate (excluding high pre- or post-policy built-up share)						
RD in Δ fallow rate	0.032***	0.030**	0.037***	-0.216***	-0.212***	-0.216***
(Jiangsu − comparison side)	(0.009)	(0.012)	(0.011)	(0.022)	(0.021)	(0.018)
County-pair FE	✓	✓	✓	✓	✓	✓
Kernel	Triangle	Epanech.	Uniform	Triangle	Epanech.	Uniform

Notes. Columns (1)–(3) compare Jiangsu with Anhui. Columns (4)–(6) compare Jiangsu with Shanghai as a robustness check; Shanghai faces greater provincial cropland protection pressure than Jiangsu. The samples contain cells in matched cross-border county pairs within the ±50 km corridor; the Jiangsu–Anhui samples also exclude mountain cells. Panel A uses all eligible cropland cells and is not restricted by FVC availability. Its outcome is the cell's mean cropland share in 2018–2022 minus its mean in 2012–2017. Panel B uses cells with complete FVC in both periods and restricts the sample to ex ante marginal cells with 2012–2017 mean cropland share below 0.6. Its outcome is the change in mean fallow rate between 2012–2017 and 2018–2022. Panels C and D address the possibility that urban expansion lowers FVC on ex ante marginal cells independently of agricultural use. Panel C excludes cells whose mean built-up share in 2012–2017 exceeds 0.8. Panel D excludes cells whose mean built-up share exceeds 0.8 in either 2012–2017 or 2018–2022. The running variable is signed distance to the provincial border, with positive values on the Jiangsu side. Each column is a covariate-adjusted local-linear RD-DID that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Table B.9: Cohort Difference-in-Differences in Post-Reclamation Fallow Rate at the Jiangsu–Anhui Border

	Fallow					
	Pooled		By years since reclamation s			
	(1)	(2)	$s = 1$	$s = 2$	$s = 3$	$s = 4$
Jiangsu × Post	0.157**	0.138	0.022	0.129	0.221*	0.362***
	(0.068)	(0.085)	(0.093)	(0.097)	(0.115)	(0.107)
Control mean	0.068	0.068	0.099	0.055	0.059	0.059
Cross-border county-pair FE		✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE	✓	✓				
Number of counties	20	20	20	20	20	19
Num. Obs.	1,797	1,797	541	492	433	331
R^2	0.195	0.254	0.298	0.235	0.297	0.316

Notes: Each column reports the cohort-difference-in-differences coefficient on $\text{Jiangsu}_i \times \text{Post}_i$ in a regression of the post-reclamation fallow indicator $\mathbf{1}\{\text{FVC}_{i,t_0(i)+s} < 0.20\}$ on Jiangsu_i, $\text{Post}_i = \mathbf{1}\{\text{cohort}_i \geq 2018\}$, and their interaction. The unit of observation is a 500 m grid cell–years-since-reclamation pair (i, s), with $s = 1, \ldots, 4$ years after the cell's first reclamation event $t_0(i)$. The sample uses the FVC-complete matched-pair corridor underlying Table B.8 Panels B–D before their marginal-cell restrictions: cells in the ±50 km Jiangsu–Anhui corridor inside cross-border county pairs, excluding mountain cells and requiring non-missing pre- and post-policy FVC. The sample is further restricted to cells reclaimed at least once during 2013–2022. Columns (1)–(2) report the pooled cohort DiD coefficient, with cohort and years-since-reclamation FE in (1) and adding cross-border county-pair FE in (2). Columns (3)–(6) split the sample by years since reclamation, $s = 1, \ldots, 4$, with cross-border county-pair and cohort FE. The cohort FE absorb the pre- vs. post-2017 cohort trend, so the interaction coefficient identifies the difference between the post-2017 Jiangsu fallow excess and the corresponding pre-2017 cross-side gap. The Post main effect is mechanically absorbed by the cohort FE and is therefore omitted. Standard errors are clustered at the county level and reported in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.10: Robustness: Alternative Definitions of Land-Pressure Exposure

	(1) Cropland Share	(2) Soil suitability	(3) Fallow
Panel A: All Land Types as Available Land			
Pressure × Post	0.006***	−0.071***	0.078***
	(0.002)	(0.024)	(0.026)
Grid & Year FE	✓		
County & Cohort FE		✓	✓
Years-since-reclamation FE			✓
Cluster level	County	County	County
Number of counties	64	52	54
Num. Obs.	3,743,982	939	3,243
R^2	0.981	0.452	0.230
Panel B: Alternative Definition of Core-Urban Districts			
Pressure × Post	0.003**	−0.051*	0.084**
	(0.002)	(0.028)	(0.032)
Grid & Year FE	✓		
County & Cohort FE		✓	✓
Years-since-reclamation FE			✓
Cluster level	County	County	County
Number of counties	64	52	54
Num. Obs.	3,743,982	939	3,243
R^2	0.981	0.451	0.234

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: This table examines the robustness of the main results to alternative definitions of cropland protection pressure. In Panel A, *available land* is redefined to include all land types, rather than excluding water bodies and impervious surfaces as in the baseline specification. In Panel B, the classification of *core-urban districts* is tightened by coding only fully core districts as core, while partially core districts are grouped with non-core districts. Column (2) measures the suitable-land share using the 1995 soil map: source pixels with suitability scores of at least three are classified as suitable before area-weighted aggregation to 500-meter grids. Cropland specifications include grid and year FE; suitability specifications include county and reclamation-cohort FE; fallowing specifications additionally include years-since-reclamation FE. No specification includes additional county-level controls. Standard errors are clustered by county.

Table B.11: Land Quality and Fallowing of Newly Reclaimed Land

	Outcome: Fallow	
	(1) Soil suitability score (1–5)	(2) Soil suitability
Pressure × Post	0.099***	0.088***
	(0.030)	(0.025)
Land quality × Pressure × Post	−0.010**	−0.037**
	(0.005)	(0.018)
County & Cohort FE	✓	✓
Years-since-reclamation FE	✓	✓
Cluster level	County	County
Number of counties	52	52
Num. Obs.	3,185	3,185
R^2	0.234	0.234

Notes: This table reports cohort DID estimates using newly reclaimed cropland data. The dependent variable indicates annual fractional vegetation coverage below 0.20 in years 1–4 after reclamation. Treatment is county cropland protection pressure interacted with an indicator for reclamation from 2018 onward. Column (1) interacts treatment with the continuous 1995 soil suitability score (1–5). Column (2) uses the suitable-land share: source-pixel indicators equal to one for scores of at least three are averaged within each 500-meter grid using area weights. The threshold is applied before aggregation, so this grid-level share ranges from zero to one. All specifications include county, reclamation-cohort, and years-since-reclamation FE. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.12: Economic Growth Burden and Land Misallocation

	Rural cropland		Urban construction land expansion			
	(1) Cropland share	(2) 2023 all-suitable grid	(3) New construction count	(4) Farmland-origin count	(5) New construction area	(6) Farmland-origin area
Pressure × Post × economic burden	0.006*	-0.853***	137.199***	139.005***	267.707**	261.115**
	(0.003)	(0.034)	(26.532)	(25.322)	(95.222)	(86.557)
	[0.358]	[0.138]	[0.306]	[0.316]	[0.340]	[0.301]
Grid FE	✓					
Prefecture FE	✓	✓	✓	✓	✓	✓
Year/cohort FE	✓	✓	✓	✓	✓	✓
Party-secretary controls	✓	✓	✓	✓	✓	✓
Observations	3,050,309	748	116	116	116	116
R^2	0.985	0.343	0.691	0.714	0.604	0.637

Notes: This table reports DDD estimates of whether economic growth burden amplifies land misallocation. The coefficient of interest is Pressure × Post × Economic burden, where Pressure is the prefecture-level cropland protection measure and Post equals one after 2017. Economic burden equals one when a prefecture's predicted annual-growth shortfall is positive and ranks in the top tercile among Jiangsu prefectures in the same year. Columns (1)–(2) report rural cropland outcomes for newly reclaimed non-core cropland; column (2) retains the legacy binary outcome, equal to one only when the 2023 suitable-soil share equals one; fractional shares are coded zero. Columns (3)–(6) report prefecture-year urban-core construction-land transaction outcomes for contracts signed after March 31; counts are numbers of transactions and areas are measured in hectares. Columns (3) and (5) include transactions whose recorded land source is new construction land. Columns (4) and (6) further restrict these transactions to those whose centroid lies in a 500-meter grid containing cropland in the previous year; this is a proxy rather than a parcel-footprint measure. All specifications include the outcome-specific FE listed in the table, all lower-order DDD terms, and party-secretary controls. Parentheses report conventional standard errors clustered by prefecture, and stars are based on them. Square brackets report two-sided null-imposed, studentized Rademacher wild-cluster bootstrap p-values based on 9,999 random draws at the prefecture level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.13: Pooled Policy Effects with Beginning-of-Year Reservoir Controls

	Parsimonious (1)	With reservoir controls (2)
Panel A. Cropland share		
Pressure × Post	0.390***	0.310***
	(0.125)	(0.108)
Grid-cell FE	✓	✓
Calendar-year FE	✓	✓
Observations	3,135,060	3,135,060
Number of counties	58	58
R^2	0.984	0.984
Panel B. Soil suitability (1995)		
Pressure × Post	−5.667**	−5.023**
	(2.220)	(2.327)
County FE	✓	✓
Reclamation-cohort FE	✓	✓
Observations	938	938
Number of counties	51	51
R^2	0.451	0.453
Panel C. Fallow status		
Pressure × Post	6.048***	5.738***
	(1.687)	(1.796)
County FE	✓	✓
Reclamation-cohort FE	✓	✓
Years-since-reclamation FE	✓	✓
Observations	3,240	3,240
Number of counties	53	53
R^2	0.229	0.230
Remaining suitable share		✓
Depletion		✓

Notes: Entries report pooled Pressure × Post coefficients in percentage points per one-standard-deviation increase in cropland-protection pressure. Post equals one in 2018–2021, using calendar year in Panel A and reclamation cohort in Panels B and C. Column (1) re-estimates the parsimonious specification on the same complete-case sample as Column (2). Panel A measures grid-level cropland share. Panel B uses the harmonized 1995 soil measure: the within-grid area-weighted share of soil components with suitability scores of at least three. Panel C measures fallow status, defined by FVC below 0.20, during years one through four after reclamation. Remaining suitable share uses the same threshold-before-aggregation soil measure and is suitable land area in the beginning-of-year remaining reservoir divided by that reservoir's total area, multiplied by 100. Depletion is beginning-of-year cropland area within the fixed 2001 reservoir divided by its initial area, multiplied by 100. The fixed reservoir mask and depletion definition follow Appendix Section D; the quality control uses the harmonized suitable-land share. For Panels B and C, they are measured at the beginning of the reclamation year. Reservoir controls are available through 2021. County-clustered standard errors are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Additional Results

C.1 Seasonality of Fallow Land Use

Fallow land remains uncultivated when crops would normally be grown. An annual FVC-based measure may overstate fallow if low FVC reflects sparse vegetation in non-growing months rather than non-cultivation during the growing season. I therefore examine monthly FVC and estimate month-specific effects to test whether the fallow associated with cropland protection pressure appears when crops should be growing.

To establish Jiangsu's normal crop calendar before the policy intervention, I combine pre-policy vegetation data with agronomic evidence on the province's planting seasons. The China Cropping Pattern maps confirm that multiple cropping is widespread and that double cropping is the dominant cropping pattern in Jiangsu Province.

Consistent with this double-cropping calendar, Panel A of Figure A.13 displays two high-canopy periods in the pre-policy FVC profile. FVC rises in April and May with the winter wheat canopy, falls in June as wheat is harvested and fields transition to rice, and rises again from July through September with the rice canopy.[20] The same two-peak pattern appears for all cropland and for the rice and wheat rotation subset. Together, the pre-policy FVC profile and the crop-calendar evidence indicate that April–May and July–September are the high canopy windows, when winter wheat and rice, respectively, produce the highest vegetation cover.

I then test whether newly reclaimed land in high-pressure counties is more likely to be fallow during these high canopy windows. I estimate month-specific effects using the cohort DID specification in Model 3, with county, reclamation-cohort, and years-since-reclamation fixed effects.

Panel B of Figure A.13 reports monthly fallow-rate profiles and cohort DID estimates. The upper two plots show mean fallow rates for post-policy and pre-policy reclaimed cohorts separately in low- and high-pressure counties. In both pressure groups, post-policy reclaimed land is more likely to be fallow than the pre-policy benchmark throughout the year. The labels report these increases during the main growing seasons. They range from 6.5 to 13.4 percentage points in low-pressure counties and from 12.6 to 18.8 percentage points in high-pressure counties. The increase is therefore larger in high-pressure counties in each of the five high-canopy months. This contrast is the descriptive counterpart of the policy effect of interest, suggesting that tighter cropland constraints amplify the post-policy increase in fallow when active cultivation should produce dense crop cover. The lower plot tests the same hypothesis using continuous pressure and the cohort DID specification in Model 3. The estimates are positive in all five high-canopy months and statistically significant in April, July, and September, whereas the lean-season estimates are small and insignificant and June is negative.

Together, these results show that the FVC-based fallow effect captures reduced cultivation

[20] In 2015, rice and wheat account for 81 percent of harvested area and co-occur in 78 percent of harvest-positive grids. Crop-pattern maps classify 86 percent of non-fallow area as double cropped in 2015–2017. Rice and wheat rotation is therefore the principal benchmark, but not the only double-cropping system.

during the main growing seasons rather than normal seasonal variation in vegetation cover.

C.2 Crop Patterns and Fallow Land Use

The seasonal evidence shows that fractional vegetation cover-based fallow increases when crop canopies should be dense. A remaining explanation is that different crops or cropping frequencies produce lower vegetation cover on cultivated land. I use annual cropping-pattern maps to distinguish this explanation from lower detected cultivation.

The China Cropping Pattern maps (hereafter ChinaCP) classify crop activity at a 500-meter resolution from 2015 through 2021. Each pixel-year receives one mutually exclusive crop-pattern code.[21] Within each reclaimed grid, I use CLCD to identify pixels that become persistent cropland, assign them the ChinaCP category observed in each post-reclamation year, and aggregate these matches into grid-event shares. The shares partition newly converted pixels into detected cultivation, the ChinaCP fallow category, and ChinaCP unclassified. Detected cultivation is further divided into single, double, and triple cropping. All shares use the number of persistent newly converted pixels in the grid-event as their denominator; hence the frequency components sum to detected cultivation and the full partition to one.

I estimate Model 3 separately for the first three years after reclamation. The specification interacts continuous county-level cropland protection pressure with the post-policy cohort indicator, absorbs county, reclamation-cohort, and years-since-reclamation fixed effects, and clusters standard errors by county.

Panel A of Figure A.14 suggests a delayed reduction in detected cultivation on newly reclaimed land in more pressured counties. In the first two post-reclamation years, I find no significant differences in detected cultivation or assignment to the ChinaCP fallow category across counties with different levels of policy exposure. By the third year, however, detected cultivation falls by 8.9 percentage points and the ChinaCP fallow share rises by 5.5 points. Because another observed crop would remain part of detected cultivation, these offsetting changes indicate that more land is classified as fallow rather than shifted to another crop category.

Panel A speaks to a temporary crop-establishment explanation. Under this explanation, reclaimed land is initially uncultivated during preparation, so the fallow difference should peak immediately after reclamation and then decline. The estimates show the opposite: no significant difference in the first two years, followed by lower detected cultivation and more ChinaCP fallow in the third. Thus, observed fallow is not simply a short-lived consequence of land preparation.

Conditional on detected cropland, Panel B asks whether stronger policy exposure changes the shares of newly reclaimed land under single, double, and triple cropping. The estimates show no clear differences: only the first-year decline in double cropping is marginally significant. The remaining frequency estimates are statistically insignificant.

[21] ChinaCP codes combine cropping frequency and crop identity where available. The mutually exclusive classes are fallow; single maize, rice, wheat, or other crops; other double cropping; rice-maize; wheat-maize; double rice; wheat-rice; and triple cropping. Pixels without a valid code are labeled ChinaCP unclassified in this analysis.

Panel C further decomposes the single- and double-cropping components into nine crop-specific ChinaCP classes. None of these coefficients is statistically significant at the 5 percent level in any of the first three post-reclamation years. Several estimates are marginally significant at the 10 percent level, but none remains significant after applying the Benjamini–Hochberg false discovery rate adjustment across the nine classes within each event year. The estimates therefore provide no evidence that stronger policy exposure shifts cultivation toward a particular observed crop or crop rotation.

The timing evidence in Panel A and the composition evidence in Panels B and C provide little support for either a short-lived establishment phase or a shift toward other observed crops or cropping frequencies. They are instead consistent with the main result: stronger policy exposure is associated with lower realized cultivation on newly reclaimed land.

D Reservoir Constraints and Reclamation Choices

This appendix details the historical-reservoir decomposition in Section 6.1.3. The exercise uses the 1995 suitable-land share for both reclaimed land and the remaining reservoir. Source pixels are classified as suitable when their score is at least three, before area-weighted aggregation to 500-meter grids. The accounting sample is at the county–cohort level; the exercise is not a decomposition of the main grid-level treatment coefficient.

D.1 Reservoir definitions

The fixed initial reservoir $\mathcal{R}_{c,2001}$ consists of eligible land that is not cropland in 2001. The 500-meter main mask excludes baseline cropland and impervious land, pre-policy urban cores, and open-water interiors beyond the retained 500-meter water fringe. Let a_j denote land area and $C_{j,t-1}$ its cropland status at the end of year $t-1$. The factual reservoir available at the beginning of year t is

$$\mathcal{R}^{\text{rem}}_{c,t-1} = \{j \in \mathcal{R}_{c,2001} : C_{j,t-1} = 0\}.$$

Land leaves this pool while it is cropland and re-enters after conversion back to noncropland. Thus depletion is net of subsequent cropland loss, rather than cumulative gross reclamation. The remaining pool contains both suitable and unsuitable land.

Let $R_{c0} = \sum_{j\in\mathcal{R}_{c,2001}} a_j$ and $R_{c,t-1} = \sum_{j\in\mathcal{R}^{\text{rem}}_{c,t-1}} a_j$. Let $G^{95}_j \in [0,1]$ denote the suitable-land share of the 500-meter grid containing land unit j: the area-weighted mean of source-pixel indicators for soil scores of at least three. The threshold is applied before aggregation, rather than to the mean grid score. Each land unit inherits its grid's share. The three predictors, measured in percentage points, are

$$S_{c,t-1} = 100\frac{\sum_{j\in\mathcal{R}^{\text{rem}}_{c,t-1}} a_j G^{95}_j}{R_{c,t-1}}, \qquad \text{remaining suitable share,} \tag{6}$$

$$D_{c,t-1} = 100\left(1 - \frac{R_{c,t-1}}{R_{c0}}\right), \qquad \text{depletion,} \tag{7}$$

$$I_{ct} = 100\frac{A_{ct}}{R_{c,t-1}}, \qquad \text{reclamation intensity.} \tag{8}$$

Here A_{ct} is the area first reclaimed in cohort t. The available land-cover record ends in 2022, so these persistent-reclamation cohorts end in 2021. The fixed reservoir contains 107,931 grids. To retain the original geographic sample, the 693 grids without mapped soil coverage contribute zero suitable area, as in the original construction; they contain 11,036 hectares of initial reservoir land.

D.2 Historical reference and aggregate accounting

The outcome is the percentage of newly reclaimed land classified as suitable by the 1995 soil measure. I estimate the following model at the county–cohort level using pre-policy cohorts

from 2002–2017:

$$Q_{ct} = \alpha_c + \lambda_t + \beta S_{c,t-1} + \delta D_{c,t-1} + \gamma I_{ct} + \varepsilon_{ct}. \tag{9}$$

This model uses three predictors: the remaining suitable share and depletion at the beginning of the year, and current reclamation intensity. These variables predict the percentage of newly reclaimed land in each county and year classified as suitable by the 1995 soil measure. Estimation weights each observation by reclaimed area and clusters standard errors by county. The full model uses 421 county–cohort observations in 51 counties. I also estimate simpler models using only the remaining suitable share, or the remaining suitable share and depletion. Appendix Table D.1, Panel A, reports the full model.

Let $\bar{Q}_c^{\text{pre}}$, $\bar{S}_c$, $\bar{D}_c$, and $\bar{I}_c$ be reclaimed-area-weighted county means over the historical period. The reference applies the estimated slopes to departures from those means:

$$\widehat{Q}_{ct}^{\text{ref}} = \bar{Q}_c^{\text{pre}} + \widehat{\beta}(S_{c,t-1} - \bar{S}_c) + \widehat{\delta}(D_{c,t-1} - \bar{D}_c) + \widehat{\gamma}(I_{ct} - \bar{I}_c). \tag{10}$$

For each county, the prediction starts from the average suitability of land reclaimed before the policy. I then adjust this starting value for changes in the predictors, using their estimated pre-policy relationships with reclaimed-land suitability. Each model uses only the predictors included in its historical regression. The prediction asks how newly reclaimed land quality would change relative to the county's historical average, based only on changes in reservoir conditions and reclamation intensity. They use the observed remaining suitable share, depletion, and reclamation intensity for each post-policy year. Actual reclamation quantities and the actual remaining land pool are therefore unchanged.

For each post-policy county–cohort, define

$$\begin{aligned} L_{ct}^{\text{total}} &= \bar{Q}_c^{\text{pre}} - Q_{ct}, \\ L_{ct}^{\text{frontier}} &= \bar{Q}_c^{\text{pre}} - \widehat{Q}_{ct}^{\text{ref}}, \\ L_{ct}^{\text{selection}} &= \widehat{Q}_{ct}^{\text{ref}} - Q_{ct}. \end{aligned} \tag{11}$$

For each county and reclamation year, the total quality decline is the difference between the county's pre-policy average suitability and the suitability of land actually reclaimed. The frontier component is the decline predicted by the historical model. The selection component is the additional shortfall of actual suitability below that prediction. These two components sum to the total decline. I average the losses across 92 county–cohort observations in 37 counties over the post-policy period, 2018–2021, giving more weight to cohorts with more reclaimed land. On this basis, the suitable share falls from a historical average of 48.622 percent to an observed 38.466 percent, a decline of 10.156 percentage points. In the full specification, the frontier component is 2.718 points and the selection wedge is 7.438 points, or 73.2 percent of the total. Each component's share is its average loss divided by the average total loss. Panel B reports the decomposition also shown in the main-text table.

D.3 Pressure-related quality losses

For each component $k \in \{\text{total}, \text{frontier}, \text{selection}\}$, I estimate

$$L_{ct}^{k} = \alpha_{c}^{k} + \lambda_{t}^{k} + \tau^{k}(P_{c} \times \text{Post}_{t}) + u_{ct}^{k}, \tag{12}$$

where P_c is standardized county pressure and post denotes cohorts 2018–2021. The three regressions use identical observations, weights equal to the number of newly reclaimed 500-meter grids, county and cohort FE, and county-clustered inference. Unlike the detailed reclaimed-area weights in the historical models and aggregate accounting, these weights measure reclaimed-grid counts. Their coefficients therefore satisfy $\widehat{\tau}^{\text{total}} = \widehat{\tau}^{\text{frontier}} + \widehat{\tau}^{\text{selection}}$. The estimable sample contains 188 county–cohort observations in 37 counties. These coefficients capture the differential post-policy loss per standard deviation of pressure, not the nationwide effect of the policy.

Panel C reports the three additive references and a fourth reference allowing the historical remaining-suitability slope to vary with pressure. The pressure main effect is absorbed by county FE. Its reference is

$$\widehat{Q}_{ct}^{\text{ref},P} = \bar{Q}_{c}^{\text{pre}} + (\widehat{\beta} + \widehat{\theta} P_{c})(S_{c,t-1} - \bar{S}_{c}).$$

Panel C shares are $100\widehat{\tau}^{k}/\widehat{\tau}^{\text{total}}$. Reported standard errors condition on the estimated historical-reference coefficients.

Panel C shows that the selection wedge accounts for 87.5–99.7 percent of the additional quality loss in higher-pressure counties and is statistically significant at the 5 percent level across all four specifications. The total loss is 6.200 percentage points per standard deviation of pressure. In the full additive specification, the selection wedge is 6.179 points and the frontier component is 0.021 points. None of the four frontier estimates is statistically distinguishable from zero. Relative to the historical benchmark, the residual selection component accounts for most of the pressure-related decline; the exercise does not separately identify a causal effect of distorted selection.

Table D.1: Historical-Reservoir Decomposition with Depletion and Reclamation Intensity

Panel A. Historical quality prediction, 2002–2017

Predictor	Estimate (1)	Standard error	p-value
Remaining suitable share	3.439**	1.683	0.046
Depletion	−0.018	0.426	0.966
Reclamation intensity	0.943	1.713	0.584
County FE		✓	
Reclamation-cohort FE		✓	
Reclaimed-area weights		✓	
County–cohort observations		421	
Number of counties		51	
R^2		0.748	
Within R^2		0.031	
Cluster level		County	

Panel B. Aggregate accounting, 2018–2021

Reference specification	Total loss	Frontier component	Selection wedge	Frontier share (%)	Selection share (%)	Reference quality
Remaining suitable share	10.156	2.413	7.743	23.8	76.2	46.209
Remaining suitable share + depletion	10.156	2.204	7.952	21.7	78.3	46.418
Remaining suitable share + depletion + reclamation intensity	10.156	2.718	7.438	26.8	73.2	45.904

Panel C. Pressure-induced pooled decomposition

Reference predictors	Total loss (1)	Frontier component (2)	Selection wedge (3)	Frontier share (%)	Selection share (%)
Remaining suitable share	6.200***	0.280	5.919***	4.5	95.5
	(2.026)	(0.460)	(2.128)		
Remaining suitable share + depletion	6.200***	0.273	5.927***	4.4	95.6
	(2.026)	(0.471)	(2.133)		
Remaining suitable share + depletion + reclamation intensity	6.200***	0.021	6.179***	0.3	99.7
	(2.026)	(0.388)	(2.076)		
Remaining suitable share × pressure	6.200***	0.778	5.422**	12.5	87.5
	(2.026)	(0.637)	(2.216)		
County FE			✓		
Reclamation-cohort FE			✓		
Reclaimed-grid-count weights			✓		
Observations		188 in each regression			
Number of counties		37 in each regression			
R^2 (rows 1–4)	0.360; 0.360	0.557; 0.556	0.337; 0.336		
	0.360; 0.360	0.586; 0.596	0.334; 0.330		
Cluster level			County		

Notes: All three predictors are measured in percentage points. Suitable-land shares apply the score-at-least-three indicator to source pixels in the 1995 soil map before area-weighted aggregation to each 500 m grid. Remaining suitable share is suitable area divided by beginning-of-year remaining reservoir area. Depletion is beginning-of-year cropland area within the fixed 2001 reservoir divided by its initial area. Reclamation intensity is current reclaimed area divided by beginning-of-year remaining reservoir area. Each ratio is multiplied by 100. The fixed reservoir contains eligible noncropland under the main mask and permits land to re-enter after cropland loss. Uncovered soil-map grids retain zero suitable area. Historical models use 2002–2017 cohorts, county and cohort FE, reclaimed-area weights, and county-clustered standard errors. Panel B uses reclaimed-area-weighted averages over 92 county–cohort observations in 37 counties during 2018–2021. Panel C uses reclaimed-grid-count weights on the unchanged 2012–2021 manuscript cohort sample. Rows 1–3 use additive reference predictors; row 4 interacts remaining suitable share with standardized pressure. The pressure main effect is absorbed by county FE. References apply historical slopes to deviations from county pre-policy means. Panel C reports pressure × post coefficients, where post denotes 2018–2021. Total loss equals frontier plus selection; positive values denote quality loss. Standard errors in parentheses condition on estimated historical-reference coefficients. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Agricultural Losses and Land-Conveyance Revenue

This appendix documents the calculations in Section 6.3. The land-conveyance comparison places agricultural losses on a common per-hectare monetary scale; it does not estimate the net benefits of conversion or the value of food security.

E.1 Annual Agricultural Losses

The annual monetary value of one administrative hectare of newly reclaimed cropland follows from the agricultural production function. [22] The resulting value expression is

$$v^{ha}(F, Q) = \kappa(1 - F)Q^{\Omega}. \tag{13}$$

Here F is the fallow share and Q is land quality. Intuitively, part of an administrative hectare remains fallow, for example because land-use rights are not leased or transferred to farmers or agricultural operators. This share generates no agricultural output. The remaining share $1-F$ enters cultivation. For the cultivated portion, agricultural value rises with predetermined land quality Q, with elasticity Ω after farmers optimally adjust non-land inputs. The scalar κ captures common agricultural productivity, output prices, input costs, and the conversion from physical output to monetary value. Following Adamopoulos et al. (2024), the calibration sets $\Omega = 0.615$.[23]

To link agricultural value to pressure, we combine the value expression with the reduced-form relationships between pressure and the two agricultural margins. Let z denote standardized cropland protection pressure. The reduced-form analysis gives $F(z) = \bar{F} + \hat{\beta}_F z$ and $Q(z) = \bar{Q} + \hat{\beta}_Q z$, where $\hat{\beta}_F$ and $\hat{\beta}_Q$ are the corresponding estimates. Since $\bar{v}^{ha} = \kappa(1 - \bar{F})\bar{Q}^{\Omega}$, the value of an administrative hectare at pressure z can be written without separately observing κ:

$$v^{ha}(z) = \bar{v}^{ha} \frac{1 - F(z)}{1 - \bar{F}} \left(\frac{Q(z)}{\bar{Q}} \right)^{\Omega}.$$

Differentiating this expression gives the marginal agricultural cost of pressure:

$$MC^{ag} \equiv - \left. \frac{\partial v^{ha}(z)}{\partial z} \right|_{z=0} = \bar{v}^{ha} \left[\frac{\hat{\beta}_F}{1 - \bar{F}} - \Omega \frac{\hat{\beta}_Q}{\bar{Q}} \right]. \tag{14}$$

[22] Following Chen et al. (2023), suppose physical output on cultivated land is $y(q, l, x) = a[(ql)^{\theta} x^{1-\theta}]^{\gamma}$, where q is land quality, l is land input, and x collects labor, capital, and intermediate inputs. A farmer chooses the endogenous input x to maximize $p\, y(q, l, x) - c_x x$. The first-order condition gives $x^*(q, l) = [pa(1 - \theta)\gamma / c_x]^{1/[1-(1-\theta)\gamma]} (ql)^{\theta\gamma/[1-(1-\theta)\gamma]}$. Substituting x^* back into the production function yields $y^*(q, l) = A(ql)^{\Omega}$, where $\Omega = \theta\gamma/[1-(1-\theta)\gamma]$ and A collects constants. This derivation gives the monetary value of the cultivated portion of reclaimed land. Setting $l = 1$ for a cultivated hectare and absorbing monetary conversion into κ gives cultivated value κQ^{Ω}. An administrative hectare also includes the fallow share F, which contributes zero agricultural value; multiplying cultivated value by $1 - F$ gives equation (13).

[23] The calibration uses $\theta = 0.533$ and $\gamma = 0.750$, which imply $\Omega = \theta\gamma/[1 - (1 - \theta)\gamma] = 0.615$.

Table 8 reports the exact annual discrete loss from a one-standard-deviation increase in pressure:

$$\ell_{1sd}^{ag} = \bar{v}^{ha}\left[1 - \frac{1-(\bar{F}+\hat{\beta}_F)}{1-\bar{F}}\left(\frac{\bar{Q}+\hat{\beta}_Q}{\bar{Q}}\right)^{\Omega}\right]. \tag{15}$$

Here ℓ_{1sd}^{ag} is measured in RMB per hectare per year. Q is the continuous HWSD v2 suitable-area share used in Table 1. The calibration uses that table's mean (0.573217) and coefficient (-0.031571), multiplying the coefficient by the pressure standard deviation (1.104211). The suitable-area share is interpreted as a productivity index; this is a calibration assumption, not an estimated mapping from soil quality into output. The monetary baselines are annual value added per cropland hectare, averaged over 2012–2016 for primary industry and available 2013–2016 observations for agriculture, after adjustment to 2017 prices using China's consumer price index. These are value-added losses.

Table E.1: Calibration Inputs for the Agricultural Economic Magnitude

Input	Value	Source or interpretation
Pressure standard deviation	1.104211	Standardized across non-core counties.
Baseline fallow share, $\bar{F}$	0.197828	Pre-2018 reclamation cohorts, post-reclamation years 1–4.
Fallow effect, $\hat{\beta}_F$	0.060919	Corrected cohort DiD.
Baseline suitable-land share, $\bar{Q}$	0.573217	HWSD v2 suitable-area share; Table 1.
Suitability effect, $\hat{\beta}_Q$	-0.034862	Table 1 coefficient -0.031571, multiplied by the pressure standard deviation.
Quality elasticity, Ω	0.615	Production-function calibration; $\theta = 0.533$, $\gamma = 0.750$.
Agricultural baseline (RMB/ha/year)	31,155.634	Available 2013–2016 county-years; 232 observations.
Primary-industry baseline (RMB/ha/year)	49,856.068	2012–2016 county-years; 316 observations.
Exact annual loss share (%)	11.0921	Equation (15).
Agricultural marginal cost (RMB/ha/year)	3,531	Equation (14).
Primary-industry marginal cost (RMB/ha/year)	5,651	Equation (14).
Reclaimed area (ha)	14,624.133	Sum of qualifying 2018–2022 grid-year area increments.
Pressure-weighted area (ha)	8,283.565	Positive standardized pressure weights.
Benchmark capitalization rate	5.8%	Yu (2026); sensitivity: 3%, 5%, 7%.
Nearby transactions / area (ha)	712 / 1,887.97	Same-year, five-kilometer newly added construction-land conveyances.

Notes: Monetary values are in 2017 RMB. This table lists inputs to a calibration. The corrected fallow coefficient has county-clustered standard error 0.016860, $N = 3,244$, 54 counties, and $R^2 = 0.232$; it includes county, cohort, and years-since-reclamation fixed effects. The HWSD calibration uses the mean and coefficient from Table 1. Suitability is the area-weighted share of soil components scoring at least three. Baseline monetary values are area-weighted value added per cropland hectare. Other inputs retain their available precision.

E.2 Persistence, Capitalization, and Provincial Aggregation

The benchmark rate, $r = 0.058$, follows Yu (2026), Appendix D.3 and E.3, who uses China's average lending rate over 2000–2015. Because this rate is not explicitly inflation-adjusted, its application to constant-price losses is a calibration assumption. Capitalized losses are

$$PV_{ha}^{ag}(r) = \sum_{s=1}^{\infty} \frac{\ell_{1sd}^{ag}}{(1+r)^s} = \frac{\ell_{1sd}^{ag}}{r}. \tag{16}$$

The sum excludes the current-period flow included in Yu's valuation. Permanent protection does not establish permanent productivity losses; persistence beyond the sample remains an assumption. The rate alternatives are 7%, Yu's median rural-household borrowing rate, and 3% and 5%, used by Adamopoulos et al. (2024) to value Chinese farmland income over 30 years. Here all horizons remain infinite.

Let G_{ct} contain 25-hectare grids whose cropland share rises from below 0.60 to above 0.70 in year t. County reclaimed area and positive standardized pressure are

$$A_c = 25 \sum_{t=2018}^{2022} \sum_{i \in G_{ct}} (\text{CropShare}_{it} - \text{CropShare}_{i,t-1}), \qquad z_c^+ = \max\{z_c, 0\}.$$

Provincial losses are

$$\ell^{JS} = \sum_c A_c z_c^+ \ell_{1sd}^{ag}, \qquad PV_{2022}^{JS}(r) = \ell^{JS}/r. \tag{17}$$

Linear scaling by z_c^+ is an aggregation approximation. One unmatched county (8.202 hectares) receives zero weight. Area sums grid-year transitions, without verifying unique parcels or permanent-basic-farmland designation. The valuation covers future losses on the 2018–2022 cohorts.

Table E.2: Sensitivity of Perpetual Losses to the Capitalization Rate

	Per-hectare present value (RMB thousand/ha)		Jiangsu present value (RMB million)	
Rate	Agricultural	Primary industry	Agricultural	Primary industry
3%	115.2	184.3	954.2	1,527.0
5%	69.1	110.6	572.5	916.2
5.8% (baseline)	59.6	95.3	493.6	789.8
7%	49.4	79.0	408.9	654.4

Notes: All values assume constant annual losses over an infinite horizon. Present values are measured at end-2022 in 2017 RMB, with the first future loss at end-2023. Agricultural denotes the agricultural value-added baseline. The 5.8% benchmark and 7% alternative follow Yu (2026); 3% and 5% are the endpoints used in the farmland-income valuation exercise of Adamopoulos et al. (2024). Only the rate varies across rows.

E.3 Nearby Land-Conveyance Revenue

The Jiangsu China Land Market Network records cover 2018–2022. Eligible conveyances have positive payments and areas and originate from newly added construction land. Allocations, leases, existing construction land, stock-library supply, and explicitly nonurban uses are excluded; unspecified construction uses remain. Each transaction enters once, even when matched to multiple reclaimed grids. Identical item-link repetitions and one link with conflicting area records are removed.

For nominal payment P_j, area a_j in hectares, and contract year t_j, revenue per hectare is

$$\bar{P}^{ha}_{2017} = \frac{\sum_j P_j \, CPI_{2017}/CPI_{t_j}}{\sum_j a_j}, \tag{18}$$

using the World Bank's China CPI series. Recorded payments are converted from RMB 10,000 to RMB before deflation.

Comparables cover 439 of 1,688 reclamation events and 22.5% of reclaimed area; unmatched areas receive no imputed price. Reported coordinates are treated as WGS84, with distances computed in UTM zone 50N, although the original datum and geocoding precision are undocumented. In 2022, 52.3% of otherwise eligible priced conveyances lack valid coordinates. Twenty-four transactions share one coordinate and account for 19.6% of benchmark receipts. These coverage and location limitations constrain the benchmark's representativeness.

F Marketization and the Incidence of Cropland Protection

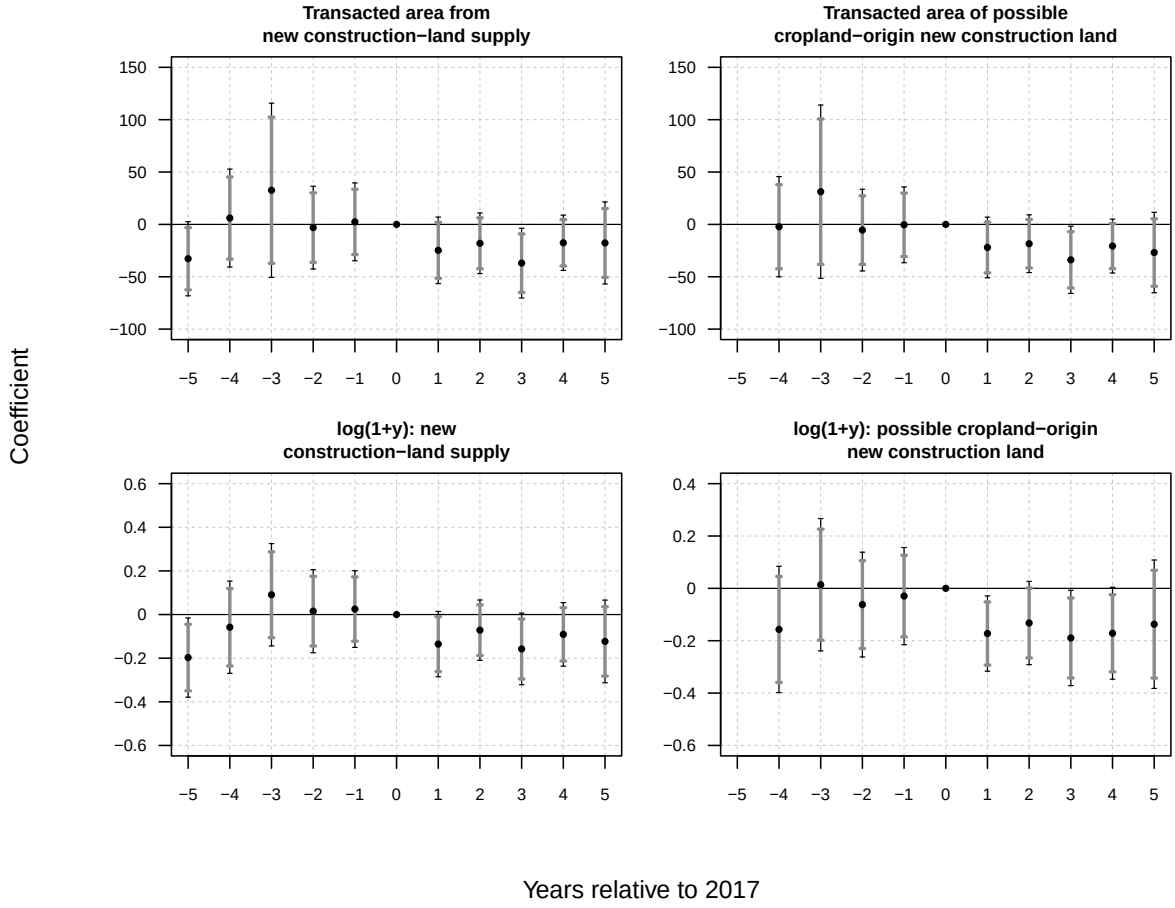
An important policy question is how to reduce the cropland quality and productivity losses induced by binding local land constraints. When enforcement tightens in land-scarce jurisdictions, meeting the compensation obligation through local reclamation pushes expansion toward less suitable plots. Quota trading offers an alternative: a jurisdiction can purchase supplementary-cropland indicators created elsewhere rather than meet the full obligation through local reclamation.

Jiangsu operates a provincial platform for such transfers (Department of Natural Resources of Jiangsu Province, 2023) (see Appendix Section G.3 for the earlier provincial rule). Yet this adjustment margin is costly. Local reclamation fees range from RMB 30 to 50 per square meter, whereas public platform results commonly report transaction prices of RMB 420 per square meter for paddy indicators and RMB 345 for non-paddy indicators (Jiangsu Provincial Real Estate Development Center, 2026). This gap raises a policy-design question: why does the trading system sustain prices far above direct reclamation costs instead of making quota purchases a more feasible and widely used compliance margin?

One hypothesis is that quota prices reflect the opportunity cost of reallocating cropland protection across space. Trading can spare a land-scarce buyer from reclaiming lower-quality plots, but the supplying jurisdiction must commit its own land to cropland protection. This commitment can restrict future urban expansion and the associated development opportunities. Consistent with this mechanism, Yu (2026) shows that cropland protection constraints restrict urban land supply and generate aggregate welfare losses. Lower transfer prices could therefore reduce quality losses in high-pressure buyers while concentrating cropland protection and its development costs in land-abundant suppliers. High prices may compensate suppliers for this spatial incidence, creating an equity-efficiency trade-off across jurisdictions.

To assess this opportunity-cost channel, I ask whether counties that absorb larger compensation obligations from core urban districts subsequently supply less construction land. Appendix Figure F.1 shows a relative decline in construction-land transaction supply after 2017 among higher-pressure counties. Appendix Table F.1 reports similar estimates for total transacted construction land, new construction-land supply, and possible cropland-origin new construction land. In the outer ring of the pre-policy built-up area, a one-standard-deviation increase in cropland protection pressure reduces these outcomes by approximately 12 to 16 percent after 2017. These estimates show that absorbing a larger cropland protection burden carries an urban-development cost. The same opportunity cost provides a rationale for high quota prices: suppliers must be compensated not only for creating replacement cropland but also for the development opportunities forgone when land is committed to cropland. High prices protect supplying jurisdictions from bearing this burden without compensation, but they also limit quota trading's ability to relax binding land constraints and reduce quality losses in high-pressure buyers.

Figure F.1: Event-Study Estimates for Construction-Land Transaction Supply



Notes: This figure plots county-level event-study estimates for realized primary-market construction-land transaction records in Jiangsu from 2012 to 2022. The estimating equation interacts county-level cropland protection pressure with year indicators, omitting 2017, and includes county and year FE. The top panels use area outcomes in hectares; the bottom panels use $\log(1 + y)$, where y is the corresponding hectare outcome. The left panels restrict to transacted area from new construction-land supply. The right panels further restrict to transactions in possible cropland-origin new construction land, defined as new-supply transactions whose centroid falls in a 500 m grid cell with any cropland cover in the previous-year land-cover data. Thin bars denote 95% confidence intervals; thick bars denote 90% confidence intervals. Standard errors are clustered by county.

Table F.1: Construction-Land Expansion Responses to Pressure

	Total transacted construction-land area		Transacted area from new construction-land supply		Transacted area of possible cropland-origin new construction land	
	Whole county (1)	20km outer ring (2)	Whole county (3)	20km outer ring (4)	Whole county (5)	20km outer ring (6)
Pressure × Post	-0.113**	-0.150**	-0.134*	-0.180**	-0.141*	-0.154*
	(0.056)	(0.058)	(0.077)	(0.085)	(0.078)	(0.091)
County FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
County linear trends	✓	✓	✓	✓	✓	✓
Observations	704	704	704	704	640	640
Number of counties	64	64	64	64	64	64
R^2	0.538	0.750	0.459	0.672	0.541	0.689

Notes: The unit of observation is a Jiangsu county-year from 2012 to 2022. The sample excludes fully core-urban districts, retaining non-fully-core districts. Outcomes are $\log(1+y)$, where y is the county-year sum of transaction area in hectares before the log transformation. Total transacted construction-land area sums all realized primary-market construction-land transaction records. Transacted area from new construction-land supply restricts to parcels whose source category indicates newly converted construction land. Transacted area of possible cropland-origin new construction land further restricts to those new-construction-source parcels whose centroid lies in a 500m grid cell with cropland cover in the previous year; the 2012 proxy is omitted because 2011 grid-level land cover is unavailable. Within each outcome pair, the first column uses all county transactions and the second uses transactions outside the pre-policy built-up patch and within 20 km of that patch's edge. The outer-ring columns focus on urban expansion at the margin, which typically proceeds outward from the existing built-up patch. The coefficient is Pressure × Post. All specifications include county and year FE and county-specific linear trends. Standard errors are clustered by county. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

G Policy and Source Documentation

This section reports the policy documents, government materials, news accounts, and historical reports cited collectively as Appendix G in the main text. Each entry states the institutional fact, mechanism, or numerical benchmark for which the source is used.

G.1 National Policy and Land-Use Rules

- **Cropland occupation–compensation.** State Council of the People's Republic of China (1998), *Implementation Regulations of the Land Administration Law of the People's Republic of China*, Decree No. 256, effective January 1, 1999. This regulation establishes the national occupation–compensation requirement and the one-for-one replacement rule.
- **National land-use plan.** State Council of the People's Republic of China (2006), *Outline of the National Overall Land Use Plan (2006–2020)*. This plan provides the basis for the national cropland-retention target and the hierarchy of land-use controls.
- **The 2017 Opinions.** Central Committee of the Communist Party of China and State Council (2017), *Opinions on Strengthening Cropland Protection and Improving the Balance of Cropland Occupation and Compensation*, January 9, publicly released January 23, State Council Gazette, No. 5. The same policy underlies the discussion of tighter cropland-protection accountability after 2017. Section I.3 links cropland protection and the occupation–compensation system to domestic grain-producing capacity, broad self-sufficiency in cereals, and secure supplies of staple food. Sections I.1–I.2 stress quantity, quality, and ecological protection together. These objectives concern physical supply security and productive capacity, not merely agriculture's monetary contribution to GDP. Official text.
- **Land-conveyance revenue.** General Office of the State Council (2006), *Notice on Regulating the Administration of Revenue and Expenditure from the Conveyance of State-Owned Land-Use Rights*, Document No. 100, effective January 1, 2007. This notice supports the discussion of land-conveyance receipts as local fiscal revenue and their permitted uses.
- **Construction-land conveyance.** Ministry of Land and Resources (2007), *Provisions on the Assignment of State-Owned Construction Land-Use Rights through Invitation to Bid, Auction, and Listing*, Order No. 39, effective November 1, 2007. These provisions establish the public bid, auction, and listing procedures described for construction land.
- **Rural land contracting.** Standing Committee of the National People's Congress (2018), *Rural Land Contracting Law of the People's Republic of China*, amended December 29, 2018 and effective January 1, 2019. This law supports the discussion of household contracting, transferable operating rights, and restrictions on changing agricultural use.
- **Land-use controls.** Standing Committee of the National People's Congress (2019),

Land Administration Law of the People's Republic of China, amended August 26, 2019 and effective January 1, 2020. This law provides the land-use categories, planning requirements, and approval process for converting agricultural land.

G.2 Cropland Quality and Monitoring

- **Cropland-quality concerns.** Guanghua Wang (2023), *Earnestly Strengthen Cropland Protection and Reform to Improve the Cropland Occupation–Compensation Balance System*, *People's Daily*, October 10, 2023, p. 10. This account provides evidence on low-quality reclamation, ecological damage, fallowing, and the later loss of newly created cropland.
- **High-standard farmland construction standard.** State Administration for Market Regulation and Standardization Administration of China (2022), *Well-facilitated Farmland Construction—General Rules*, GB/T 30600–2022, issued March 9, 2022 and effective October 1, 2022. This national standard specifies plot improvement, field-road, irrigation, drainage, and other infrastructure requirements for mechanized and large-scale farming.
- **High-standard farmland upgrading.** Ministry of Agriculture and Rural Affairs (2022), *Guiding Opinions on Promoting the Reconstruction and Upgrading of High-Standard Farmland*, Nong Jian Fa [2022] No. 5, November 9, 2022. This guidance calls for appropriately consolidating and leveling plots and improving field roads to support mechanized operations.
- **National monitoring rules.** Ministry of Agriculture (2016), *Measures for the Survey, Monitoring, and Evaluation of Cropland Quality*, issued June 21, 2016 and effective August 1, 2016. These rules require monitoring sites to represent local soil types and cropping systems.
- **Jiangsu monitoring guidance.** Department of Agriculture and Rural Affairs of Jiangsu Province (2021), *Opinions on Effectively Strengthening Cropland-Quality Monitoring and Evaluation in Jiangsu Province*, Su Nong Ye [2021] No. 37. This guidance supports the description of Jiangsu's fixed monitoring network and minimum coverage requirements.
- **Nantong implementation.** Nantong Municipal Bureau of Agriculture and Rural Affairs (2024), *Notice on Further Strengthening Cropland-Quality Monitoring and Evaluation*, Tong Nong Fa [2024] No. 9, January 24, 2024. This local rule motivates the use of distance to a monitoring station as a proxy for exposure to quality oversight.

G.3 Provincial Constraints and Indicator Trading

- **Jiangsu.** General Office of the Jiangsu Provincial People's Government (2021), *Jiangsu Province's 14th Five-Year Plan for Natural Resources Protection and Utilization*, Su Zheng Ban Fa [2021] No. 41, August 9, 2021. This plan supports the claim that Jiangsu faces tight land availability and competing development and protection demands.

- **Zhejiang.** Zhejiang Online (2024), *Balancing Cropland Occupation and Compensation: Zhejiang Reorganizes Its Cropland-Protection Strategy*, December 27, 2024. This account describes the difficulty of satisfying compensation obligations within a high-pressure coastal province.
- **Guangdong.** Department of Land and Resources of Guangdong Province (2018), *Guangdong Territorial Plan (2016–2030)*. This plan provides a further example of land-use conflicts in a high-pressure coastal province.
- **Interim trading measures.** General Office of the Jiangsu Provincial People's Government (2014), *Interim Measures for Trading Supplementary-Cropland Indicators in Jiangsu Province*, General Office Document No. 10. These measures establish the earlier provincial basis for cross-jurisdiction trading of supplementary-cropland indicators.
- **Online trading rules.** Department of Natural Resources of Jiangsu Province (2023), *Detailed Rules for Online Trading of Provincial Supplementary-Cropland Indicators in Jiangsu Province*, February 2023. These rules establish the current online procedures for provincial indicator trading.
- **Transaction results.** Jiangsu Provincial Real Estate Development Center (2019–2026), *Public Results of Online Supplementary-Cropland Indicator Transactions*, Jiangsu Land Market Network, accessed May 24, 2026. These results provide the transaction-price benchmarks used in the paper: RMB 420 per square meter for paddy indicators and RMB 345 for non-paddy indicators.

G.4 Historical Reclamation-Cost Benchmarks

- Ramzi M. Sansur (1995), *Environment and Development Prospects in the West Bank and Gaza Strip*, United Nations Conference on Trade and Development, UNCTAD/ECDC/SEU/8. Its estimate of about $100 per dunum for previously cultivated but abandoned land, compared with $180–200 for never-cultivated marginal land, supports the use of cultivation history as a proxy for reclamation cost.
- H. H. Wooten and Margaret R. Purcell (1949), *Farm Land Development: Present and Future, by Clearing, Drainage, and Irrigation*, U.S. Department of Agriculture, Bureau of Agricultural Economics, Circular No. 825. Its estimate of $15–30 per acre for lightly vegetated, recently abandoned fields, compared with $95–160 for woodland converted to crops, provides a second cost benchmark for that interpretation.

G.5 Permanent Protection of Consolidated Farmland

- *National Land Consolidation Plan (2016–2020)* (2017), Guo Tu Zi Fa [2017] No. 2. Chapter III, Section 5, requires cropland consolidated under high-standard farmland construction to be designated as permanent basic farmland and protected against construction occupation. This requirement motivates the infinite-horizon valuation in Section 6.3. Official plan.